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A Standard Introduction to Non-Standard Logics

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The Publisher

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The harmony of the world is made manifest in Form and Number, and the heart and soul and all the poetry of Natural Philosophy are embodied in the concept of mathematical beauty.

– D'Arcy Wentworth Thompson

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Preface

This isn't the preface of the book.

The author

Introduction

This course provides a technical introduction to formal languages in logic. Formal languages play a central role in logic, philosophy, linguistics, and computer science, as they provide precise tools for representing structure, meaning, and inference. The aim of the course is to develop a clear understanding of how languages can be formally specified, manipulated, and compared, and how different formal systems capture different kinds of expressive power.

We begin with the *basic building blocks* of formal languages. This includes the notions of characters and strings, the distinction between use and mention, the role of quotation marks, and the operation of concatenation. We also discuss how a single object may be associated with multiple names, and introduce the idea of an expression as a well-formed string that carries meaning within a language. These foundational notions will be treated with care, as they underlie all later developments in the course.

The second part of the course focuses on *standard formal languages*. After introducing the necessary set-theoretic tools – such as sets, tuples, and formal definitions – we present the general concept of a formal language. We then study a range of canonical examples, including syllogistic languages, sentential (or propositional) languages, first-order languages, and modal languages. These systems illustrate how increasing syntactic resources allow us to express increasingly complex kinds of reasoning.

In the third part, we move beyond these familiar frameworks to consider *generalised standard languages*. Here we examine systematic extensions of standard languages, exploring how their syntax and semantics can be modified while preserving a clear formal structure.

Finally, we turn to *non-standard languages*. These include higher-order languages, plural languages, multi-sorted languages, and other systems that depart more radically from the standard first-order paradigm. Studying such languages helps to clarify both the limitations of standard frameworks and the conceptual costs and benefits of greater expressive power.

Aims of the Course

By the end of this course, students should be able to:

- understand and precisely apply the basic notions of characters, strings, expressions, and formal definitions;

- ▶ distinguish clearly between use and mention, and correctly handle quotation and concatenation;
- ▶ define and work with standard formal languages and their syntactic components;
- ▶ compare different formal languages in terms of expressive power and structure;
- ▶ appreciate the motivations for, and consequences of, moving beyond standard first-order languages;
- ▶ read, construct, and critically assess formal systems used in logic and related disciplines.

Part I

THE BASICS

1.1 Types and Tokens

Expressions. Ideas, reasoning, information are all communicated by means of language. The objects of language by which we communicate them are called *expressions*.

Notion 1.1 (Expression). An **expression** is a reproducible pattern of symbols that can be written or printed or displayed or uttered.

Letters, numerals (names for numbers), digits (one-character numerals), words, sentences, paragraphs, for instance, count as expressions, and so do made-up symbols or any other combination with no established meaning. In what follows, we will focus on printable or displayable expressions, rather than on utterable ones.

Type, token, and occurrence. A distinction may be made between a concrete appearance of an expression and its abstract shape, that is, between an *expression token* and an *expression type*.

Notion 1.2 (Token). An **expression token** is a physical instance of an expression, e.g., a printing of the expression on a page.

Notion 1.3 (Type). An **expression type** is the abstract entity that expression tokens instantiate.

For instance, each of the following three lines display distinct tokens of the same expression type:

bacaba
bacaba
bacaba

Unless otherwise specified, when we speak of an expression, we will refer to its type rather than to a token.

A common but highly misleading conflation is that between an expression token and an *expression occurrence*.

Historical note: Types and Tokens in Typography

In typography, *types* are the basic moulds from which text is imprinted (see figure). By contrast, a *token* is a particular, concrete mark printed from that mould.



Notion 1.4 (Occurrence). An **expression occurrence** is the appearance of an expression type within an expression token or within an expression type.

For instance, in the expression 'bacaba' (whether considered as this specific token or as the type that this token instantiates), there is one occurrence of the letter (type) 'c', there are two occurrences of the letter 'b', and there are three occurrences of the letter 'a'. In other words, the letters 'a', 'b', and 'c' occur one, two, and three times, respectively, in the word 'bacaba'.

Task 1.1 (Tokens and types). Consider the sentence:

Aristotle was a philosopher who lived in Ancient Greece.

Answer the following questions:

1. Which letter types occur in the sentence?
2. How many letter types does it contain?
3. How many tokens correspond to each of letter type?
4. How many letter tokens does it contain in total?

Treat each letter design as a distinct expression type – so that, e.g., the expressions 'a' and 'A' would count as different types.

To answer **task 1.1.1**, we treat uppercase and lowercase letters as distinct character types. Thus, the distinct letter types in our sentence are the expressions 'A', 'G', 'a', 'c', 'd', 'e', 'h', 'i', 'l', 'n', 'o', 'p', 'r', 's', 't', 'v', and 'w'. That is, we have 17 letter types, which answers **task 1.1.2**.

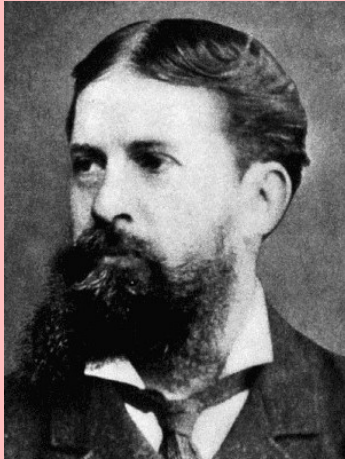
To answer **task 1.1.4**, we count the times each letter type identified occurs in our sentence (see table on the next page). Finally, to answer **task 1.1.4**, we add up these values, giving a total of 47.

Well-formed and ill-formed expressions. Not all expressions can perform a communicative function within a language. Each language – whether natural or artificial – has certain conventions,

Solution to task 1.1.

1. The distinct letter types in the sentence are: 'A', 'G', 'a', 'c', 'd', 'e', 'h', 'i', 'l', 'n', 'o', 'p', 'r', 's', 't', 'v', 'w'.
2. There are 17 letter types in total: 2 uppercase letters and 15 lowercase letters.
3. The number of tokens corresponding to each letter type is indicated in the table.
4. The total number of letter tokens equals 47.

Types	Tokens
e	7
i	5
o	4
h, l, n, r, t	3
A, a, c, p, s, w	2
G, d, v	1

Historical note: Type-Token Distinction in Logic

Charles S. Peirce (1839–1914)

The distinction between token and type was first introduced by Charles Sanders Peirce in a paper entitled 'Prolegomena to an apology for pragmatism' (1906, pp. 505–6):

There will ordinarily be about twenty *thes* on a page, and of course they count as twenty words. In another sense of the word "word", however, there is but one word "the" in the English language; and it is impossible that this word should lie visibly on a page. ... Such a definitely significant Form, I propose to term a *Type*. A Single event which happens once and whose identity is limited to that one happening ... I will venture to call a *Token*.

such as rules of syntax, that determine whether a given expression can fulfil that function. A symbol fulfilling such conventions counts as a *well-formed expression* of that language, while one which does not is an *ill-formed expression*.

Names, phrases, syntagms, sentences, and the like all count as well-formed expressions. Moreover, some symbols that lack meaning in English – like, e.g., the phrase 'the colour of my favourite number' – may also count as well-formed expressions in English.

However, many expressions – like, e.g., as 'favourite my the the my', 'AKARaaaA', and '\$)&(((' – do not qualify as well-formed expressions in English. Indeed most expressions are ill-formed rather than well formed with respect to any language.

Now, the distinction between *well-formed* and *ill-formed expressions* is quite specific to particular languages rather than to language in general. For this reason, we shall not explore it in depth for now.

1.2 Meaning and Quotation

Meaningful expressions. So far, we have spoken about expressions as such, which may or may not be well-formed. However, not every expression – even a well-formed one – is meaningful.

Consider, for example, the following expression:

the colour of the natural numbers.

Although grammatically well-formed in English, this expression does not refer to anything, since neither numbers nor sets of numbers have colours.

There are many different theories about what it means for an expression to have a meaning. Some theories connect meaning with mental ideas, others with use in language, and still others with truth-conditions.

We cannot settle these questions here. Instead, we will focus on a simpler idea: *when an expression is meaningful, it stands for something*. The thing a meaningful expression stands for is its *referent*.

Notion 1.5 (Meaningful expression). A **meaningful expression** stands for something, which is its **referent**.

The referent of an expression may be a physical object, such as a planet or a person; it may be a mathematical object, such as a number or a set. Some have even suggested that expressions may refer to fictional entities, such as a character in a story, or even impossible entities, such as a round square.

In natural language, the referent of an expression may vary because a single expression can have various meanings. For instance, the expression 'bank' may refer to a financial institution, or to the side of a river. Likewise, the expression 'bat' may refer to a flying mammal, or to a piece of equipment used in certain sports.

In ordinary language, ambiguity of this kind is very common, and usually the intended referent is determined by the larger expression within which the original expression occurs.

In contrast, the formal languages that we will study are designed to avoid such ambiguity. Each meaningful expression in a formal language is introduced with a fixed meaning, so that it always stands for one referent, regardless of the context, and not for different ones, which may vary according to the context.

For this reason, in what follows we will speak as if every meaningful expression has exactly one referent.

Task 1.2. Determine the referent of each of the expressions:

1. Earth's natural satellite
2. 2^3
3. Sherlock Holmes' flatmate
4. the largest prime number
5. the freezing point of water
6. the present king of France
7. The Beatles' first album
8. the number of letters in the word 'logic'

If an expression has no referent, state so.

Historical note: The Theory of Supposition

William of Ockham
(ca. 1287–1347)

It is difficult to attribute the introduction of the notions of *use*, *mention*, *standing for*, and *reference*, as we understand them today, to a particular source. Rather, they appear to have emerged gradually, with antecedents traceable to Aristotle.

A milestone in their development was reached in William of Ockham's *theory of supposition*, formulated by in his *Summa Logicae* (*Sum of Logic*, [ca. 1323](#)). The following is an extract:

Personal supposition, in general, occurs when a term stands for (*supponit pro*^a) the thing it signifies...

Material supposition is when a term does not stand signifi-
catively, but instead stands either for something spoken
or written. This is clear in 'man is a name', where the term
'man' stands for itself, and yet does not signify itself. (I.64)

^a More accurate translations prefer 'supposits for'.

Use and mention. We have determined to speak as if every meaningful expression had exactly one referent. There is nevertheless one notable exception we want to make to this rule.

Sometimes, we wish to speak about expressions themselves rather than about their referents. For example, we might wish to employ the word 'Berlin' not to speak about its referent, the city Berlin, but about the word 'Berlin', which contains six letters.

In such cases, the word 'Berlin' is not being *used* to express its referent, but is instead *mentioned* as a linguistic object.

Notion 1.6 (Use). A meaningful expression is **used** within another when the former occurs within the latter standing for its referent.

Notion 1.7 (Mention). Something is **mentioned** within a meaningful expression when an expression occurs within the latter standing for the former.

Note that, the notion of *mention* applies to any object whatsoever, including, e.g., objects in the real world and expressions. Instead, the notion of *use* applies only to meaningful expressions. Thus, we cannot use an object in this sense, as objects have no referent – though they can be the referent of an expression.

To exemplify, consider the following sentences:

Alice is a great musician. (7)

Alice has five letters. (8)

Solution to task 1.2.

1. the Moon
2. 8
3. John H. Watson or
no referent
4. no referent
5. 0° Celsius
6. no referent
7. the album *Please
Please Me*, of 1963
8. 5

In sentence (7), the name 'Alice' is being used to talk about Alice, the person. In sentence (8), instead, the name 'Alice' is being mentioned: we are talking about the word itself rather than using it to speak about its referent, (the person) Alice. Alice herself is only mentioned in (7). In fact, she cannot be used in the sense of **notion 1.7**, because Alice is not an expression – her name is.

Moreover, we used the symbols enclosed in parentheses in the previous paragraph – namely, '(7)' and '(8)', with parentheses included – to mention the expressions that appear to the left of their first occurrence in this section – namely, the sentences 'Alice is a great musician' and 'Alice has five letters'.

Also note that the enumerated expressions in **task 1.2** are being mentioned (through themselves), while the expressions in its solution are being used to express their referent.

Task 1.3 (Use and mention). For the sentence below, determine whether the underlined expression is being used or mentioned at that particular occurrence.

1. Socrates is mortal.
2. Socrates has three syllabi.
3. Dogs is the plural of dog.
4. Dogs bark.
5. Athens begins with the letter A.
6. Athens is the name of a Greek city.
7. Aristotle is not short.
8. The name of Aristotle is not short.

In **sentence 1.3.1**, the name 'Socrates' refers to the philosopher, thus being used. In contrast, in **sentence 1.3.2**, the name 'Socrates' is not employed to speak about the person but about the word itself, and it is thus being mentioned through itself.

A similar pattern appears in the next pair. In **sentence 1.3.4**, the word 'Dogs' mentions a kind of animal, thus being used. But in **sentence 1.3.3**, it refers to a word in English, thus being mentioned.

The word 'Athens' is mentioned rather than used in both **sentences 1.3.5** and **sentence 1.3.6**. Note, however, that although Athens, the city, is mentioned in **sentence 1.3.6**, we do not do so via the name 'Athens', but via the phrase 'a Greek city'. Hence, we are not using the name 'Athens' in **sentence 1.3.6**.

In **sentence 1.3.7**, the name 'Aristotle' has an ambiguous reference. The sentence may be interpreted either as using the name 'Aristotle' to say that the philosopher was small in size, or to say of that the

Historical note: Use and mention

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Author (XXXX–XXXX)

philosopher's name is brief in length. Hence, depending on the interpretation the name 'Aristotle' may be used or mentioned.

Now, in **sentence 1.3.8**, the phrase 'the name of Aristotle' refers to an expression, i.e., the name 'Aristotle'. However, the occurrence of the name 'Aristotle' within it refers to the philosopher. Hence, we use the name 'Aristotle' to mention Aristotle, but also mention the name 'Aristotle' by using the phrase 'the name of Aristotle'.

Quotation. The underlined expressions in **task 1.3** sometimes denote their referents and are sometimes used to denote themselves, depending on the sentence. More generally, expressions in natural language can be used either to convey their usual meanings or to refer to themselves as linguistic expressions.

Unfortunately, as **sentence 1.3.7** illustrates, it is not always easy to determine which meaning is intended. We have avoided those ambiguities by *using* expressions such as 'the word', 'the name', and 'the phrase', followed by the *mentioned* expression, enclosed in quote marks. Alternatively, we have introduced the mentioned expressions by means of a colon, placing it on a new, centred line.

From now on, we will explicitly adopt the *quote marks* notation.

Solution to task 1.3.

Used: 1, 4, 6, 8.

Mentioned: 2, 3, 5.

Ambiguous: 7.

Notation 1.8 (Quotation). An expression can be mentioned via another expression formed by: first, a left quote mark, next, the original expression, and, finally, a right quote mark.

For instance, to mention the first letter of the alphabet we use 'a', and to mention this latter expression we use "a", and so forth.

To explain this notation, I will use a visual trick: the sub-expression inside the quote marks of some quoted expressions will be coloured in order to show exactly what is being mentioned.

'Queen are legendary' mentions Queen and uses 'Queen'. "Queen' has five letters' mentions 'Queen' and uses "Queen". To mention Queen we use 'Queen', to mention 'Queen' we use "Queen", and to mention "Queen" we use "'Queen'". 'Queen' has five letters. "Queen" has seven characters: a left quote mark, five letters, and a right quote mark. Finally, note that 'Queen' names and mentions Queen, a legendary British rock band.

If this feels tricky, you are not alone, so take your time to grasp it. And even if you feel confident, pause and reflect again! Quotation is one most difficult things to grasp in logic, so patience pays off.

Task 1.4. Rewrite the sentences of **task 1.3** using quote marks. (Pay attention to non-underlined expressions too.)

The only sentences to modify are those mentioning expressions, that is: **sentences 1.3.2, 1.3.3, 1.3.5, 1.3.6**. All their underlined expressions should be placed between one pair of quote marks.

Additionally, **sentence 1.3.3** should enclose 'dog' in quote marks, since 'dogs' is not said to be the plural of some particular dog. Similarly, **sentence 1.3.5** should enclose 'A' in quote marks, as we are referring to the letter itself, and not to any referent that might be ascribed to 'A' in this or any other context.

Solution to task 1.4.

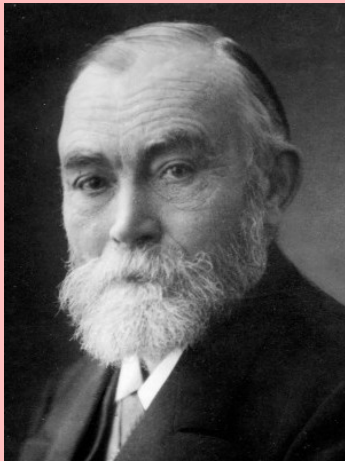
1. Socrates is mortal.
2. 'Socrates' has three syllabi.
3. 'Dogs' is the plural of 'dog'.
4. Dogs bark.
5. 'Athens' begins with the letter 'A'.
6. 'Athens' is the name of a Greek city.
7. Aristotle is not short. 'Aristotle' is not short.
8. The name of Aristotle is not short.

Quote marks make it easy to refer to the empty expression.

Notion 1.9 (Empty expression). The **empty expression** is the expression which contains no symbols.

Notation 1.10 (Empty expression). We mention the empty expression with **"**. Thus, **"** (i.e., the nothing between the quote marks) is the empty expression.

The empty expression is not the same as the blank space ' '.

Historical note: Quote marks

Gottlob Frege (1848–1925)

The modern use of quote marks in logic appears to have been first clearly articulated by Gottlob Frege in his paper ‘Über Sinn und Bedeutung’ (On sense and reference, 1892, p. 23):

When using words in the ordinary way, we wish to speak about their reference [*Bedeutung*]. But it may also happen that we wish to speak about the words themselves... We then have signs of signs. In writing, in such a case, we enclose the word-forms in quote marks. A word-form standing within quote marks must therefore not be taken in its ordinary reference. If we wish to speak of the sense [*Sinn*] of an expression ‘A’, we can do so simply by using the phrase ‘the sense of the expression ‘A’.

The name of the name of the name. Sometimes, we do not want to speak about the name of Socrates – that is, about ‘Socrates’ – but about the name of ‘Socrates’ – that is, about the name of the name of Socrates. Moreover, we may want to speak about the name of the name of the name of Socrates, or about the name of the name of the name of the name of Socrates. and so forth.

In such cases, remember that we can just add around our expression another pair of quote marks, as in “Socrates”, and another pair of quote marks, as in ““Socrates””, and so forth.

Task 1.5. Decide whether the following are true or false:

1. Aristotle is a philosopher.
2. Aristotle has nine characters.
3. Aristotle’s name has nine characters.
4. ‘Aristotle’ has nine characters.
5. ‘Aristotle’ is a philosopher.
6. “Aristotle” has nine characters.
7. “Aristotle” has eleven characters.
8. ““Aristotle”” is the name of the name of “Aristotle”.

Under the convention just proposed, it is clear that sentences 1.5.1, 1.5.3, 1.5.4, and 1.5.7 are true. In fact, note that sentences 1.5.3 and 1.5.4 mean the same, as Aristotle’s name is ‘Aristotle’.

Sentence 1.5.6 is clearly false. The expression referred to here is “Aristotle” and not ‘Aristotle’, since removing one pair of quote marks around “Aristotle” results in the former expression rather than in the latter.

Sentence 1.5.8 is also true, since, in our convention, the name of x is ' x '. Hence, the name of the name of x would be " x ". Thus, if we replace ' x ' with "'Aristotle'", we obtain what **sentence 1.5.8** says.

Finally, **sentences 1.5.2** and **1.5.5** may be regarded as false or nonsensical, depending on the account of truth we assume.

1.3 Sentences and Propositions

Sentences. We have used and mentioned several sentences so far, but we have not yet explained what is a sentence.

Notion 1.11 (Sentence). A **sentence** is an expression that is complete in the sense that it expresses a statement.

The following expressions are all examples of sentences:

'The door is closed.'	($\top$)
'Is the door closed?'	($\oslash$)
'Close the door!'	($\perp$)

Each of these expressions is complete in the sense that it suffices (in appropriate contexts) to perform a full communicative act.

Sentence ($\top$) is *declarative*, as it declares that the door is closed, which may be true or false. Sentence ($\oslash$) is *interrogative*, as it poses the question whether the door is closed, which has a specific true answer. Finally, sentence ($\perp$) is *imperative*, as it commands that the door be closed, which may or may not be carried out.

In each case, nothing further needs to be added in order to understand what is being said, asked, or demanded. In this sense, we might say that a sentence is, in general, an expression conveying a complete piece of information.

Sentences and propositions. Our first focus will be on declarative sentences. However, we will not consider any sentence in declarative form, but only *propositional sentences*, that is sentences expressing something that can be true or false or *propositions*.

Notion 1.12 (Proposition). A **proposition** is something that can be true or false.

Notion 1.13 (Propositional sentence). A **propositional sentence** is a sentence that stands for a proposition.

Remember that two expressions formed by different symbols are different even if they share the same meaning. Thus, two sentences may be different even if they express the same proposition. Take,

Solution to task 1.5.

True: 1, 3, 4, 7, 8.

False: 6.

Nonsensical or false: 2, 5.

Historical note: Judgement and the movements of the soul

A. Arnauld (1612–1694)

The classification of sentence types into declarative, interrogative, and imperative has a precursor in the Port-Royal's *Grammaire générale et raisonnée* (*General and Rational Grammar*, 1660, p. 19), by Antoine Arnauld and Claude Lancelot:

And thus the most important distinction concerning what takes place in our mind is that we may consider there the object of our thought, and the form or manner of our thought, of which the principal one is judgment. But we must also include conjunctions, disjunctions, and other similar operations of our mind, as well as all the other movements of our soul, such as desires, command, interrogation, etc.

for instance, the following three different descriptive sentences which express one and the same proposition:

‘Hamlet was written by William Shakespeare.’

‘William Shakespeare is the author of Hamlet.’

‘Hamlet is a work by William Shakespeare.’

Propositions may be thought as expressing states of affairs. However, those states of affairs need not concern particular facts about the world we live in. They may also concern, e.g., general laws about it or even mathematical theses about any possible world.*

In any case, note that it is necessary for a sentence in declarative form to be meaningful in order to express a proposition.

*: Some even argue that there are propositions about fictional objects, but we will leave those aside.

Task 1.6 (Propositional sentences). Determine which of the following sentences are propositional.

1. The Moon orbits the Earth.
2. Is it going to rain today?
3. All squares have four sides.
4. Please pass the salt.
5. The current king of France is bald.
6. If it has rained, then the floor is wet.
7. Munich is the capital of Germany.
8. This sentence has five words.

Sentences 1.6.1, 1.6.3, and 1.6.7 are straightforward cases: they are propositional sentences stating facts about the world or mathematics, and their truth-values can be clearly assessed.

Sentence 1.6.6 is more interesting, as it asserts that the floor is wet, but conditionally upon it having rained. Hence, it is a propositional sentence, although it has a conditional structure.

In contrast, **sentences 1.6.2** and **1.6.4**. **Sentence 1.6.2** is interrogative, as it asks a question. **Sentence 1.6.4** is imperative, instructing the reader to perform an action.

Finally, **sentences 1.6.5** and **1.6.8** have the form of propositional sentences but it is not clear that they express propositions.

Sentence 1.6.5 appears declarative, yet it refers to a non-existent entity, the current king of France, so it is not clear whether we should regard it as being false or meaningless. In the latter case, it would not express a proposition.

Similarly, **Sentence 1.6.8** makes a claim about itself. It is not clear that sentences as such should be regarded as meaningful at all. On the other hand, it seems clear that the sentence is saying something true. Thus, whether this counts as a propositional sentence or not will depend on philosophical considerations.

Solution to task 1.6.

1. Propositional sentence (proposition about the Solar system)
2. Interrogative sentence
3. Propositional sentence (proposition about geometry)
4. Imperative sentence
5. Sentence of declarative form, which may express a proposition depending on our interpretation of it.
6. Propositional sentence (conditional proposition)
7. Propositional sentence (false proposition about the world)
8. Sentence of declarative form, which may express a proposition depending how we understand truth.

True and false sentences. The sentence-proposition distinction is philosophically important, and becomes even more subtle if we introduce the notion of a *statement*. Important as these distinctions are, however, they will not play a role in this textbook.

For convenience, we will also say that a sentence can be true or false given the conditions stated below.

Notion 1.14 (True and false sentence). A propositional sentence is true if the proposition it stands for is true. A propositional sentence is false if the proposition it stands for is false.

This approach is useful, as the concepts of truth and falsehood will be defined in relation to formulae in a formal language, which are the analogues of sentences in natural language.

Note that the statement of **notion 1.14** is compatible with there being a sentence of declarative form which is neither true nor false, as it would fail to express a proposition.

Task 1.7. Classify the following sentences into: true (t), false (f), and expressing no proposition (n).

1. Paris is the capital of France.
2. The Earth is flat.
3. Some triangles have four sides.
4. Bruce Wayne is Batman.
5. Bruce Wayne is Superman.
6. The round square is a square that is round.
7. There is a present king of France.
8. The present king of France is bald.
9. This sentence is in English.
10. Colourless green ideas sleep furiously.
11. If humans are mortal, then they are mortal.
12. If humans are mortal, then they are not.
13. Either humans are mortal or are not mortal.
14. Humans are mortal and are not mortal.
15. This sentence has exactly six words.
16. This sentence has more than ten words.
17. This sentence is true.
18. This sentence is false.
19. This sentence expresses a proposition.
20. This sentence expresses no proposition.

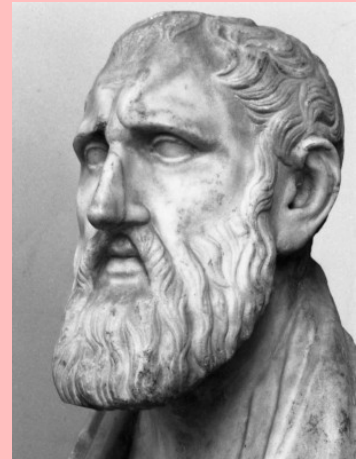
Some sentences in **task 1.7** clearly express propositions, and these propositions can be straightforwardly evaluated as true or false; for instance, **sentences 1.7.1, 1.7.2, and 1.7.3**.

Some other sentences are true or false logic. For instance, the **sentence 1.7.13** is true, since it exhausts the only two possibilities concerning humanity and mortality, while **sentence 1.7.14** is false because it asserts and denies the same property simultaneously. Likewise, **sentence 1.7.11** is logically true. As for **sentence 1.7.12**, it is debatable whether it is logically false. However, it happens to be

Historical note: The sayables

The Stoic concept of the *sayable* (λεκτόν) is probably the earliest antecedent of our modern notion of a proposition. Sextus Empiricus (ca. 160–ca. 210) reports about this in his *Πρὸς λογικούς* (*Against the Logicians*, 170–ca. 190, VII, 11–2):

some placed the true and false in the region of the thing signified, others in that of utterance, others in that of the motion of thought. And the Stoics stood for the first opinion, saying that three things were connected with one another, the thing signified and the signifier and the object. Of these the signifier is the utterance (e.g., the utterance “Dion”); the thing signified is the actual state of affairs revealed by it, and which we apprehend as it subsists in our thought ...; and the object is the externally existing thing (e.g., Dion himself). And of these, two are bodies, i.e. the utterance and the object, while one is incorporeal, i.e. the state of affairs signified and sayable, which is true or false.



Zeno of Citium
(ca. 334–ca. 262 BCE),
founder of Stoicism

false, because humans are indeed mortal, and it does not follow from this that they are also not mortal.

Next we have some sentences expressing descriptions about themselves, as expressions. For example, we can see [sentences 1.7.15](#) and [1.7.16](#) are true and false, respectively, if we count the words in the sentences themselves. [Sentence 1.7.9](#) is true for similar reasons, as it correctly describes the language in which it is written.

Now we find sentences concerning fictional and impossible objects. These sentences either fail to express a proposition, or are true or false depending on the properties such object would have in their respective fictional and impossible worlds. Thus, as hopefully most readers know, [sentences 1.7.4](#) and [1.7.5](#) would be true and false, respectively.[†] As for [sentence 1.7.6](#), it is quite difficult to think of a world where a round square can exist. But if it does, [sentence 1.7.6](#) would be true in it. We might also say that the sentence is just false, as it states the existence of an impossible entity.

We also have a couple of sentences involving – not impossible but – non-existent entities. Thus, [sentence 1.7.7](#) is false, as it clearly states that a king of France currently exists, which is not the case. Instead, [sentence 1.7.8](#) does not openly say this, but seems to presuppose it. As with [sentence 1.7.6](#), if we understand this as implying that a current king of France is said to exist, then [sentence 1.7.8](#) is false. Else, these sentences fail to express propositions.

[Sentence 1.7.10](#) is a clear case of a sentence failing to express a proposition. [Sentences 1.7.17–1.7.20](#) also seem to fall within this group. However, if you think about them, you might see that they may be regarded true or false. After all, if [sentences 1.7.16](#) and [1.7.16](#) express propositions, why cannot [sentences 1.7.17–1.7.20](#)?

[†]: However, see J. M. DeMatteis' *Superman: Speeding Bullets* (DC Comics, Elseworlds, 1993). In this fictional world, Bruce Wayne might be said to be Superman.

1.4 Placeholders and Quasi-Quotation

Solution to task 1.8.

t: 1, 9, 11, 13, 15

f: 2, 3, 7, 12, 14, 16

n: 10

t or n: 4

f or n: 5, 6, 8

n, or neither t nor f: 17, 19

n, or both t and f: 18, 20

∗: Most authors use, depending on the kind of expression, a specific selection of italicised Roman or Greek letters, such as: ‘A’, ‘v’, ‘φ’, ‘ψ’, ‘μ’. See, for example, the historical note on page 23. In this case, we usually call these expressions *meta-variables*.

†: Make sure you understand that it is the symbols within the quote marks – rather than these symbols plus the quote marks – that will serve as placeholders, and why.

Placeholders. We have introduced a simple device to speak about any *specific expression* we want. With it, we can say things like: ‘dog’ and ‘kid’ are nouns in English; ‘tiger’ has five characters and “tiger” has seven; etc. This device, however, is not enough for speaking about an *unspecified expression*.

For instance, we may want to speak about an expression *e* which has five letters and three numerals, or about a word *w* with two consonants and three syllables, without specifying an exact expression or word having those features. Rather, we want to speak generally about expressions having those features. In both cases, ‘*e*’ and ‘*w*’ serve as *expression placeholders* – or *placeholders*, for short.

Notion 1.15 (Placeholder). An **expression placeholder** is an expression which stands for an unspecified expression.

So, if we use a placeholder, like ‘*w*’ above, to speak about an unspecified word, we say that ‘*w*’ stands for a word, or that *w* is a word – in the latter case, we do not use quote marks.

We will use a specific selection of symbols as placeholders.∗ We introduce, first, a special kind of placeholders.

Notation 1.16 (Basic placeholder). The type-writer font letters:

‘A’, ‘B’, ‘C’, ‘D’, ‘E’, ‘F’, ‘G’, ‘H’, ‘I’, ‘J’, ‘K’, ‘L’, ‘M’,
‘N’, ‘O’, ‘P’, ‘Q’, ‘R’, ‘S’, ‘T’, ‘U’, ‘V’, ‘W’, ‘X’, ‘Y’, ‘Z’,
‘a’, ‘b’, ‘c’, ‘d’, ‘e’, ‘f’, ‘g’, ‘h’, ‘i’, ‘j’, ‘k’, ‘l’, ‘m’,
‘n’, ‘o’, ‘p’, ‘q’, ‘r’, ‘s’, ‘t’, ‘u’, ‘v’, ‘w’, ‘x’, ‘y’, ‘z’,

as well as:

‘1’, ‘2’, ‘3’, ‘4’, ‘5’, ‘6’, ‘7’, ‘8’, ‘9’, ‘0’,
‘1’, ‘2’, ‘3’, ‘4’, ‘5’, ‘6’, ‘7’, ‘8’, ‘9’, ‘0’,

will serve as **basic placeholders** and, hence, as **placeholders**.[†]

Having already some placeholders available, we may reformulate the statement of **notation 1.8** more briefly and nicely as follows:

Fact 1.17. *e* can be mentioned via an expression formed by: first, a left quote mark, next, *e*, and, finally, a right quote mark.

However, it would be a mistake to reformulate it as follows:

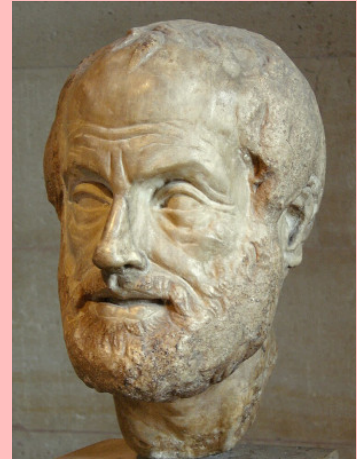
Mistake 1.18. An expression *e* can be mentioned via ‘*e*’.

In **mistake 1.18**, the first occurrence of ‘*e*’ functions as a placeholder for an unspecified expression. Its second occurrence, however,

Historical note: Schematic letters

Placeholders originate in schematic letters, which function in much the same way as our placeholders. The earliest recorded use of such devices is found in the logical works of Aristotle. See, for instance, the following passage of his *Ἀναλυτικὰ Πρώτερα* (*Prior Analytics*, ca. 351 BCE, I 2, 25a20–26):

if A belongs to some of the Bs, then necessarily B belongs to some of the As. (For if it belongs to none, then neither will A belong to any of the Bs.) But if A does not belong to some B, it is not necessary for B also not to belong to some A (e.g. if B is animal and A man: for man does not belong to every animal, but animal belongs to every man).



Aristotle (384–322 BCE)

appears within quote marks in “e”, which mentions the very variable ‘e’ rather than an unspecified quoted expression.

Task 1.8. Let *i* be an unspecified adjective containing at least one vowel and at most three consonants. What of the following is true, false, or undetermined about *a* and ‘*a*’?

- | | |
|---|--|
| 1. <i>a</i> is a word. | 15. <i>a</i> stands for a long word. |
| 2. ‘ <i>a</i> ’ is a word. | 16. ‘ <i>a</i> ’ stands for a short word. |
| 3. <i>a</i> is a long word. | 17. <i>a</i> is a numeral. |
| 4. ‘ <i>a</i> ’ is a long word. | 18. ‘ <i>a</i> ’ is a numeral. |
| 5. <i>a</i> is a short word. | 19. <i>a</i> stands for a numeral. |
| 6. ‘ <i>a</i> ’ is a short word. | 20. ‘ <i>a</i> ’ stands for a numeral. |
| 7. <i>a</i> has at least one vowel. | 21. <i>a</i> is a placeholder. |
| 8. ‘ <i>a</i> ’ has at least one vowel. | 22. ‘ <i>a</i> ’ is a placeholder. |
| 9. <i>a</i> has at least two vowels. | 23. <i>a</i> stands for a placeholder. |
| 10. ‘ <i>a</i> ’ has at least two vowels. | 24. ‘ <i>a</i> ’ stands for a placeholder. |
| 11. <i>a</i> has four consonants. | 25. “ <i>a</i> ” stands for a placeholder. |
| 12. ‘ <i>a</i> ’ has four consonants. | 26. “ <i>a</i> ” stands for an expression that stands for a placeholder. |
| 13. <i>a</i> stands for a word. | |
| 14. ‘ <i>a</i> ’ stands for a word. | |

First, since *a* is stipulated to be an adjective, which means it is a word. Hence, 1.8.1 is true. By contrast, ‘*a*’ is not a word but a placeholder, so 1.8.2 is false.

Next, we cannot know how large *a* is from the given information that it contains at least one vowel and at most three consonants. Hence, it is undetermined whether 1.8.3 and 1.8.5 are true. With regard to ‘*a*’, as it is not a word, 1.8.4 and 1.8.6 are just false.

By assumption, *a* has at least one vowel, so 1.8.7 is true. As for ‘*a*’, it in fact contains exactly one vowel, namely ‘*a*’, so 1.8.8 is also true. As for 1.8.10, ‘*a*’ clearly does not have two vowels, making it false. The placeholder ‘*a*’, however, contains no consonants at all, so 1.8.12 is false. Now, 1.8.9 is undetermined, as we do not know that *a* has more than one vowels – although it is likely, given that it has three consonants. However, 1.8.11 is false, since *a* has at most three consonants, which makes it impossible for it to have four.

Now, being *a* an adjective, it does not stand for a word – it rather stands for something like a *property*. Therefore, 1.8.13 and 1.8.15 are false. By contrast, ‘*a*’ is a placeholder for an adjective, so it stands for an adjective and, hence, for a word, which makes 1.8.14 true. Whether ‘*a*’ stands for a short word cannot be determined with the information given, which makes 1.8.16 undetermined.

Since *a* is an adjective, it is neither a numeral nor a placeholder, so 1.8.17 and 1.8.21 are false; nor does it stand for a numeral or a placeholder, so 1.8.19 and 1.8.23 are also false. For the same reason, ‘*a*’ stands for an adjective, which means it does not stand for a placeholder or for a numeral, so 1.8.24 and 1.8.20 are false. Now, since ‘*a*’ is a placeholder, it is not a numeral, so 1.8.22 is true and 1.8.18 is false.

Finally, we mention expressions in quote marks, like “*a*”, which is obtained by placing quote marks around ‘*a*’. Thus, “*a*” mentions and, hence, stands for the placeholder ‘*a*’, which makes 1.8.25. However, “*a*” stands for ‘*a*’, which is a placeholder, but which does not stand for a placeholder. Hence, 1.8.26 is false.

Solution to task 1.8.

True: 1, 7, 8, 14, 22, 25.

False: 2, 4, 5, 10, 11–15, 17–21, 23, 24, 26.

Undetermined: 3, 6, 9, 16.

Quasi-quotation. Suppose we wish to speak about an unspecified letter enclosed in braces, like ‘{*A*}’, ‘{*a*}’, ‘{*B*}’, ‘{*b*}’, etc. Previous to introducing our quote marks notation, we might describe it as an expression of the form:

{*L*}, where *L* is a letter.

However, it is not completely clear what are the braces doing there. They may be part of the form of the expression, as we wish. However, we could also read them as delimiting the very form of the expression, so that it is clear that the comma is not part of it.

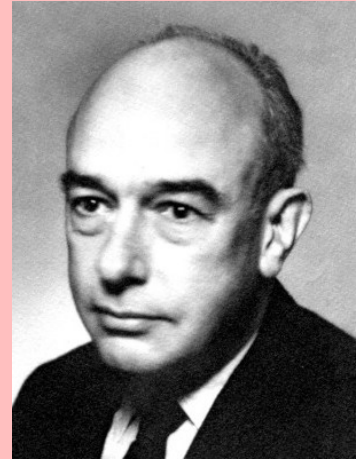
Using quote marks, we might try again and describe our expression as having the form:

‘{*L*}’, where *L* is a letter.

Historical note: Quasi-Quotation

Quasi-quotation was introduced by Willard Van Orman Quine in his book *Mathematical Logic* (1940, pp. 35–6) as follows:

There is need also of a convenient way of speaking of specific contexts of unspecified expressions: speaking, e.g., of the result of enclosing the unspecified expression μ in parentheses... Note that quotation is not available here. The quotation... ' (μ) ' designate[s] only the specific expression[] therein depicted, containing the specific Greek letter[] ' μ '... Reference to the intended contexts of the unspecified expressions ... will be accomplished by a new notation of *corners*, thus $\ulcorner(\mu)\urcorner$. Because of the close relationship which it bears to quotation, this device may be called *quasi-quotation*. It amounts to quoting the constant contextual background[], ' $(\)$ ' ... and imagining the unspecified expression[] μ ... written in the blanks.



W. V. O. Quine (1908–2000)

But this is even worst, since " $\{L\}$ " mentions a specific expression – which contains the specific symbol ' L '. It does not denote an unspecified letter occurring between braces. In order to denote this, we need a more flexible kind of quotation.

Notation 1.19 (Quasi-quotation). An expression formed by:

first, ' $\ulcorner$ ', next, e , and, finally, ' $\urcorner$ '

functions as the expression resulting from enclosing e in simple quote marks, but with any placeholder occurring in e still functioning as a placeholder.

In other words, *quasi-quotation* behaves like normal quotation, except that occurrences of placeholder expressions inside it are not taken as mentions of the placeholder itself. Instead, they maintain their condition of placeholders, thus standing for unspecified expressions within the quasi-quoted expression.

In particular, when a placeholder occurs alone inside a quasi-quotation, as in ' $\ulcorner e \urcorner$ ', the effect is exactly the same as using the placeholder ' e ' itself. In such a case, the quasi-quotation adds nothing: it simply stands for whatever expression ' e ' stands for.

Using this notation, we may refer to our unspecified letter enclosed in braces as an expression of the form:

$\ulcorner\{L\}\urcorner$, where L is a letter. ($\urcorner$)

Thus, in ' $\ulcorner\{L\}\urcorner$ ', the symbol ' L ' is not being mentioned – if anything, we are mentioning an arbitrary letter L . Therefore, our quasi-quotation ' $\ulcorner\{L\}\urcorner$ ' succeeds to mention an unspecified expression enclosed in braces ' $\{L\}$ '. The further condition that L be a letter makes ($\urcorner$) mention an arbitrary letter enclosed in braces.

Task 1.9. Rewrite the expression form described by each item below using placeholders and quasi-quotation.

1. Two letters connected by a hyphen, i.e., by ‘-’.
2. One numeral and one letter, in that order.
3. A lower-case letter, two upper-case letters, and an asterisk, in that order.
4. A lower-case letter, two upper-case letters, and any expression, in that order.
5. Expressions representing basic arithmetic operations (addition, subtraction, multiplication, and division) between natural numbers.
6. Expressions representing arbitrary mathematical operations between the numbers 1 and 2, in that order.
7. Expressions representing basic addition and multiplication between the numbers 1 or 2.
8. Categorical statements, that is, expressions of the form: category-term copula category-term, such as ‘Man is mortal’.

For each placeholder you use, specify whether it stands for a letter, a numeral, a digit, a category term, an operator, etc.

In the first four tasks, we have simple sequences of letters, numerals, or symbols. In [task 1.9.1](#), we have two letters *l* and *m* connected by ‘-’, so its form is $\ulcorner l-m \urcorner$. In [task 1.9.2](#), a numeral *n* precedes a letter *l*, so its form is $\ulcorner nl \urcorner$. In [task 1.9.3](#), we have a lower-case letter *a* followed by two upper-case letters *A*, *B*, and ‘*’, so its form is $\ulcorner aAB* \urcorner$. [Task 1.9.4](#) is similar, but we have an arbitrary expression $\bullet$ at the end instead of ‘*’, so its form is $\ulcorner aAB\bullet \urcorner$.

Note that, in [tasks 1.9.2](#) and [1.9.4](#), the corner marks are optional, since they enclose only placeholders.

In the next three of tasks, we have mathematical expressions of the form $\ulcorner a \heartsuit b \urcorner$, with different conditions on their placeholders. In [task 1.9.5](#), *a* and *b* are numerals standing for natural numbers, and $\heartsuit$ is either ‘+’, ‘-’, ‘×’, or ‘÷’.[‡] In [task 1.9.6](#), $\heartsuit$ is a mathematical operator, *a* is the numeral ‘1’ and *b* is ‘2’. Hence, our expression is also of the form $\ulcorner 1 \heartsuit 2 \urcorner$, where $\heartsuit$ is a mathematical operator. In [task 1.9.7](#), *a* and *b* are either ‘1’ or ‘2’, and $\heartsuit$ is either ‘+’ or ‘×’.

Finally, [task 1.9.8](#) presents the type of expression that has been most widely studied throughout history: categorical expressions, whose form is $\ulcorner s \heartsuit p \urcorner$, where *s* and *p* are terms, and $\heartsuit$ is a copula.

Note that, in [tasks 1.9.5–8](#), the corner marks are not optional, as there are spaces contained within them. In [1.9.8](#) we have two occurrences of the normal space ‘ ’, while in the others we have two occurrences of the thin space ‘ ’.

[‡]: Hence, we might also say that our expression is of one of the forms $\ulcorner a + b \urcorner$, $\ulcorner a - b \urcorner$, $\ulcorner a \times b \urcorner$, $\ulcorner a \div b \urcorner$.

Solution to task 1.9.

The form of the expression in each subtask above is:

1. $\lceil l-m \rceil$, where l and m are letters.
2. $n1$ (or, what is the same, $\lceil n1 \rceil$), where n is a numeral and 1 is a letter.
3. $\lceil aAB^* \rceil$, where a is a lower-case letter, and A, B, C are upper-case letters.
4. $aAB\textcircled{\scriptsize\bullet}$ (or $\lceil aAB\textcircled{\scriptsize\bullet} \rceil$), where a is a lower-case letter, and A, B are upper-case letters.
5. $\lceil a \heartsuit b \rceil$, where a, b are numerals standing for natural numbers and $\heartsuit$ is '+', '-', '×', or '÷'.
6. $\lceil a \heartsuit b \rceil$, where $\heartsuit$ is a mathematical operator, a is (the numeral) '1', and b is '2'; or
 $\lceil 1 \heartsuit 2 \rceil$, where $\heartsuit$ is a mathematical operator.
7. $\lceil a \heartsuit b \rceil$, where a, b are either '1' or '2', and $\heartsuit$ is either '+' or '×'.
8. $\lceil s \heartsuit p \rceil$, where s, p are terms and $\heartsuit$ is a copula (note the separating spaces).

Note the spaces contained within 5–8.

Limit cases of placeholders. You may have noticed that, in [task 1.9.6](#), we describe a as *the* numeral '1' and b as *the* numeral '2', which are quite specific expressions. This seems to contradict the idea that placeholders stand for unspecified expressions.

This, however, is not problematic: it represents a limit case in the use of a placeholder. There are two such limit cases to consider. On one extreme, a placeholder may have no restriction, standing for any expression whatsoever. On the other extreme, a placeholder may be so restricted that it can take exactly one value, as in [task 1.9.6](#). In [task 1.9.7](#), e.g., we approach this latter limit, as a and b are restricted to only two possibilities, namely '1' and '2'.

Note, however, it is not the placeholder itself that is making the specification. Rather, it is the further conditions we provide about

Historical note: Restrictions

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Euclid (XXXX–XXXX)

it that imposes it. Hence, even in this limit cases, placeholders remain expressions standing for unspecified expressions.

1.5 Schema and Instance

Schema and instance. An expression with placeholders is no *specific expression*. It is rather kind of a template or *schema* from which specific expressions may be obtained.

Notion 1.20 (Specific expression). A **specific expression** is an expression in which no placeholder occurs.

Notion 1.21 (Schematic expression). A **schematic expression or schema** is an expression in which at least one placeholder occurs, which may be restricted to a certain type of expressions.

Consider, for example, the schema:

‘the father of n’ (T)

From this, we can obtain specific expressions such as:

‘the father of Plato’, (O)

‘the father of Aristotle’, (II)

‘the father of the Moon’, (E)

‘the father of fo rehtaf eht’. (Ω)

*: Moreover, ‘{A}’, ‘{a}’, etc. are *instances* of schema (T) on p. 23.

Each of these expressions obtained is an *instance* of schema (T).*

Notion 1.22 (Instance). A specific expression resulting from replacing all placeholders in a schema with suitable specific expressions, where every occurrence of the same placeholder is replaced with the same expression, is an **instance** of that schema.

Our statement of **notion 1.22** prevents a schema from being an instance of another. When we consistently replace the placeholders of a schema with suitable specific expressions or schemas, the resulting schema is called a *subschema* of the original one.

Notion 1.23 (Subschema). A schema is a **subschema** of another schema when all instances of the former are also instances of the latter. Moreover, it is a **proper subschema** when the latter has at least one instance which is not an instance of the former.

†: Plural of ‘schema’.

It trivially follows that all schemata[†] are subschemata of themselves, but not proper subschemata of themselves.

Historical note: Schemata

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Author (XXXX-XXXX)

Fact 1.24. Each schema is a subschema of itself.

Fact 1.25. No schema is a proper subschema of itself.

Now, in specifying a schema such as (Υ) , we would normally want (δ) and (Π) to count as instances, but not (Θ) and (Ω) . We can do so by further specifying the schema, so that restrictions are placed on its placeholders.

Consider, for instance, the following variation of schema (Υ) :

‘the father of n ’, where n is the name of a person. (Υ')

Note that (Π) and (δ) are instances of schema (Υ) as much as of (Υ') . Expressions (Θ) and (Ω) , instead, do not count as instances of (Υ') , since ‘the Moon’ and ‘fo rehtaf eht’ are not names of persons. Also note that schema (Υ') is a subschema of (Υ) .

To make schemata easier to state, we fix the following notation.

Notation 1.26. ‘ $E : c$ ’ expresses a schema whose instances are all those expressions resulting from replacing the placeholders in ‘ E ’ with expressions satisfying the conditions stated by ‘ c ’.

Thus, we might rewrite schema (Υ') equivalently as follows:

the father of $n : n$ is the name of a person (Υ'')

Note that we do not enclose the expression to the left of ‘:’ in corner marks as such quasi-quotation is here implicit.

Schemata are very useful devices for speaking about the form common to two or more expressions, which may be different.

Consider, for instance, the sentences:

Peter is clumsy.
Nancy is smart. ($\mathfrak{M}$)

These two sentences are instances of the sentence schema:

n is a . : n is a proper name, a is an adjective ($\mathfrak{A}$)

Note, however, that given the conditions stated to the right of ':', the following do not qualify as instances of ($\mathfrak{A}$):

Plato is Aristocles.
Martin is.
Nancy is not clumsy.
Smart is Nancy. ($\mathfrak{M}$)

Note that, although expressions ($\mathfrak{M}$) and ($\mathfrak{M}$) have the same meaning, only the former is an instance of schema ($\mathfrak{A}$). This is because an expression is an instance of a schema not by virtue of its meaning, but solely by virtue of its linguistic form.

Task 1.10 (Schemata). Consider the following schemata:

- e loves f . : e and f are any expression (a)
- e loves f . : e and f are proper names (b)
- e loves f . : e is a human name, f is a deity name (c)
- e loves e . : e is any expression (d)
- e loves e . : e is a proper name (e)
- e loves e . : e is a human name or a deity name (f)

For each schema above, determine:

- (i) whether it is a subschema of (a)–(f); and
- (ii) whether any of the expressions below is an instance of it.[‡]

[‡]: All human names and deity names are proper names. Moreover, although some deity names, like 'Zeus', may sometimes be used as human names, we will treat them as if they were not human names.

- | | |
|--------------------------|-----------------------|
| 1. Peter loves Laura | 6. She loves you |
| 2. Peter loves Peter | 7. Odin loves music |
| 3. Horus loves Horus | 8. music loves music |
| 4. Today loves yesterday | 9. And I love her |
| 5. Horus loves Anna | 10. She loves herself |

First, note that all schemata are subschemata of (a). Hence, all expressions but 1.10.9 are instances of schema (a), because instead of the inflection 'loves', which is required by our schema, we have 'love'. In fact, 1.10.9 is an instance of no schema above.

Second, no expression with different expressions to the left and right of 'loves' is an instance of schemata (d)–(f). This includes 1.10.10, even though 'she' and 'herself' share the same referent.

Third, (d) and (e) are proper subschemata of (a) and (b), respectively. Hence, all instances of schemata (d) and (e) are also instances of (a) and (b), respectively, but not the other way around.

Finally, however, schema (f) is not a subschema of (c), as the latter's placeholders have different conditions. Hence, not all instances of (f) are instances of (c). This is the reason why 1.10.5 is not an instance of schema (c).

Solution to task 1.10.

(i) Each cell indicates whether the schema listed in the first column of its row is a subschema of the schema listed in the top row of its column.

	(a)	(b)	(c)	(d)	(e)	(f)
(a)	Yes	No	No	No	No	No
(b)	Yes	Yes	No	No	No	No
(c)	Yes	Yes	Yes	No	No	No
(d)	Yes	No	No	Yes	No	No
(e)	Yes	Yes	No	Yes	Yes	No
(f)	Yes	Yes	No	Yes	Yes	Yes

A schema indicated above as subschema of *another* schema, is also its proper subschema.

(ii) Each cell indicates whether a numbered expression is an instance of a schema.

	(a)	(b)	(c)	(d)	(e)	(f)
1	Yes	Yes	No	No	No	No
2	Yes	Yes	No	Yes	Yes	Yes
3	Yes	Yes	No	Yes	Yes	Yes
4	Yes	No	No	No	No	No
5	Yes	Yes	No	No	No	No
6	Yes	No	No	No	No	No
7	Yes	No	No	No	No	No
8	Yes	No	No	Yes	No	No
9	No	No	No	No	No	No
10	Yes	No	No	No	No	No

Expressing schemata. There is an important difference between specific expressions and schemata in terms of their meaning.

A specific expression is either meaningful or not. If it is meaningful, then it can stand for a specific referent – although it might be philosophically problematic to determine it. Thus, a specific expression either is meaningful and stands for its referent or it is unmeaningful and stands for nothing.

To exemplify, consider again the following expressions:

'the father of Plato',	(ϝ)
'the father of Aristotle',	(ϝ)
'the father of the Moon',	(Ϟ)
'the father of fo rehtaf eht'.	(Ϟ)

Expressions (ϝ) and (ϝ) stand for something, Aristotle's father (i.e., Nicomachus) and Plato's father (i.e., Ariston), respectively. Expressions (Ϟ) and (Ϟ), instead, lack meaning – at least a literal one – in English and, hence, have no referent.

A schema, on the other hand, though it may be regarded as meaningful in some sense, cannot stand for a specific referent in the way a specific expression can. Now, as we saw above, we can

Historical note: Schematic axioms

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make a schema more specific the more restrictions we place on its placeholders, as we did in (Υ'').

the father of n : where n is the name of a person (Υ''')

Of course, even with these restrictions, schema (Υ''') does not succeed to stand for a specific referent. Consider, however, the following schema:

the father of n is a relative of n
 : n is the name of a person. ($\times'$)

Not only is it clear that schema ($\times'$) is in some sense meaningful, but it also seems to have some kind of referent. Whoever utters ($\times'$) seems to be stating that the father of a person (with a name) is a relative of that very person, which is indeed true.

Schema ($\times'$) expresses each expression obtained by replacing ' n ' with a proper name in 'the father of n is a relative of n '. This might idea be generalised as follows.

Notion 1.27 (Expressing a schema). To **express a schema** functions as expressing each of its instances.

In other words, stating a schema is a way of expressing many – often infinitely many – things at once.

However, this does not mean, e.g., that a sentence schema is true when all its instances are true and false otherwise. A sentence schema cannot be said to be true or false. All we can say is that all or some of its instances are true or false.

Open expressions. Consider the following schema:

$$n \times n = 2 \times n : n \text{ stands for a number.} \quad (\textcircled{\text{f}})$$

Since 'n' stands for a number, schema $(\textcircled{\text{f}})$ functions as expressing, of each number, that multiplying it by itself equals twice that number – which is not true in general. However, a mathematician would not express this in this way, but rather as follows:

$$x \times x = 2x, \text{ where } x \text{ is a number.} \quad (\textcircled{\text{w}})$$

Note that, in schema $(\textcircled{\text{f}})$, 'x' is a number variable, i.e., it stands for a number. In $(\textcircled{\text{w}})$, instead, 'n' is a placeholder standing for a number expression. Hence, while $(\textcircled{\text{f}})$ is schematic in the sense of **notion 1.21**, formula $(\textcircled{\text{w}})$ is not, it is a specific expression.

However, $(\textcircled{\text{w}})$ may be said to be schematic in another sense. First, note that 'x' is an expression that stands for an unspecified number, that is, is a number variable.

Notion 1.28 (Variable). A **variable** is a symbol that serves as a placeholder for arbitrary elements of a given domain.

Now, the rules of the mathematical system in which $(\textcircled{\text{f}})$ can be expressed should allow us to replace 'x' with any expression that stands for a number. Specific expressions containing variables are not quite common in natural language, but are more proper of mathematics.

Sometimes, we make the variables specific enough for the expression to have a specific meaning, as when we say:

$$\text{For all numbers } x, x \times x = 2x. \quad (\textcircled{\text{v}})$$

$$\text{For some number } x, x \times x = 2x. \quad (\textcircled{\text{v}})$$

Thus, $(\textcircled{\text{v}})$ is false, since 3 is a number and it is not the case that $3 \times 3 = 2 \times 3$. On the other hand, $(\textcircled{\text{v}})$, since 0 is a number, and it is the case that $0 \times 0 = 2 \times 0$.

What about $(\textcircled{\text{w}})$? Can we say of this expression that it is true or false? Not quite, since, although $(\textcircled{\text{w}})$ is a specific expression, it does not stand for a specific proposition, but for an arbitrary one of a certain form. Expressions of these kinds are *open expressions*.

Notion 1.29 (Open expression). An **open expression** is a specific expression which does not have a specific referent, but an arbitrary one within some constraints.

Thus, $(\textcircled{\text{w}})$ is an open expression whose referent is an arbitrary proposition of the form ' $n \times n = 2n$ ', where n stands for a number.

Historical note: Open expressions

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This shows the sense in which ($\approx$), and open expressions in general, may also be thought of as schemata. For this reason, we will also use **notation 1.26** for representing open expressions.

Notation 1.30. $\ulcorner E : c \urcorner$ expresses an open expression whose variables are constrained by the conditions stated by $\ulcorner c \urcorner$.

Thus, ($\approx$) may be equivalently expressed as follows:

$$x \times x = 2x : x \text{ is a number.}$$

1.6 Definiendum and Definiens

Definitions. So far, we have been introducing some notions and notation but in a not very rigorous way. In fact, we lack a notation for introducing notions and notations rigorously. In order to address this, we will introduce a concept and a notation for *definitions*.

Notion 1.31 (Definition). A **definition** is an expression that stipulates the meaning of a yet unmeaningful term as used within an expression, the **definiendum**, to be equal to the meaning of an already meaningful expression, the **definiens**.

Note that the term being defined is not necessarily the same as the definiendum, as we will see later.

In this textbook, we will take for *an already meaningful expression* one whose meaning we can understand by virtue a notion, a notation, or a definition previously stated. Definitions will always be introduced in their standard notation.

Notation 1.32 (Standard form of definitions). $\ulcorner U \stackrel{\text{DF}}{=} M \urcorner$ expresses a definition whose definiendum is U and whose definiens is M.*

*: As with [notation 1.26](#), we need not enclose the definiendum or the definiens in corner or quote marks, as these are implicit.

For instance, Aristotle is not only known by the name ‘Aristotle’, but also with other names, like for instance, ‘the Stagirite’. Assuming that we know the meaning of the former but not of the latter, we might establish the meaning of ‘the Stagirite’ via:

$$\text{the Stagirite} \stackrel{\text{DF}}{=} \text{Aristotle} \quad (\Uparrow)$$

Definition $(\Uparrow)$ does not quite say that Aristotle is the same person as the Stagirite. It rather stipulates the meaning of ‘the Stagirite’ to be the same as the meaning of ‘Aristotle’. In other words, definition $(\Uparrow)$ stipulates that, by using ‘the Stagirite’ we mention the same person we would mention by using ‘Aristotle’, i.e., Aristotle.

Fact 1.33. The definiens of a definition is not stipulated to be the meaning of the definiendum, but rather to express the same meaning as the definiendum.

Once a definition is stipulated, the definiendum and definiens become interchangeable under some conditions.

Fact 1.34 (Definiendum introduction). Let $\ulcorner U \stackrel{\text{DF}}{=} M \urcorner$ be a stated definition and e an expression where M is used. If i is an expression obtained from e by replacing one or more occurrences of M among those occurrences in which M is used (rather than mentioned) with corresponding occurrences of U, then e and i have the same meaning.

Historical note: Definiendum and definiens

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Fact 1.35 (Definiendum elimination). Let $\ulcorner U \stackrel{\text{DF}}{=} M \urcorner$ be a stated definition and e an expression where U is used. If i is an expression obtained from e by replacing one or more occurrences of U among those occurrences in which U is used (rather than mentioned) with corresponding occurrences of M , then e and i have the same meaning.

The condition that the occurrence of M be used rather than mentioned prevents us from replacing expressions within quote marks. This is essential, since those replacements could result in expressions with different meanings.

Thus, for instance, the following two sentences have the exact same meaning by virtue of definition ($\ulcorner$).

Plato was the master of the Stagirite. ($\textcircled{\text{S}}$)

Plato was the master of Aristotle. ($\textcircled{\text{A}}$)

However, note that the following two expressions do not have the same meaning even after the stipulation of definition ($\ulcorner$).

The word ‘the Stagirite’ has nine letters. ($\textcircled{\text{S}}$)

The word ‘Aristotle’ has nine letters. ($\textcircled{\text{A}}$)

The reason is that, whereas the expressions in the definiendum and definiens of ($\ulcorner$) are used in ($\textcircled{\text{S}}$) and ($\textcircled{\text{A}}$), respectively, they are not in used in neither ($\textcircled{\text{S}}$) nor ($\textcircled{\text{A}}$), but just mentioned. A fact which corroborates that latter two sentences do not have the same meaning is that ($\textcircled{\text{S}}$) is false, while ($\textcircled{\text{A}}$) is true.

To finish, note that an already stated definition has already fixed the meaning of its definiendum. Hence, it can no longer be redefined.

Fact 1.36. Once a definition has been stated, its definiendum cannot be used as the definiendum of a new definition.

Schematic definitions. Definition ($\Uparrow$) is fairly simple, as it just stipulates the equivalence in meaning of two quite specific expressions. We may nevertheless want to define not just a specific expression, but of many expressions of the same form.

Suppose, for instance, that we want to define ‘complex’ as ‘not simple’. Following the pattern of ($\Uparrow$), we may try:

$$\text{complex} \stackrel{\text{DF}}{=} \text{not simple} \quad (\mathbb{M})$$

Thus, the following expressions would have the same meaning:

Logic is complex.
Logic is not simple.

But so would the following expressions, which we did not intend:

2i is complex.
2i is not simple.

Clearly, these two sentences differ in meaning because ‘complex’ has a different sense when combined with numerals than when combined with names of subjects. This latter sense is the one we wanted to capture with ($\mathbb{M}$), and it can only be established if we somehow restrict the expressions it may be combined with.

In order to do so, we need to employ a special kind of definition.

Notion 1.37 (Schematic definition). A **schematic definition** is a schema all of whose instances are definitions.

Following **notion 1.27**, to state a definition schema functions as stating all the definitions which are its instances.

Thus, with a schematic definition, we can stipulate expressions of the form ‘s is complex’ to mean the same as ‘s is not simple’, where s stands for a subject matter. We do this as follows:

$$\begin{aligned} \text{s is complex} &\stackrel{\text{DF}}{=} \text{s is not simple} \\ &: \text{s stands for a subject matter} \quad (\underline{\underline{=}}) \end{aligned}$$

The following are instances of definition schema ($\underline{\underline{=}}$):

informatics is complex $\stackrel{\text{DF}}{=}$ informatics is not simple
linguistics is complex $\stackrel{\text{DF}}{=}$ linguistics is not simple
mathematics is complex $\stackrel{\text{DF}}{=}$ mathematics is not simple

In fact, definition schema ($\underline{\underline{=}}$) results in stipulating a definition of the form ‘s is complex $\stackrel{\text{DF}}{=}$ s not simple’ for each expression s

Historical note: Identity symbol

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denoting a subject. This amounts to defining ‘complex’ as ‘not simple’ just when applied to expressions naming subjects.

We can use schematic definitions where more than one placeholder occurs. In particular, it will be useful to introduce a few shorthands for common logical expressions in English.

Notation 1.38 (Abbreviation of logical expressions).

$\neg A \stackrel{\text{DF}}{=} \text{It is not the case that } A.$

$A \wedge B \stackrel{\text{DF}}{=} A \text{ and } B.$

$A \vee B \stackrel{\text{DF}}{=} A \text{ or } B \text{ or both } A \wedge B.$

$A \Rightarrow B \stackrel{\text{DF}}{=} \text{If } A, \text{ then } B.$

$A \Leftrightarrow B \stackrel{\text{DF}}{=} A \text{ if and only if } B.$

$\forall v A \stackrel{\text{DF}}{=} \text{For all } v, \text{ it is the case that } A.$

$\exists v A \stackrel{\text{DF}}{=} \text{For some } v, \text{ it is the case that } A.$

$\exists! v A \stackrel{\text{DF}}{=} \text{For exactly one } v, \text{ it is the case that } A.$

: A, B are sentences or open sentences,
 v is a variable.

†: That is, assuming the expressions and open expressions they contain are meaningful.

Thus, the following expressions are now meaningful[†]:

$\neg \text{Hank is a criminal.}$

$\text{Gus is a father} \vee \text{Mike is a grandfather.}$

$\text{Walter is a teacher} \wedge \text{Jesse is a student.}$

$\text{Lalo is the boss} \Rightarrow \neg \text{Nacho is the boss.}$

$\text{Walter is a criminal} \Leftrightarrow \text{Jesse is a criminal.}$

$\exists x \text{ Hank captured } x.$

$\forall x x \text{ knows Gus.}$

It is important to note here that the symbols ' $\Rightarrow$ ', ' $\Leftarrow$ ', ' $\mathbb{W}$ ', ' $\Rightarrow$ ', ' $\Leftrightarrow$ ', ' $\mathbb{W}$ ', and ' $\exists$ ' introduced in **notation 1.38** are not the definienda[‡] of the definitions there stipulated, but the terms defined. ‡: Plural of 'definiendum'.

Their meanings are, of course, stipulated by that definition, but by means of defining expressions of a certain form in which they occur. Therefore, **notation 1.38** does not fix the meaning of any the following expressions:

The teacher is either Ana $\mathbb{W}$ Bruce.
Jimmy $\Leftarrow$ Chuck are brothers.

Another useful abbreviation will be the following:

Notation 1.39 (Abbreviation of identity).

$A = B \stackrel{\text{DF}}{=} A$ is identical to B
 $A \neq B \stackrel{\text{DF}}{=} \neg A = B$
: A, B are meaningful expressions.

This means that '=' will be understood in the sense of expressing the referents of two given meaningful expressions are identical, or equal, or the same.

Thus, the following expressions are now meaningful:

Walter = Heisenberg.
the author of *Demian* = the author of *Der Steppenwolf*.
 $1 + 2 = 4$.

All the expressions above are specific sentences, the latter of which is false.

Other expressions equivalent to ' $A = B$ ', in some contexts, include: ' A is equal to B ', ' A equals B ', ' A is the same as B ', ' A and B are the same', ' A is B ' (an exception is when A is a subject term and B is a predicate term), among others.

Furthermore, if two expressions are stated to be equal by definition, then their referents are identical.

Fact 1.40. $A \stackrel{\text{DF}}{=} B$ is a stated definition $\Rightarrow A = B$.

Historical note: Piecewise definitions

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§: Plural of 'definiens'.

Piecewise definitions. Schematic definitions allow for definitions which establish different definientia[§] for a given term, depending on the expressions that accompany it in the definiendum. We call these *piecewise definitions*.

Notion 1.41 (Piecewise definition). A **piecewise definition** defines a term by stating different conditions under which different definientia apply.

For instance, we may want to define the expression 'n is a parent' as 'n is a father' if n stands for a man, and as meaning 'n is a mother' if n stands for a woman. We can do this by stating two separate definitions:

$$n \text{ is a parent} \stackrel{\text{DF}}{=} n \text{ is a father} : n \text{ stands for a man} \quad (\mathfrak{M}_1)$$

$$n \text{ is a parent} \stackrel{\text{DF}}{=} n \text{ is a mother} : n \text{ stands for a woman} \quad (\mathfrak{M}_2)$$

The parallel statement of definitions $(\mathfrak{M}_1)$ and $(\mathfrak{M}_2)$ does not contradict **fact 1.36** in any way. Note that $(\mathfrak{M}_1)$ and $(\mathfrak{M}_2)$ have no instance in common, so they do not assign different definientia to the same definiens.

Piecewise functions are also useful for defining some well-known mathematical functions. For instance, the following defines the absolute value function:

$$|n| \stackrel{\text{DF}}{=} n : n \text{ stands for a number greater or equal than } 0, \quad (\times_1)$$

$$|n| \stackrel{\text{DF}}{=} -n : n \text{ stands for a number smaller than } 0. \quad (\times_2)$$

Again, note that the definitions $(\times_1)$ and $(\times_2)$ have no instances in common, so there is no conflict with **fact 1.36**.

To close, note that our piecewise definition $(\times)$ is not expressed as mathematical definitions usually are. In a normal mathematical

definition, we would use open expressions like the following:

$$|x| \stackrel{\text{DF}}{=} x \quad : x \geq 0, \quad (\mathcal{D}_1)$$

$$|x| \stackrel{\text{DF}}{=} -x \quad : x < 0. \quad (\mathcal{D}_2)$$

Since ‘ x ’ is a number variable, the rules of the mathematical system for which this expression is defined should allow us to replace ‘ x ’ with any expression standing for a number that satisfies the conditions stated after ‘:’.

Explicit and recursive definitions. In all definitions presented so far, a new term is used within the definiendum, which is stipulated to be equal in meaning to the definiens. The definiens assigned contains nothing but old terms. In particular, the defined term is not used within the definiens.

Definitions of this kind are called *explicit definitions*.

Notion 1.42 (Explicit definition). An **explicit definition** is one in which the term being defined (in the definiendum) is not used within the definiens.

The restriction that the term being defined is not used within the definiens seems quite sensible, as we want to avoid that the following be regarded as a proper definition.

$$n \text{ is plok} \stackrel{\text{DF}}{=} n \text{ is plok} \quad (\approx)$$

The problem with $(\approx)$ is that it is useless as a definition. The meaning of the defined term ‘plok’ is not being set, because the definiens is not a meaningful expression at all.

Not all definitions must be explicit, though. There are cases in which it is useful to have the term defined appearing in the definiens. Consider, for instance, the following definition, corresponding to the mathematical definition of factorial[¶]:

$$0! \stackrel{\text{DF}}{=} 1 \quad (\mathcal{D}_1)$$

$$(n)! \stackrel{\text{DF}}{=} (n-1)! \times (n) \quad : n \text{ stands for a natural number.} \quad (\mathcal{D}_2)$$

¶: Note that, in the usual convention, 0 is not a natural numbers.

Not that, although in $(\mathcal{D}_2)$ we used the defined term ‘!’, this does not generate a problem like $(\approx)$ does. The reason is that in $(\mathcal{D}_2)$ we are not merely repeating the term being defined, but providing a way to reduce each new case to cases previously defined.

First, the meaning of ‘0!’ is fixed by $(\mathcal{D}_1)$. Second, by $(\mathcal{D}_2)$, the meaning of ‘1!’ depends on the meaning of ‘(1-1)!’, i.e., of ‘0!’. Next, the meaning of ‘2!’ depends on that of ‘1!’, and so forth. Thus, the meaning of each expression of the form ‘ $n!$ ’, where n stands for 0 or a natural number, is determined by referring back to previous numerals, ultimately reaching the base case ‘0!’.

Thus, although the defined symbol occurs within the definiens of each instance of $(\succ_2)$, definition $(\succ)$ is problematic, because the meaning of each instance can be traced back to expressions whose meanings are already fixed. Definitions of these kinds are known as *recursive* or *inductive definitions*.

Notion 1.43 (Recursive definition). A **recursive** aka **inductive definition** stipulates the meaning of a term starting from **base cases** and then defining **inductive cases** depending on those base cases and the other inductive cases.

Thus, in the definition of factorial above, $(\succ_1)$ corresponds to the base case, while $(\succ_2)$ corresponds to the inductive case.

Historical note: Fibonacci numbers

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Clause-based recursive definitions. Not all recursive definitions are easily stated in the form of (⋈). Suppose that we want to define the term ‘ancestor’ so that x is an ancestor of y in case x is a parent of y , or x is a grandparent of y , or x is a great-grandparent of y , or x is a great-great-grandparent of y , etc.

Here, we do not have one specific case from which to start. In definition (⋈), our base case was about ‘0’, but here, the base case consists is about all expressions expressing a parenthood relationship. Typically, we use a clause-based inductive definition, which may be described as follows.

Notion 1.44. A clause-based recursive definition consists of:

1. One or more *base clauses*, which introduce the simplest instances of the expression being defined.
2. One or more *inductive clauses*, which specifies how new instances arise from previously established ones.
3. A *final clause*, which states that all cases of the defined term are obtained from the base clauses with finitely many applications of the inductive clauses.

Let us see how a recursive definition may be given of ‘ancestor’.

Definition ȳ. Where ‘ x ’ and ‘ y ’ are variables standing for living beings, the term ‘ancestor’ is recursively defined by the following clauses:

1. x is a parent of $y \Rightarrow x$ is an ancestor of y . (*base clause*)
2. x is an ancestor of z and z is an ancestor of $y \Rightarrow x$ is an ancestor of y . (*inductive clause*)
3. No x is an ancestor of y unless this can be obtained by finitely many applications of 1 and 2. (*final clause*)

The base clause introduces the most immediate case of ancestry, namely that of parenthood. The inductive clause then extends the relation step by step: to x ’s parents, and to x ’s parents’ parents, and to x ’s parents’ parents’ parents¹. The final clause ensures that no additional cases are admitted as ancestors.

¹ And to x ’s parents’ parents’ parents’ parents.

Recursive definitions and definientia. Note that definition (ȳ) is not schematic in the sense of **notion 1.37**, since it contains no placeholders. As it contains variables, one might call it an open definition – and, thus, schematic in a sense.

Moreover, definition (ȳ) does not have the typical definiendum-definiens structure. Clearly, the expression ‘ x is an ancestor of y ’ is

the definiendum. However, where is its definiens? In order to see it more clearly, we might reformulate (¶) as follows:

$$\begin{aligned} \text{a is an ancestor of b} &\stackrel{\text{DF}}{=} \text{a is a parent of b} \vee \\ &\exists x (\text{a is an ancestor of } x \wedge \\ &\quad x \text{ is an ancestor of b}) \\ &: \text{a, b stand for living beings.} \quad (\text{¶}') \end{aligned}$$

Note that, unlike (¶), definition (¶') is schematic and has an easily identifiable definiens for each definitia of its instance definitions, all of which share the same form.

Moreover, note that, despite appearances, definition Equation ¶' is inductive, as its definiens states both the base case (to the left of '∨') and its inductive case (to its right).

2.1 Objects and Collections

In earlier sections, we introduced expressions and discussed the meanings they convey. Meaningful expressions refer to *objects*.

Notion 2.1 (Object). An **object** is anything to which a meaningful expression may refer.

For example, a personal name refers to a person, a numeral refers to a number, a quoted expression refers to an expression. All these are examples of objects.

Often, we must refer to several objects collectively rather than individually. To do so, we group objects into collections.

Notion 2.2 (Collection). A **collection** is a selection of objects, which are its **elements**.

To name collections, I will normally use expressions written starting bold with upper case letters.

Notation 2.3 (Names for collections). Expressions of the form $\mathbf{A}e$, where $\mathbf{A}$ is a bold upper-case letter and e is any expression containing letters or numerals will serve as names for collections.

Examples of names for collections following this pattern are:

$\mathbf{A}$, $\mathbf{B}e_1$, $\mathbf{ZZ}_A$, $\mathbf{AB}$, $\mathbf{In}^1$, $\mathbf{Aa}$, $\mathbf{Stuff}$, $\mathbf{Stuff}$.

The following, instead, will not name collections:

$\mathbf{A}$, $\mathbf{B}e_1$, $\mathbf{ZZ}_A$, $\mathbf{AB}$, $^1\mathbf{In}$, $\mathbf{aA}$, $\mathbf{Stuff}$, $\mathbf{Stuff}$.

Historical note: The notion of a collection

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One of the exceptions to the convention above will be my use of the double-struck upper case letters ' $\mathbb{B}$ ', ' $\mathbb{N}$ ', ' $\mathbb{W}$ ', ' $\mathbb{Z}$ ', ' $\mathbb{Q}$ ', ' $\mathbb{R}$ ', and ' $\mathbb{C}$ ' to denote the sets of boolean, natural, whole, integer, rational, real, and complex numbers, respectively.

Collections come in two kinds: *sets* and *tuples*. In the following, I will provide intuitive and informal account of these notions to be used through this textbook.

2.2 Sets and members

A *set* is a collection in which we disregard any order or repetitions concerning its elements aka members.

Notion 2.4 (Set). A **set** is a collection in which the elements are neither ordered nor repeated.

The sign ' $\in$ ' stands for the *set membership* relation.

Notation 2.5 (Set membership). The expression ' $e \in S$ ', where e stands for an object and S stands for a set, is an abbreviation of ' e is a member of S '. Moreover, ' $e \notin S$ ' abbreviates ' $\neg e \in S$ '.

Thus, for example, ' $a \in A$ ' means that a is a member of A , and ' $a \notin A$ ' means that a is not a member of A .

Task 2.1. Abbreviate the following expressions using **notations 2.5 and 1.39**.

1. 1 is a natural number.
2. A is the set of natural numbers.
3. 1 is not 0.
4. 0 is not a natural number.
5. 1 is not the set of natural numbers.
6. Natural numbers and real numbers are not the same.
7. The letter ' a ' is in the set A .
8. The set of natural numbers is in the set of rational numbers.

Hint: A collection can be an element of another collection.

Note first that the expressions in this task are quite different from those used when the notation was introduced. They are all formulated by means of the verb *to be*, which has several meanings in natural language, and which is represented by means of different logical and set-theoretical notions depending on the context. This task is an opportunity to see these different uses.

Historical note: The notion of a set

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∗: Both expressions are true under the usual convention for natural numbers. In particular, 0 is placed in the set of *whole numbers*, usually represented by $\mathbb{W}$ or $\mathbb{N}_0$, which consists of all natural numbers together with 0.

In Expressions 2.1.1 and 2.1.4, ‘is’ indicates that an object falls under a certain collection. Saying that 1 is a natural number, or that 0 is not a natural number means to place (or exclude) these numbers within a collection. Thus, this use of ‘is’ is rendered by the membership relation, and these expressions would be abbreviated as $1 \in \mathbb{N}$ and $0 \notin \mathbb{N}$, respectively.*

A different use appears in Expressions 2.1.2, 2.1.3, and 2.1.6. Here, ‘is’, ‘is not’, and ‘are’ express sameness or difference between objects. In these cases, the appropriate notion is identity, rather than membership. Hence, these expressions should be abbreviated as $\mathbf{A} = \mathbb{N}$, $1 \neq 0$ and $\mathbb{N} \neq \mathbb{R}$, respectively.

Expressions 2.1.5, 2.1.7, and 2.1.8 are trickier.

Rather than saying, like Expression 2.1.1, that 1 is not a natural number Expression 2.1.5 says 1 is not the (set of) natural numbers. That is, the number 1 and the natural numbers are not the same object. This expression is true, and should be translated as $1 \neq \mathbb{N}$.

Expression 2.1.7 is not saying that the object a is in the set $\mathbf{A}$. It rather says that the letter ‘ a ’ – that is, a linguistic sign – is in the set $\mathbf{A}$. This requires us a way to distinguishing between a sign and its referent, which in this case we do through quote marks.

Finally, Expression 2.1.8 is quite ambiguous and may be understood in at least two ways.

First, it may be understood as stating that $\mathbb{N}$ is an element of $\mathbb{R}$, which should be represented with $\mathbb{N} \in \mathbb{R}$. This statement is false, since, although all the natural numbers are real numbers, the set of natural numbers is not itself a number and, hence, it cannot be a member of $\mathbb{R}$ nor of any other set of numbers.

Second, it may be understood as stating that $\mathbb{N}$ is a part of or is *included* in $\mathbb{R}$. This would be true if understood in the sense that all the natural numbers are real numbers. We cannot represent this

understanding with the notation introduced so far, but we will acquire the notation needed in [definition 2.11](#).

Solution to task 2.1.

- | | |
|--|---|
| 1. 1 is a natural number. | Abbreviation: $1 \in \mathbb{N}$ |
| 2. A is the set of natural numbers. | Abbr.: $\mathbf{A} = \mathbb{N}$ |
| 3. 1 is not 0. | Abbr.: $1 \neq 0$ |
| 4. 0 is not a natural number. | Abbr.: $0 \notin \mathbb{N}$ |
| 5. 1 is not the set of natural numbers. | Abbr.: $1 \neq \mathbb{N}$ |
| 6. Natural numbers and real numbers are not the same. | Abbr.: $\mathbb{N} \neq \mathbb{R}$ |
| 7. The letter ' <i>a</i> ' is in the set A . | Abbr.: ' <i>a</i> ' $\in \mathbf{A}$ |
| 8. The set of natural numbers is in the set of rational numbers. | Possible abbreviation: $\mathbb{N} \in \mathbb{Q}$ |

A notable set is the *empty set*, which has no elements.

Notion 2.6 (Empty set). The **empty set** is the set with no members.

Notation 2.7 (Empty set). ' $\{\}$ ' denotes the empty set.

Describing sets. There are two usual ways to describe the contents a set. For the purposes of this explanation, let **A** be a set containing the numbers 1, 2, 3, and nothing else as members.

The first way is by enumeration, that is, by listing all of its elements between braces.

Notation 2.8 (Set by enumeration). ' $\lceil \{e\} \rceil$ ', where *e* stands for an object, denotes the set having that object and nothing else as a member. Moreover, ' $\lceil \{1, e\} \rceil$ ', where *l* is a comma-separated list of expressions standing for objects and *e* stands for an object, denotes the set having all those objects and nothing else as members.

The order in which the elements are listed in this notation is irrelevant, and so is whether the elements are repeated.

For instance, the expression ' $\{1, 2, 3\}$ ' adequately describes **A**, which entails that $\mathbf{A} = \{1, 2, 3\}$. Hence:

$$\mathbf{A} = \{1, 2, 3\} = \{2, 1, 3\} = \{2, 1, 2, 3\} = \{3, 2, 1, 1, 2, 3\}.$$

This notation is suitable for describing small, finite sets, but not for large or infinite sets.

Historical note: Set-builder notation

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Author (XXXX–XXXX)

The second way is by extension, that is, by specifying a property common to all its members. For instance, **A** may be correctly described as the set of all natural numbers which are smaller than 4. To express this, we use the *set-builder notation*.

Notation 2.9 (Set-builder). The expression $\lceil \{t : A\} \rceil$, where t stands for an unspecified object and A is a propositional sentence, denotes the objects denoting by t which make A is true.

Thus, to describe **A** in the terms above, we write:

$$\mathbf{A} = \{n : n \in \mathbb{N} \ \& \ n < 4\}.$$

In words, **A** is equal to the set of all n such that n is a natural number ($n \in \mathbb{N}$) and n is smaller than 4 ($n < 4$).

A more elegant way to put it is as follows:

$$\mathbf{A} = \{n \in \mathbb{N} : n < 4\}.$$

In words, **A** is equal to the set of all natural numbers n ($n \in \mathbb{N}$) such that n is smaller than 4 ($n < 4$). In this case, the expression to the left of ‘:’ indicates the domain from which the elements of the set are drawn, while the expression to the right specifies the particular property that these elements must satisfy.

Not that an expression like ‘ $\mathbf{A} = \{1, 2, 3\}$ ’ is a *propositional sentence* stating that **A** is the set which contains the numbers 1, 2, 3, and nothing else, which may be true or false.

Sometimes, we do not want to make an assertion about the identity between two sets, but rather to provide a **definition** of set name in terms of our notation. In such cases, instead of the equality symbol ‘=’, we use the definition symbol ‘ $\stackrel{\text{def}}{=}$ ’.

Task 2.2. Determine the elements of the sets defined below.

- $$\begin{aligned} \mathbf{A} &\stackrel{\text{DF}}{=} \{x : x \text{ is or was a Pope}\}. \\ \mathbf{B} &\stackrel{\text{DF}}{=} \{x : x \text{ is a member of the Beatles}\}. \\ \mathbf{C} &\stackrel{\text{DF}}{=} \{x : x \text{ is the name of a member of the Beatles}\}. \\ \mathbf{D} &\stackrel{\text{DF}}{=} \{x : x \text{ is an uppercase vowel in English}\}. \\ \mathbf{E} &\stackrel{\text{DF}}{=} \{x \in \mathbb{N} : x \geq 10 \wedge x \leq 20\}. \\ \mathbf{F} &\stackrel{\text{DF}}{=} \{x \in \mathbb{R} : x \geq 10 \wedge x \leq 20\}. \\ \mathbf{G} &\stackrel{\text{DF}}{=} \{m + n : m, n \in \mathbb{N} \wedge m, n < 10 \wedge n = 2m\}. \\ \mathbf{H} &\stackrel{\text{DF}}{=} \{x : x \in \mathbf{B} \wedge x \text{ is alive in 2025}\}. \\ \mathbf{I} &\stackrel{\text{DF}}{=} \{x \in \mathbf{B} : x \text{ is a member of Wings}\}. \\ \mathbf{J} &\stackrel{\text{DF}}{=} \{x \in \mathbb{N} : x \leq 10 \wedge x \geq 20\}. \end{aligned}$$

We first have sets whose members are people.

A collects all those who are or have been Popes – figures such as Saint Peter, John Paul II, Benedict XVI, and Francis. **B** also collects people, but in this case to the four members of the Beatles: John Lennon, Paul McCartney, George Harrison, and Ringo Starr.

H and **I** are closely related to **B**. The former contains those Beatles who are still alive in 2025 (Paul McCartney and Ringo Starr), whereas the latter singles out the Beatle who was also part of the band Wings (Paul McCartney).

The next sets concern words and letters.

C contains the names of the Beatles – that is, the strings ‘John Lennon’, ‘Paul McCartney’, ‘George Harrison’, and ‘Ringo Starr’ – not John Lennon, Paul McCartney, George Harrison, and Ringo Starr themselves.

D contains the uppercase vowels of English: ‘A’, ‘E’, ‘I’, ‘O’, and ‘U’. However, A, E, I, O, and U are not members of **D**. This answer assumes that ‘A’, ‘E’, ‘I’, ‘O’, and ‘U’ are not used – as it occurs in ordinary written language – as names for ‘A’, ‘E’, ‘I’, ‘O’, and ‘U’. We do not want to do that in this textbook.

Next, we have some numerical sets.

E and **F** both include all numbers greater than or equal to 10 and less than or equal to 20, the former restricted to natural numbers and the latter extended to real numbers.

G consists of sums of two natural numbers below 10 such that one is twice the other; this gives the values: $1 + 2 = 3$, $2 + 4 = 6$, $3 + 6 = 9$, and $4 + 8 = 12$.

Finally, **J** has no members, since no natural number can be simultaneously less than 10 and greater than 20.

Historical note: The membership symbol

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Author (XXXX–XXXX)

Solution to task 2.2.

- ▶ Saint Peter, ..., Francis, Leo XIV $\in \mathbf{A}$.
- ▶ John Lennon, Paul McCartney, George Harrison, Ringo Starr $\in \mathbf{B}$.
- ▶ 'John Lennon', 'Paul McCartney', 'George Harrison', 'Ringo Starr' $\in \mathbf{C}$.
- ▶ 'A', 'E', 'I', 'O', 'U' $\in \mathbf{D}$.
- ▶ 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 $\in \mathbf{E}$.
- ▶ All members of $\mathbf{D}$ and all real numbers between them are members of $\mathbf{F}$.
- ▶ 3, 6, 9, 12 $\in \mathbf{G}$.
- ▶ Paul McCartney, Ringo Star $\in \mathbf{H}$.
- ▶ Paul McCartney $\in \mathbf{I}$.
- ▶ $\mathbf{J}$ has no members.

Paradoxical sets. Using the set builder notation does not commit us to regard as mathematically or philosophically acceptable to form a set from any statements.

To explain this point, remember that sets are objects too and, hence, can also be members of sets. Next, note that some member might be said to be members of themselves. Thus for instance, the set of mathematical entities is a mathematical entity, so it would be a member of itself. Other sets are clearly not members of themselves, like the set of living beings, which is no living being.

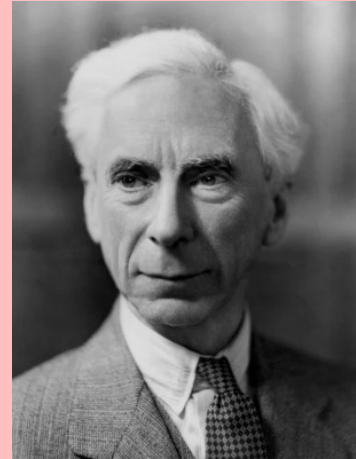
Now, consider the purported set:

$$\mathbf{R} = \{\mathbf{S} : \mathbf{S} \notin \mathbf{S}\}$$

Historical note: Russell's paradox

Though first noticed by Ernst Zermelo, Russell's paradox was first published by Bertrand Russell in his *Principles of Mathematics* (1903, §100). Here is how he explains it in a later work:

[N]ormally a class is not a member of itself. Mankind, for example, is not a man. Form now the assemblage of all classes which are not members of themselves. This is a class: is it a member of itself or not? If it is, it is one of those classes that are not members of themselves, i.e. it is not a member of itself. If it is not, it is not one of those classes that are not members of themselves, i.e. it is a member of itself. Thus of the two hypotheses—that it is, and that it is not, a member of itself—each implies its contradictory. (1919, p. 136)



B. Russell (1872–1970)

In other words, $\mathbf{R}$ is the set of all sets which are not members of themselves.

Now assume that $\mathbf{R} \in \mathbf{R}$. Consequently, $\mathbf{R}$ is a member of itself, which means that $\mathbf{R} \notin \mathbf{R}$. Since this contradicts with our assumption $\mathbf{R} \in \mathbf{R}$, we must conclude that $\mathbf{R} \notin \mathbf{R}$. However, this means that $\mathbf{R}$ is not a member of itself, which entails that $\mathbf{R} \in \mathbf{R}$.

We can then see that, at least in this reasoning, the existence of $\mathbf{R}$ is paradoxical, as it entails a contradiction. Thus, we should probably consider that no set $\mathbf{R}$, as described above, exists.

This paradox is known as *Russell's paradox*.

The lesson to take from this, at least for now, is that the set-builder notation must not be taken as license to make-up set in terms of a given expression at the left of ':'. It just allows us to speak about a set, if it exist, through this notation.

2.3 Operations and Relations on Sets

Union, intersection, difference. Given some sets, we can apply various operations to them in order to describe associated sets. The following defines the more standard such operations on sets:

Definition 2.10 (Set operations).

$$A \cup B \stackrel{\text{DF}}{=} \{x : x \in A \text{ } \mathbb{W} \text{ } x \in B\} \quad (\text{Union})$$

$$A \cap B \stackrel{\text{DF}}{=} \{x : x \in A \text{ } \mathbb{M} \text{ } x \in B\} \quad (\text{Intersection})$$

$$A \setminus B \stackrel{\text{DF}}{=} \{x : x \in A \text{ } \mathbb{M} \text{ } x \notin B\} \quad (\text{Difference})$$

: A, B stand for sets.

*: In some notations, we use ' $\subset$ ' for the *subset* relation and ' $\subsetneq$ ' for the *proper subset* relation (see below).

Subset, superset. Now, sets may be related in certain ways. Importantly, when a set contains all the members of another set, the latter is a *subset* of the former, and the former is a *superset* of the latter.*

Definition 2.11 (Subset and superset).

$$A \subseteq B \stackrel{\text{DF}}{=} \forall x (x \in A \Rightarrow x \in B). \quad (\text{Subset})$$

$$A \supseteq B \stackrel{\text{DF}}{=} B \subseteq A. \quad (\text{Superset})$$

: A, B stand for sets.

Extensionality. An important intuitive principle about sets is that they are defined solely by their *extension*, that is, by the elements they contain. Hence, if any two sets have the same elements, they are actually one and the same set.

†: More typically, we would express this axiom as follows:

$$\forall x (x \in A \Leftrightarrow x \in B) \Leftrightarrow A = B$$

: A, B stand for sets.

This principle is known as the *axiom of extensionality*, which we might express (a bit unorthodoxly) with the following axiom schema[†]:

Axiom 2.1 (Extensionality).

$$A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A : A, B \text{ stand for sets.}$$

‡: The notation ' ∇ ' is idiosyncratic.

More relations between sets. Other important relations include the following[‡]:

Definition 2.12 (Set relations).

$$A \subset B \stackrel{\text{DF}}{=} A \subseteq B \wedge A \neq B \quad (\text{Proper subset})$$

$$A \supset B \stackrel{\text{DF}}{=} B \subset A \quad (\text{Proper superset})$$

$$A \nabla B \stackrel{\text{DF}}{=} A \cap B = \{\} \quad (\text{Disjointness})$$

: A, B stand for sets.

Power set. As we have seen, set can also contain other sets as members. For example:

$$\{\{1, 2\}, \{a, b\}, \{3, 4\}, \{'a', 'za'\}\}.$$

A whole kind of sets are constructed via the **power set** operation. The power set of A , written ' $\wp A$ ' or ' 2^A ', is the set of all subsets of A .

Definition 2.13 (Power set).

$$\wp A \stackrel{\text{DF}}{=} \{X : X \subseteq A\}$$

: A stands for a set, X is an unspecified set.

Historical note: Naïve set theory

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Author (XXXX-XXXX)

Note that the empty set, $\{\}$, is a subset of any set and, hence, an element of the power set of any set.

Corollary 2.14. For all sets A , $\{\} \subseteq A$.

Proof. Let A be an arbitrarily chosen set. Note that $\{\}$ has no elements. Hence, all $x \in \{\}$ are such that $x \in A$ (and $x \notin \{\}$, and whatever we want). This, by [definition 2.12](#), entails that $\{\} \subseteq A$. Since A was arbitrarily chosen, this holds for all sets A . $\square$

Corollary 2.15. For all sets A , $\{\} \in \wp A$.

Proof. Let A be an arbitrarily chosen set. By [Corollary 2.14](#), we know that $\{\} \subseteq A$, which, by [definition 2.13](#), entails that $\{\} \in \wp A$. Since A was arbitrarily chosen, this holds for all sets A . $\square$

Task 2.3. Consider the following sets:

$$\begin{aligned} A &= \{n \in \mathbb{N} : 0 \leq n < 4\}, & B &= \{n \in \mathbb{N} : 1 < n \leq 4\}, \\ C &= A \cap B, & D &= A \cup B, \\ E &= B \setminus A, & F &= C \cup E. \end{aligned}$$

- For each pair A, B of these sets, determine whether: (i) $A \subseteq B$, (ii) $A = B$, (iii) $A \subset B$, (iv) $A \not\subset B$.
- Now consider their respective power sets $\wp A, \wp B, \wp C, \wp D, \wp E, \wp F$. Do these relations still hold for these sets too?

First, note that, in general, for all sets A , we have: $A \subseteq A$, $A = A$, not $A \subset A$, and not $A \not\subset A$. As for the relationships between sets with

different names, it is be useful to determine their exact contents:

$$\begin{aligned} \mathbf{A} &= \{n \in \mathbb{N} : 0 \leq n < 4\} = \{1, 2, 3\}, \\ \mathbf{B} &= \{n \in \mathbb{N} : 1 < n \leq 4\} = \{2, 3, 4\}, \\ \mathbf{C} &= \mathbf{A} \cap \mathbf{B} = \{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}, \\ \mathbf{D} &= \mathbf{A} \cup \mathbf{B} = \{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}, \\ \mathbf{E} &= \mathbf{B} \setminus \mathbf{A} = \{1, 2, 3\} \setminus \{2, 3, 4\} = \{4\}, \\ \mathbf{F} &= \mathbf{C} \cup \mathbf{E} = \{2, 3\} \cup \{4\} = \{2, 3, 4\} = \mathbf{B}, \end{aligned}$$

We can now easily determine the relations between **A**, **B**, **C**, **D**, **E**, **F**.

Regarding their power sets, things are a bit different. In particular, note that, whereas $\mathbf{A} \pitchfork \mathbf{E}$, this does not hold for their power sets. By [Corollary 2.14](#), the empty set $\{\}$ is an element – not just a subset – that $\wp \mathbf{A}$ and $\wp \mathbf{E}$ have in common, that is, $\wp \mathbf{A} \cap \wp \mathbf{E} = \{\{\}\}$.

Solution to task 2.3.

1. The tables below show how the sets are related. For example, in the $\subseteq$ -table, each cell at row **A** and column **B** contains ‘yes’ if $\mathbf{A} \subseteq \mathbf{B}$, and ‘no’ otherwise.

$\subseteq$	A	B	C	D	E	F
A	yes	no	no	yes	no	no
B	no	yes	no	yes	no	yes
C	yes	yes	yes	yes	no	yes
D	no	no	no	yes	no	no
E	no	yes	no	yes	yes	yes
F	no	yes	no	yes	no	yes

$=$	A	B	C	D	E	F
A	yes	no	no	no	no	no
B	no	yes	no	no	no	yes
C	no	no	yes	no	no	no
D	no	no	no	yes	no	no
E	no	no	no	no	yes	no
F	no	yes	no	no	no	yes

$\subset$	A	B	C	D	E	F
A	no	no	no	yes	no	no
B	no	no	no	yes	no	no
C	yes	yes	no	yes	no	yes
D	no	no	no	no	no	no
E	no	yes	no	yes	no	yes
F	no	no	no	yes	no	no

$\pitchfork$	A	B	C	D	E	F
A	no	no	no	no	yes	no
B	no	no	no	no	no	no
C	no	no	no	no	no	no
D	no	no	no	no	no	no
E	yes	no	no	no	no	no
F	no	no	no	no	no	no

2. Most of these relationships remain exactly the same for the power sets. However, this does not hold for the disjointness relationship, because, for all sets **A**, **B**, we have $\{\} \in \wp \mathbf{A} \cap \wp \mathbf{B}$.

2.4 Numerals and Numbers

Digits and Numerals. In this section, we want to define a special kind of expressions: *numerals*. These are names for numbers, e.g., ‘0’, ‘2’, ‘55’, ‘100’, etc. Note, however, that ‘00’, ‘01’, ‘000989’ are not usually regarded as numerals.

A way to do so is by defining the term ‘is a numeral’ via a clause-based inductive definition to be predicated of all expressions having the form of numerals.

In order to fix our base cases, we will first explicitly define the term 'digit' to cover numerals consisting of exactly one character.

$$\begin{aligned} d \text{ is a digit} &\stackrel{\text{DF}}{=} d = '0' \vee d = '1' \vee d = '2' \vee d = '3' \vee d = '4' \vee \\ &\quad d = '5' \vee d = '6' \vee d = '7' \vee d = '8' \vee d = '9' \quad (\Upsilon) \end{aligned}$$

The definition must make sure that, in the inductive clause, the first number is not a numeral.

Task 2.4. Provide a clause-based inductive definition of 'numeral'.

Hint: Note that all digits are numerals, but no multi-digit numeral starts with '0'.

Solution to task 2.4.

Definition 8. The term 'numeral' is recursively defined by the following clauses:

1. $d \text{ is a digit} \Rightarrow d \text{ is a numeral.}$ *(base clause)*
2. $n \text{ is a numeral} \wedge n \neq '0' \wedge d \text{ is a digit} \Rightarrow nd \text{ is a numeral.}$ *(inductive clause)*
3. No e is a numeral unless it can be obtained by finitely many applications of **1** and **2**. *(final clause)*

Note that the following definition of a numeral would be incorrect:

Mistake 2.16. The term 'numeral' is recursively defined by the following clauses:

1. $d \text{ is a digit} \Rightarrow d \text{ is a numeral.}$ *(base clause)*
2. $d \text{ is a digit} \wedge d \neq '0' \wedge n \text{ is a numeral} \Rightarrow \lceil dn \rceil \text{ is a numeral.}$ *(inductive clause)*
3. No e is a numeral unless it can be obtained by finitely many applications of **1** and **2**. *(final clause)*

Task 2.5. Explain why the definition in **mistake 2.16** is wrong.

The definition is intended to generate exactly the set of numerals, that is, strings of digits that either consist of the single digit '0' or begin with a non-zero digit followed by arbitrary digits. However, the given recursive definition fails to generate all such strings.

The error lies in the direction of concatenation used in the inductive clause. According to the base clause, every digit is a numeral. Hence, the numerals initially obtained are:

'0', '1', '2', '3', '4', '5', '6', '7', '8', '9'.

Historical note: The positional system

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Author (XXXX–XXXX)

The inductive clause then states that if d is a digit different from '0' and n is a numeral, then dn is a numeral. Thus, new digits are always added to the *left* of an already existing numeral, and the added digit must be non-zero.

Because of this, it is impossible to append digits to the right of an existing numeral. In particular, numerals consisting of more than one digit and ending in '0' cannot in general be generated.

For example, consider the numeral '100'. From the base clause we obtain '0' and '1'. Using the inductive clause, we can form '10' from '1' and '0'. However, there is no way to obtain '100', because this would require appending another '0' to the right of '10', which is not permitted by the inductive clause. The clause only allows placing a non-zero digit on the left of an already constructed numeral.

More generally, once the rightmost digit of a numeral has been fixed by the base clause, it can never be changed. Hence, many valid numerals, such as '100', '250', or '40', cannot be obtained by the given rules.

Solution to task 2.5.

The definition in **mistake 2.16** is incorrect because its inductive clause only allows adding digits to the left of a numeral. Thus digits cannot be appended to the right. For example, '10' can be formed, but '100' cannot, since this would require adding a '0' to the right of '10'. Hence, not all numerals can be generated.

The clause-based definitions above can become somewhat lengthy. A more convenient way to define numerals is via defining sets of digits and then sets of numerals.

We will define two sets of digits: **Lea**, the set of *leading digits*, which includes all digits from '1' to '9', and **Dig**, the set of *digits*,

which also includes '0'. Similarly, we define two sets of numerals: $\mathbf{Num}_{\mathbb{N}}$, the set of *natural numerals*, which includes all numerals starting from '1', and $\mathbf{Num}_{\mathbb{W}}$, the set of *whole numerals*, which also includes '0'.

Task 2.6. Define the sets **Lea**, **Dig**, $\mathbf{Num}_{\mathbb{N}}$ and $\mathbf{Num}_{\mathbb{W}}$.

Hint: Take inspiration from definition (I), on page 42.

Definition 2.17 (solution to task 2.6).

$$\mathbf{Lea} \stackrel{\text{DF}}{=} \{ '1', '2', '3', '4', '5', '6', '7', '8', '9' \}$$

$$\mathbf{Dig} \stackrel{\text{DF}}{=} \{ '0' \} \cup \mathbf{Lea}$$

$$\mathbf{Num}_{\mathbb{N}} \stackrel{\text{DF}}{=} \mathbf{Lea} \cup \{ nd : n \in \mathbf{Num}_{\mathbb{N}} \ \& \ d \in \mathbf{Dig} \}$$

$$\mathbf{Num}_{\mathbb{W}} \stackrel{\text{DF}}{=} \{ '0' \} \cup \mathbf{Num}_{\mathbb{N}}$$

Note that, despite appearances, **definition 2.17** is inductive, as its definiens states both the base case (to the left of '∪') and its inductive case (to its right).

Zero and successor. Having introduced numerals as expressions, we now turn to *number*, the objects that numerals denote. For example, the numeral '0' denotes the number zero, '2' denotes the number two, '55' denotes the number fifty-five, etc. Thus, numerals are linguistic objects, whereas numbers are abstract objects.

A basic theory of numbers can be expressed with a reduced language containing just the numeral '0' and the successor operator 'S(n)', where n stands for a number.

Intuitively, '0' represents the null quantity; for instance, to say that there are 0 dogs is to say that there are no dogs. The successor operator 'S' denotes the operation of going to the next number. For example, the successor of 0 is the number one, the successor of one is two, and so on.

Historical note: Peano Axioms

G. Peano (1858–1932)

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The fundamental properties of numbers are captured by the following axioms, known as the *Peano axioms*.

Axiom 2.2 (Peano axioms). Where ' n ' is a number variable and K is an arbitrary set, the set $\mathbb{W}$ of whole numbers is such that:

1. $0 \in \mathbb{W}$.
2. $\forall n (n \in \mathbb{W} \Rightarrow S(n) \in \mathbb{W})$.
3. $\neg \exists n (n \in \mathbb{W} \wedge S(n) = 0)$.
4. $\forall m \forall n (S(m) = S(n) \Rightarrow m = n)$.
5. $(0 \in K \wedge \forall n (n \in K \Rightarrow S(n) \in K)) \Rightarrow \mathbb{W} \subseteq K$. (Induction)

Axiom 2.2 characterises all the properties of whole numbers. In particular, we can now introduce the basic arithmetic operations. The following are some important theorems of Peano axioms.

Theorem 2.18. $\forall n (n \neq 0 \Rightarrow \exists m (n = S(m)))$.

Theorem 2.19. $\forall n (n \neq S(n))$.

The *sum* operation, written $\ulcorner m + n \urcorner$ (where m, n stand for numbers) combines two numbers into a new number. Intuitively, $(m + n)$ is the result of adding n units to m . We can define addition recursively.

Definition 2.20 (Sum).

$$\begin{aligned} (m + 0) &\stackrel{\text{DF}}{=} m \\ (m + S(n)) &\stackrel{\text{DF}}{=} S(m + n) \\ &: m, n \text{ stand for numbers.} \end{aligned}$$

The operation of *multiplication*, written $\ulcorner m \cdot n \urcorner$ represents repeated addition. Intuitively, $(m \cdot n)$ is the result of adding m to itself n times. Like sum, multiplication is defined recursively.

Definition 2.21 (Multiplication).

$$\begin{aligned} (m \cdot 0) &\stackrel{\text{DF}}{=} 0 \\ (m \cdot S(n)) &\stackrel{\text{DF}}{=} (m \cdot n) + m \\ &: m, n \text{ stand for numbers.} \end{aligned}$$

We can now easily introduce the usual numerals.

Definition 2.22 (Numerals).

$$\begin{aligned} 1 &\stackrel{\text{DF}}{=} S(0), & 2 &\stackrel{\text{DF}}{=} S(1), & 3 &\stackrel{\text{DF}}{=} S(2), \\ 4 &\stackrel{\text{DF}}{=} S(3), & 5 &\stackrel{\text{DF}}{=} S(4), & 6 &\stackrel{\text{DF}}{=} S(5), \\ 7 &\stackrel{\text{DF}}{=} S(6), & 8 &\stackrel{\text{DF}}{=} S(7), & 9 &\stackrel{\text{DF}}{=} S(8), \\ 10 &\stackrel{\text{DF}}{=} S(9), & m1 &\stackrel{\text{DF}}{=} (m \cdot 10) + 1, & m2 &\stackrel{\text{DF}}{=} (m \cdot 10) + 2, \\ m3 &\stackrel{\text{DF}}{=} (m \cdot 10) + 3, & m4 &\stackrel{\text{DF}}{=} (m \cdot 10) + 4, & m5 &\stackrel{\text{DF}}{=} (m \cdot 10) + 5, \\ m6 &\stackrel{\text{DF}}{=} (m \cdot 10) + 6, & m7 &\stackrel{\text{DF}}{=} (m \cdot 10) + 7, & m8 &\stackrel{\text{DF}}{=} (m \cdot 10) + 8, \\ m9 &\stackrel{\text{DF}}{=} (m \cdot 10) + 9, & n0 &\stackrel{\text{DF}}{=} (n \cdot 10) \\ &: m \in \mathbf{Num}_{\mathbb{W}} \setminus \{ '0' \} \ \& \ n \in \mathbf{Num}_{\mathbb{W}} \setminus \{ '0', '1' \}. \end{aligned}$$

Below are some important consequences of these definitions.

Theorem 2.23. $\forall n (0 + n = n)$.

Theorem 2.24. $\forall m \forall n (m + n = n + m)$.

Theorem 2.25. $\forall l \forall m \forall n ((l + m) + n = l + (m + n))$.

Theorem 2.26. $\forall n (0 \cdot n = 0)$.

Theorem 2.27. $\forall n (1 \cdot n = n)$.

Theorem 2.28. $\forall m \forall n (m \cdot n = n \cdot m)$.

Theorem 2.29. $\forall l \forall m \forall n ((l \cdot m) \cdot n = l \cdot (m \cdot n)).$

Theorem 2.30. $\forall l \forall m \forall n (l \cdot (m + n) = (l \cdot m) + (l \cdot n)).$

Strict and non-strict order. In addition to arithmetic operations, numbers admit an *order* relationship expressing when one number is less than another.

In particular, we have a relator $\lceil m < n \rceil$, which reads ' m is less than n ', expressing the relation of *strict order*.

Definition 2.31 (Strict order for numbers).

$$(m < n) \stackrel{\text{DF}}{=} \exists x (x \in \mathbb{N} \wedge (m + x) = n) \\ : m, n \text{ stand for numbers.}$$

Intuitively, $(m < n)$ holds precisely when n can be obtained from m by adding at least 1. That is, there is at least one step of separation from m to n in the succession chain.

We may also define a relator $\lceil m \leq n \rceil$, which reads ' m is less than or equal to n ', expressing the relation of *non-strict order*. Moreover, we have the converses $\lceil m \geq n \rceil$ and $\lceil m > n \rceil$ of the relations above, which read ' m is greater than or equal to n ' and ' m is greater n ', respectively.

Definition 2.32 (Order).

$$(m > n) \stackrel{\text{DF}}{=} (n < m) \\ (m \leq n) \stackrel{\text{DF}}{=} (m < n) \vee (m = n) \\ (m \geq n) \stackrel{\text{DF}}{=} (n \leq m)$$

Intuitively, $(m \leq n)$ holds precisely when n can be obtained from m by adding some number, which may be 0. Thus, it holds between m and m itself, and between m and all its successors.

The following are consequences of the definition of order.

Theorem 2.33. $\forall n (0 \leq n)$

Theorem 2.34. $\forall n (n \leq n).$

Theorem 2.35. $\forall m \forall n (m \leq n \wedge n \leq m \Rightarrow m = n).$

Theorem 2.36. $\forall l \forall m \forall n (l \leq m \wedge m \leq n \Rightarrow l \leq n).$

Historical note: Order

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Author (XXXX–XXXX)

Theorem 2.37. $\forall m \forall n (m < n \vee m = n \vee n < m).$

Theorem 2.38. $\forall k \forall m \forall n (m \leq n \Rightarrow m + k \leq n + k).$

Theorem 2.39. $\forall k \forall m \forall n (m < n \Rightarrow m + k < n + k).$

Theorem 2.40. $\forall k \forall m \forall n (m \leq n \Rightarrow m \cdot k \leq n \cdot k).$

Theorem 2.41. $\forall k \forall m \forall n (m < n \wedge k \neq 0 \Rightarrow m \cdot k < n \cdot k).$

Theorem 2.42. $\forall m \forall n (m < n \Rightarrow S(m) \leq n).$

Axiom 2.2 together with **definitions 2.20, 2.21 and 2.32** provide the basic structure needed to carry out arithmetic within the system of whole numbers.

Order and numerals. Note that definition **definition 2.31** applies to numbers, but not between numerals. It is useful to extend this relation to numerals too, so that this relation holds between two numerals provided it holds between its denoted numbers.

However, also note that **definition 2.31** is only defined for numbers, while **definition 2.32** has no restriction whatsoever. This means that, if we extend the relation ‘>’ to numerals, the **definition 2.32** would apply to them. We can do so by defining an operation analogous to ‘S’ for numerals: the next numeral operation ‘N’.

Definition 2.43 (Next).

$$\begin{aligned} N(em) = en &\stackrel{\text{DF}}{=} (m = '0' \wedge n = '1') \vee (m = '1' \wedge n = '2') \vee \\ &\quad (m = '2' \wedge n = '3') \vee (m = '3' \wedge n = '4') \vee \\ &\quad (m = '4' \wedge n = '5') \vee (m = '5' \wedge n = '6') \vee \\ &\quad (m = '6' \wedge n = '7') \vee (m = '7' \wedge n = '8') \vee \\ &\quad (m = '8' \wedge n = '9') \\ N(fm) = en &\stackrel{\text{DF}}{=} N(f) = e \wedge m = '9' \wedge n = '0' \end{aligned}$$

On this basis, we extend the relation '>' to numerals as follows:

Definition 2.44 (Strict order for numerals).

$$\begin{aligned} (m < n) &\stackrel{\text{DF}}{=} \exists x (x \in \mathbb{N} \wedge (m + x) = n) \\ &\quad : m, n \text{ stand for numbers.} \end{aligned}$$

This results in the following desired theorem:

Theorem 2.45. Let m, n be numerals and m, n , respectively, the numbers they stand for: $m < n \Leftrightarrow m < n$.

Note that **definitions 2.43** and **2.44** not only apply numerals, but also to any expression which ends with a numeral. This will include numerals occurring as subindexes, which we will introduce in the next section.

The operation 'N' and the relation '<' can also be extended to letters, so that ' $a < b$ ', ' $b < c$ ', and so forth, also for upper-case letters. In fact, we can extended to any expressions among which we can find some intuitive ordering.

Historical note: Cardinality

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Cardinality. The *cardinality* of a set expresses the idea of how many elements the set contains. It provides a way to compare sets in terms of their size, whether they are finite or infinite.

A set **A** is said to be finite iff its elements can be listed one by one and the listing ends. The cardinality of a set is simply the number of distinct elements it has.

We write ' $|A|$ ' to denote the cardinality of **A**. For instance:

$$|\{1, 2, 3\}| = 3, \quad |\{a, b\}| = 2, \quad |\{\}| = 0.$$

Two finite sets have the same cardinality if there exists a one-to-one correspondence (a **bijection**) between their elements.

The following definition express all this:

Definition 2.46 (Cardinality).

$$|\{\}\| \stackrel{\text{DF}}{=} 0$$

$$|\{a\}| \stackrel{\text{DF}}{=} 1$$

$$|A \cup B| \stackrel{\text{DF}}{=} |A| + |B| - |A \cap B|$$

: A, B stand for sets $\wedge a$ stands for an object.

A set is *infinite* iff it is not finite, that is, iff its elements cannot be fully listed in a finite sequence. Even so, infinite sets can also be compared by their cardinalities.

A classical example is the set of natural numbers:

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}.$$

Its cardinality is denoted by the symbol ' $\aleph_0$ ' (read 'aleph-null'), which is the smallest infinite cardinality.

Definition 2.47 (Cardinality of $\mathbb{N}$). $\aleph_0 \stackrel{\text{DF}}{=} |\mathbb{N}|$.

Although we cannot go into the proof, we will state the following important facts:

Fact 2.48. $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| = \aleph_0$.

Fact 2.49. $|\mathbb{R}| > \aleph_0$.

2.5 Sub-indexes and Ellipsis

Sub-indexed placeholders. Our list of basic placeholders provided in [Section 1.4](#) seems to be sufficient to express anything we want to say about unspecified expressions. We have seventy-four in total, and I am sure I will not use more than ten placeholders in any schema used or mentioned in this textbook.

However, this does not mean that the sixty-six basic placeholders will suffice for saying all we want to say. Sometimes, we will want to speak not just about one, two, or seventy-four unspecified expressions, but about an unspecified number of them; for instance, n such unspecified expressions.

To do so, we need to be able to have infinitely many placeholders. Actually, at least one for each whole number. A way to achieve this is to take all placeholders introduced in [notation 1.16](#) and create new placeholders constructed from their concatenation with subindexed numerals.

Let us improve our definition of a placeholder. First, we define a set **BEPl**_a containing our basic expression placeholders, and then we define the set **SEPl**_a of subindexed basic placeholders. We next merge these two sets into **EPl**_a, containing all our expression placeholders.

Definition 2.50 (Placeholder).

$$\mathbf{BEPl}_a \stackrel{\text{DF}}{=} \{b : b \text{ is a basic placeholder}\}.$$

$$\mathbf{SEPl}_a \stackrel{\text{DF}}{=} \{b_n : b \in \mathbf{BEPl}_a \wedge n \in \mathbf{Num}_{\mathbb{W}}\}.$$

$$\mathbf{EPl}_a \stackrel{\text{DF}}{=} \mathbf{BEPl}_a \cup \mathbf{SEPl}_a.$$

What [definition 2.50](#) says is that any expression of the form b_n is a subindexed placeholder. Note that the placeholder 'n' in [definition 2.50](#) stands for a numeral. Thus, the 'n' to the right of 'b' and to the left of ':' denotes our numeral n but written in a smaller font size at the bottom, i.e., a subindex.

Complex placeholders. Subindexed placeholders are not always enough, or at least we can use a further extension. Sometimes, we need to write a placeholder to make explicit that some expression occurs within it. Focus in the statement of the following notation, as we will use meta-placeholders.

Notation 2.51. Let P be a placeholder symbol* and l a comma-separated list of expressions. The expression $\lceil P[l] \rceil$ is a placeholder for an expression P in which the expressions contained in the list l may occur (in no particular order). Moreover, where m is a comma-separated list of expressions with the same number of commas as l , $\lceil P[m] \rceil$ denotes the expression resulting from

*: Thus, ' P ' functions here as a meta-placeholder.

Historical note: Ellipsis

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replacing P each element of l with a corresponding element of m .

†: Except, of course, in quotations.

Ellipsis. The reader might have noticed that a symbol has not appeared in this textbook so far[†] which would have been quite used and useful in logic textbooks: the ellipsis, that is, ‘...’.

This is an informal symbol used in mathematics, and which is not normally defined neither rigorously nor informally. The following defines ‘...’ as it will be understood throughout this textbook (outside quotations).

Definition 2.52 (Ellipsis). The expression ‘...’ is recursively defined as follows:

$$\begin{aligned}
 E[N(n)] \blacktriangleright \dots \blacktriangleright E[m] &\stackrel{\text{DF}}{=} \\
 E[m] \blacktriangleright \dots \blacktriangleright E[m] &\stackrel{\text{DF}}{=} E[m] \\
 E[m] \blacktriangleright \dots \blacktriangleright E[N(n)] &\stackrel{\text{DF}}{=} E[m] \blacktriangleright \dots \blacktriangleright E[n] \blacktriangleright E[N(n)] \\
 E[m] \blacktriangleright \dots &\stackrel{\text{DF}}{=} E[m] \blacktriangleright \dots \blacktriangleright E[u] \\
 &: m, n, u \in \mathbf{Num}_{\mathbb{W}} \ \& \ m \leq n \ \& \ u \geq m.
 \end{aligned}$$

In this definition, ‘ $\blacktriangleright$ ’ is a placeholder for a separating expression, like a comma. However, nothing prevents from applying this notation in an expression of the form:

$$\ulcorner S(1) + \dots + S(n) \urcorner,$$

where ‘+’ takes the place of ‘ $\blacktriangleright$ ’.

Also note that, in the third clause of **definition 2.52**, we the first expression is of the form ‘ $\ulcorner E[N(n)] \urcorner$ ’, where $N(n)$ is strictly less than m , the numeral occurring in the last expression, i.e., in $E[m]$. In this case, this denotes the empty expression, and not a sequence in the reverse direction.

We may refer to an element of a tuple in a specific position via a *projection*.

Notation 2.55 (Projection). $\ulcorner \mathbf{T}[n] \urcorner$ denotes the n -th member of $\mathbf{T}$: $\mathbf{T}$ is a tuple $n \in \mathbf{Num}_0 \bowtie n \in \mathbb{N}$ and n stands for n .

This allows us to define a tuple via a projection. For instance:

$$\mathbf{TEv}[n] \stackrel{\text{DF}}{=} 2 \times n$$

Thus, $\mathbf{TEv}[1] = 2$, $\mathbf{TEv}[2] = 4$, $\mathbf{TEv}[3] = 6$, $\mathbf{TEv}[4] = 8$, etc.

A practical way to denote tuples with finitely many members is by enumeration.

Notation 2.56 (Tuple by enumeration).

$$\begin{aligned} \langle a_1, \dots, a_n \rangle [1] &\stackrel{\text{DF}}{=} a_1, \dots \langle a_1, \dots, a_n \rangle [n] \stackrel{\text{DF}}{=} a_n \\ &: n \in \mathbf{Num}_{\mathbb{N}}. \end{aligned}$$

Historical note: Cartesian products

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Cartesian product. Given two sets **A** and **B**, the **Cartesian product** of **A** and **B**, written '**A** × **B**', is the set of all ordered pairs with first component in **A** and second in **B**.

$$\mathbf{A} \times \mathbf{B} \stackrel{\text{DF}}{=} \{\langle a, b \rangle : a \in \mathbf{A} \ \& \ b \in \mathbf{B}\}.$$

Given sets $\mathbf{A}_1, \dots, \mathbf{A}_n$, their n -fold Cartesian product, written ' $\mathbf{A}_1 \times \dots \times \mathbf{A}_n$ ', is defined as follows:

Definition 2.57 (Cartesian product).

$$\begin{aligned} \mathbf{A}_1 \times \dots \times \mathbf{A}_n &\stackrel{\text{DF}}{=} \{\langle a_1, \dots, a_n \rangle : a_1 \in \mathbf{A}_1 \ \& \ \dots \ \& \ a_n \in \mathbf{A}_n\} \\ &: n \in \mathbf{Num}_{\mathbb{N}} \ \& \ n \text{ stands for } n \ \& \\ &\mathbf{A}_1, \dots, \mathbf{A}_n \text{ stand for sets} \ \& \ a_1, \dots, a_n \text{ are objects.} \end{aligned}$$

When we take the Cartesian product with respect to a single set, we use the following notation.

Notation 2.58.

$$\begin{aligned} \mathbf{A}^1 &\stackrel{\text{DF}}{=} \mathbf{A} \\ \mathbf{A}^{N(n)} &\stackrel{\text{DF}}{=} \mathbf{A}^n \times \mathbf{A} \\ &: \mathbf{A} \text{ stands for a set.} \end{aligned}$$

Relations. Cartesian products allow us to describe how elements of sets are associated with one another. Such associations are called *relations*. We will consider the case of relations of arity two.

Definition 2.59 (Two-place relation).

R is a relation from A to $B \stackrel{\text{DF}}{=} R \subseteq A \times B$
 $: A, B$ stand for sets.

In other words, a relation is a set of ordered pairs, which would contain the objects which stand in this relation to each other, in a given direction. Thus, if $\langle a, b \rangle \in R$, we say that ' a stands in the relation R with respect to b '.

Some basic notions associated with relations are the following.

Definition 2.60 (Domain and range of a relation).

$\text{Dom}(R) \stackrel{\text{DF}}{=} \{x \in A : \exists y (y \in B \wedge \langle x, y \rangle \in R)\}$
 $\text{Ran}(R) \stackrel{\text{DF}}{=} \{y \in B : \exists x (x \in A \wedge \langle x, y \rangle \in R)\}$
 $: R \subseteq A \times B.$

Thus, for $R \subseteq A \times B$, we have that $\text{Dom}(R)$ is the set of all elements of A which stand in the relation R to some element of B , in that order; and $\text{Ran}(R)$ is the set of all elements of B which stand in the relation R to some element of A , but in the reverse order.

Definition 2.61 (Image and preimage).

$\text{Pre}_R(X) \stackrel{\text{DF}}{=} \{x \in X : \exists y (y \in B \wedge \langle x, y \rangle \in R)\}$
 $\text{Ima}_R(Y) \stackrel{\text{DF}}{=} \{y \in Y : \exists x (x \in A \wedge \langle x, y \rangle \in R)\}$
 $: R \subseteq A \times B \wedge X \subseteq A \wedge Y \subseteq B.$

Thus, where $R \subseteq A \times B$, $X \subseteq A$, and $Y \subseteq B$, we have that $\text{Pre}_R(X)$ is the set of all elements of X which stand in the relation R to some element of B , in that order; and $\text{Ima}_R(Y)$ is the set of all elements of Y which stand in the relation R to some element of A , but in the reverse order.

Functions. A *function* may be regarded as a special kind of relation in which each element of the domain is associated with exactly one element of the codomain. Let us consider one-place functions.

Definition 2.62 (Function).

$$\begin{aligned} f : A \rightarrow B &\stackrel{\text{DF}}{=} f \subseteq A \times B \wedge \\ &\forall x (x \in A \wedge \exists! y (y \in B \wedge \langle x, y \rangle \in f)) \\ &: A, B \text{ stand for sets.} \end{aligned}$$

The expression $\ulcorner f : A \rightarrow B \urcorner$ reads ' f is a function from A to B '. We have moreover the following notation.

Notation 2.63 (Function).

$$\begin{aligned} f(a) = b &\stackrel{\text{DF}}{=} \langle a, b \rangle \in f \\ &: a, b \text{ stand for objects.} \end{aligned}$$

Reasoning and Logic

3

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3.1 Argument and Validity

Arguments and argumentations. Loosely speaking, logic studies the principles ruling a correct *argument*. Just as we express propositions linguistically through declarative sentences, we express arguments linguistically through *argumentations* or *argument-texts*.

Notion 3.1 (Argument). An **argument** is an arrangement of propositions, one of which, the **conclusion**, is indicated as following from the others, the **premises**.

Notion 3.2 (Argumentation). An **argumentation** or **argument-text** is a linguistic expression that expresses an argument.

Since an argumentation A has to somehow express an arrangement of propositions, A may be construed as articulations of sentences, one of which expresses the conclusion of the argument expressed by A , while the rest express its premises.

In natural language, we often use a *conclusion marker* to indicate that the sentence that follows expresses the conclusion of the argument expressed by A . Consider, e.g., the following argumentation:

All humans are mortal.	($\top_1$)
Socrates is human.	($\top_2$)
Therefore,	($\top_3$)
Socrates is mortal.	($\top_4$)

Sentences ($\top_1$) and ($\top_2$) express the premises of the argument expressed by ($\top$), while ($\top_4$) expresses its conclusion. Sentence ($\top_4$) is marked to express the conclusion via ($\top_3$), a common *conclusion marker* in English.

In natural language, however, we often find argumentations with no explicit conclusion marker. For instance, it would suffice for sentences ($\top_1$), ($\top_2$), and ($\top_4$) to be written or uttered in that exact order for the average English speaker to regard it as an argumentation whose conclusion is expressed by ($\top_4$).

As with sentences and propositions, we will avoid some of the distinctions between arguments and argumentations. In particular, the following terminology will be convenient.

Notion 3.3 (Premises and conclusion of an argumentation). Let A be an argumentation and let $P_1, \dots, P_n$ stand for the premises and C for the conclusion of the argument A expresses. Hence, $P_1, \dots, P_n$ are the **premises** of A and that C is its **conclusion**.

Historical note: Arguments

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Author (XXXX-XXXX)

Standard form. In natural language, the premises and conclusions of an argumentation are not always easily identifiable. They are easy to identify in argumentation ($\Uparrow$), as it is written in a way approximating the standard form.

Notation 3.4 (Argumentation in standard form). ' $P_1 \dots P_n \therefore C$ ' and ' $C \because P_1 \dots P_n$ ' both denote an argument, where C expresses the conclusion and $P_1, \dots, P_n$ express the premises.

In **notation 3.4**, ' $\therefore$ ' is a *right-conclusion marker*, because it marks the expression to its right as the one expressing conclusion. ' $\because$ ' is instead a *left-conclusion marker*, as it marks the expression to its left as the one expressing the conclusion.*

*: Most authors use only a right-conclusion marker, which is normally ' $\therefore$ '. A few other authors also use ' $\because$ ', serving as a left-conclusion marker.

Thus, argumentation ($\Uparrow$) may be expressed in standard form in any of the two following ways:

All humans are mortal. Socrates is human. $\therefore$ Socrates is mortal.
Socrates is mortal. $\because$ All humans are mortal. Socrates is human.

It is also important to note that, in rewriting a given argumentation in standard form, we have some flexibility to rewrite the expressions contained in the original argumentation.

Consider, for example, the following argumentations:

All philosophers are human.	($\forall_1$)
All humans are mortal.	($\forall_2$)
Therefore,	($\forall_3$)
all philosophers are mortal.	($\forall_4$)

Humans die.	(II_1)
Philosophers are human.	(II_2)
Hence,	(II_3)
philosophers die.	(II_4)

Since they differ textually, argumentations (II) and (II) are different. The same holds for the pairs of expressions: (II_1), (II_2); (II_2), (II_1); (II_3), (II_3); and (II_4), (II_4). However, both argumentations express the same argument, and, for each of sentence pairs above, its two sentences express the same proposition.

Hence, it would be quite acceptable, when rewriting these argumentations in standard form, to do so in the same way.

All philosophers are human. All humans are mortal.
 $\therefore$ All philosophers are mortal. (II)

Humans die. Philosophers are human.
 $\therefore$ Philosophers die. (II)

In other words, it is quite acceptable to say that (II) and (II) are both rewritings of (II) in standard form, as well as of (II). After all, what we want to do with these standard forms is to more accurately express the argument expressed by the original argumentations, and not the original argumentations themselves.

Task 3.1. Write the following argumentations in standard form:

1. If it rains, the ground will get wet. And it's raining. Ergo, the ground is wet.
2. You should drink more water. For staying hydrated improves concentration and health.
3. Since the data are incomplete, the results are unreliable, as several variables were not considered.
4. I forgot my keys. Now I cannot enter the house.
5. Solar panels generate clean energy, so installing them is beneficial. After all, electricity bills are high.
6. The museum is worth visiting. It contains rare historical artefacts and interactive exhibits.
7. Chocolate contains sugar. However, eating sugar is unhealthy. Hence, eating chocolate must also be unhealthy.
8. Thus, the movie was long but entertaining.

Let us identify sentences the premises and conclusion of each argumentation above.

Argumentations 3.1.1, 3.1.4, and, 3.1.7 are quite straightforward, as they are written almost in standard form. Thus, ‘ergo’ and ‘hence’ work as right-conclusion markers in 3.1.1 and 3.1.7. Now, although we find no explicit conclusion marker in 3.1.4, we can easily identify the conclusion by the context.

Argumentations 3.1.2 and 3.1.6 are less typical, as they begin by stating the conclusion, and the premises follow as justification. In 3.1.2, ‘for’ works as some sort of premise marker. In 3.1.6, we have no such marker but the context suggests that the second sentence is the premise.

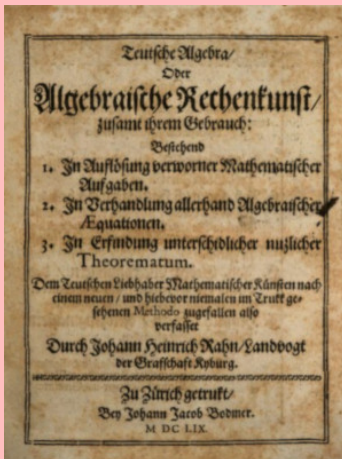
The forms of argumentations 3.1.3 and 3.1.5 are even less typical, the conclusion appearing in the middle. We have nevertheless some words indicating its premises and conclusions. Thus, in 3.1.3, ‘since’ seems to work as a premise marker, while in 3.1.5, ‘so’ seems to work as a right-conclusion marker. However, it may be debatable whether these are proper argumentations at all.

Finally, 3.1.8 does not look like a proper argumentation, as it consists of a single sentence preceded by ‘thus’ – which often does work as a conclusion marker in English – with no premises. We will, however, regard 3.1.8 as an argumentation. For although argumentations without premises are rarely – if at all – expressed in colloquial language, they are crucial in logic.

Solution to task 3.1.

Following common practice, we use $\vdash$ more often than $\therefore$ to write an argumentation in standard form.

1. If it rains, the ground will get wet. It is raining.
 $\vdash$ The ground is wet.
2. Staying hydrated improves concentration and health.
 $\vdash$ You should drink more water.
3. The data are incomplete.
 Several variables were not considered.
 $\vdash$ The results are unreliable.
4. I forgot my keys. $\vdash$ I cannot enter the house.
5. Solar panels generate clean energy.
 Electricity bills are high.
 $\vdash$ Installing solar panel is beneficial.
6. The museum contains rare historical artefacts and interactive exhibits. $\vdash$ The museum is worth visiting.
7. Chocolate contains sugar. Eating sugar is unhealthy.
 $\vdash$ Eating chocolate is unhealthy.
8. $\vdash$ The movie was long but entertaining.

Historical note: The conclusion marker

Teutsche Algebra, title page

The usual conclusion marker ' $\therefore$ ' was first introduced by Johann Heinrich Rahn (1622–1676) in his work *Teutsche Algebra oder algebraische Rechenkunst* (German Algebra or the Art of algebraic Calculation, 1659, pp. 52–3). The following is an extract:

As $a\sqrt{bc}$ is to $d\sqrt{bc}$, so a is to d

$$\therefore ad\sqrt{bc} = ad\sqrt{bc}.$$

According to the rule of proportions, the product of the two outer terms is equal to the product of the two middle terms; this is here indicated by the three little dots $\therefore$, which are called *ergo* or *therefore*.

Argumentations and validity. Note that argumentations, like sentences, are expressions conveying information which is, in some sense, complete – an argumentation expresses an argument that is either good or not.

This would suggest that argumentations fall within our notion of a sentence, and arguments within our notion of a proposition. This is, however, not the case. For arguments cannot be true or false.

Fact 3.5. Arguments cannot be true or false.

Fact 3.6. Arguments are not propositions.

Fact 3.7. Argumentations are not sentences.

What can be said to be true with respect to an argument is whether it is a good one. For instance, we can say that it is true that the argument expressed by an argumentation ' $P : \cdot C$ ' is good, e.g., in case the proposition expressed by B does follow from the proposition expressed by A .

To be more specific, there are at least two senses in which argument can be good.

Notion 3.8 (Argument validity). A **valid argument** is one whose conclusion follows from its premises.

Notion 3.9 (Argument soundness). A **sound argument** is one which is valid and whose premises are true.

We extend both notions to argumentations.

Notion 3.10 (Argumentation validity). A **valid argumentation** is one which expresses a valid argument.

Notion 3.11 (Argumentation soundness). A **sound argumentation** is one which expresses a sound argument.

Note, en passant, that the fact that something is an argument or an argumentation does not mean it is any good. It does not necessarily mean that there must be some logical connection between its premises and conclusion. It just means that it has a conclusion, and (zero or more) premises.

Logical validity. Not all arguments are valid for the same reasons. Thus, consider the following two arguments.

All swans are white. $\therefore$ No swan is black. $(\mathfrak{M})$

All humans are mortal. All Greeks are humans.
 $\therefore$ All Greeks are mortal. $(\neg)$

Both arguments seem to be valid in the sense that its conclusion follows from their premises. Argument $(\mathfrak{M})$, however, seems to be an instance of a schema which all of whose instances are valid arguments:

All h are m. All g are h. $\therefore$ All g are m $(\mathfrak{M})$

Argument $(\mathfrak{M})$, instead, does not seem to be come from an schema producing only valid arguments. It is valid because blackness and whiteness happen to be incompatible properties.

Thus, it seems that while argument $(\mathfrak{M})$ is valid just because of meaning of the terms 'black' and 'white', argument $(\neg)$ by virtue of its logical form. In other words, argument $(\neg)$ is *logically valid*.

Definition 3.12 (Logical validity). A is **logically valid** $\stackrel{\text{df}}{=}$ every B having the same logical form is valid : A, B are both arguments or both argumentations.

The problem is now to decide when a given argument is valid or not. There is more than one view on the matter, and such views are often expressed via logical systems or systems of logic.

3.2 Logical Systems

A *system of logic* aka *logical system* is a mathematical structure intended to capture how expressions, or the information they express, entails or is entailed by others. In what follows, we will use a special notation to name logical system.

Notation 3.13 (Logical system). A logical system is named with an expression starting with an upper-case letters in sans-serif font, such as: ‘A’, ‘B’, ..., ‘Y’, ‘Z’.

Formally, we usually define a system of logic as consisting of a formal language – intended to express the logical form of statements – and a relation of logical consequence defined on that language – intended to express which inferences are logically valid or not.

Using the concepts introduced so far, we may express this definition as follows.

Definition 3.14 (Logical system).

$$\langle \mathbf{FoS}, \mathbf{CnS} \rangle \text{ is a logical system} \stackrel{\text{DF}}{=} \mathbf{FoS} \subseteq \mathbf{Fo} \\ \mathbb{A} \mathbf{CnS} \subseteq \wp \mathbf{FoS} \times \wp \mathbf{FoS}$$

Here, $\langle \mathbf{FoS}, \mathbf{CnS} \rangle$ is a pair consisting of the formal language $\mathbf{FoS}$, construed as a subset of $\mathbf{Fo}$, and a relation between $\mathbf{CnS}$ between subsets of $\mathbf{FoS}$. This very pair is our logical system. Thus, if we want the formula ‘ Pa ’ to be a logical consequence of the formula ‘ $\forall x Px$ ’ according to this system, we need to define ‘ $\langle \mathbf{FoS}, \mathbf{CnS} \rangle$ ’ so that $\langle \{ \forall x Px \}, \{ Pa \} \rangle \in \mathbf{CnS}$.

So far, we have roughly presented the usual definition of a logical system. In this textbook, however, it will be convenient to analyse this notion in more detail.

In particular, for each logical system here considered, we will consider: its vocabulary $\mathbf{VoS}$; and its inference relation $\mathbf{InS}$ (not to be confused with the consequence relation). These are all collections which may be sets or tuples depending on the what is convenient for the system to be described.

The vocabulary $\mathbf{VoS}$ is just the collection of symbols we use in the system. It will be common that the vocabulary be of the form:

$$\mathbf{VoS} = \langle \mathbf{LogS}, \mathbf{NonS}, \mathbf{AuxS} \rangle$$

where $\mathbf{LogS}$ contains the logical symbols, $\mathbf{NonS}$ the non-logical symbols, and $\mathbf{AuxS}$ the auxiliary symbols of S . From these symbols in $\mathbf{VoS}$, we build the set of formulae $\mathbf{FoS}$, which is the formal language of the system.

The inference relation $\mathbf{InS}$ collects all the possible ways formulae can be arranged into inferences in S – whether those inferences are valid or not. Finally, the consequence relation $\mathbf{CnS} \subseteq \mathbf{InS}$ picks out the inferences that are actually valid in S . So while $\mathbf{InS}$ includes all possible inferences, $\mathbf{CnS}$ contains only those that count as correct according to the system.

Two further notions are closely tied to logical systems. A *derivation system*, or *calculus*, is a collection of rules that tell us how to

derive some formulae from others. An *interpretation system*, or *semantics*, assigns references to the logical symbols, and in this way determines what the formulae refer to or mean, under a given interpretation.

Let us now look at these systems in more detail using familiar logical systems as examples: the standard propositional logic, called 'S0' and the standard first-order logic, called 'S1'.

3.3 Vocabulary

Any logical system features a collection of logical symbols and a collection of non-logical symbols. We might also add a collection of auxiliary symbols, like, e.g., parentheses. The function of these symbols would be to regiment the expressions of our languages.

Thus, the vocabulary of a logical system S may be construed as a triple consisting of a collection $\mathbf{Log}_S$ of logical symbols, a collection $\mathbf{Non}_S$ of non-logical symbols, and a collection $\mathbf{Aux}_S$ of auxiliary symbols.

Many of the systems will use some of the following alphabets:

Definition 3.15 (Basic alphabets).

$$\begin{array}{ll} \mathbf{Abc} \stackrel{\text{DF}}{=} \{ 'a', 'b', 'c' \}, & \mathbf{ABC} \stackrel{\text{DF}}{=} \{ 'A', 'B', 'C' \}, \\ \mathbf{Pqr} \stackrel{\text{DF}}{=} \{ 'p', 'q', 'r' \}, & \mathbf{PQR} \stackrel{\text{DF}}{=} \{ 'P', 'Q', 'R' \}, \\ \mathbf{Xyz} \stackrel{\text{DF}}{=} \{ 'x', 'y', 'z' \}, & \mathbf{XYZ} \stackrel{\text{DF}}{=} \{ 'X', 'Y', 'X' \}, \\ \mathbf{Con} \stackrel{\text{DF}}{=} \{ '\neg', '\vee', '\wedge', '\rightarrow', '\leftrightarrow' \}, & \mathbf{Par} \stackrel{\text{DF}}{=} \{ '(', ')' \}. \end{array}$$

These collections may assume different functions depending on the logical system in which they are used. However, we might explain their usual functions as follows:

- ▶ **Abc** serves for constructing individual constants or terms;
- ▶ **ABC** serves for constructing predicate constants;
- ▶ **Pqr** serves for constructing propositional variables;
- ▶ **PQR** serves for constructing predicate constants or propositional variables;
- ▶ **Xyz** serves for constructing individual variables;
- ▶ **XYZ** serves for constructing predicate variables;
- ▶ **Con** contains logical connectives;
- ▶ **Par** contains auxiliary signs for regimentation.

There is reason why the latter two sets *contain* their symbol types, while the others *serve for constructing* them. Normally – but not always – we just need a finite amount of logical symbols, while we normally need an infinite amount of non-logical symbols. Aristotle’s canonical syllogistic is an exception to the latter.

Vocabulary of S0. The logical symbols of the standard propositional logic are logical connectives, which usually are: ‘¬’ (negation), ‘∨’ (disjunction), ‘∧’ (conjunction), ‘→’ (conditional), ‘↔’ (biconditional), which are precisely the members of **Con**.

It would be nevertheless useful to construe the set **Log_{S0}** of logical symbols of S0, whose first member consists of the one-place connective ‘¬’, and the second of all the other connectives, which are two-place connectives.

Task 3.2. Define ‘**Log_{S0}**’.

Hint: Define ‘**Log_{S0}[1]**’ and ‘**Log_{S0}[2]**’ separately.

Definition 3.16 (Logical symbols of S0 (answer to task 3.3)).

$$\begin{aligned}\mathbf{Log}_{S0}[1] &\stackrel{\text{DF}}{=} \{\neg\}, \\ \mathbf{Log}_{S0}[2] &\stackrel{\text{DF}}{=} \mathbf{Con} - \{\neg\}.\end{aligned}$$

The non-logical symbols of **Fo_{S0}** are propositional variables, which are usually represented with the lower case letters ‘*p*’, ‘*q*’, ‘*r*’ and their subindexed variants ‘*p*₁’, ‘*p*₂’, ..., ‘*q*₁’, ‘*q*₂’, ... ‘*r*₁’, ‘*r*₂’, ...

Thus, we may define our set of non-logical symbols **Non_{S0}** from **Pqr** and the set of numerals **Num_W**.

Task 3.3. Define ‘**Non_{S0}**’.

Definition 3.17 (Non-logical symbols of S0 (answer to task 3.3)).

$$\mathbf{Non}_{S0} \stackrel{\text{DF}}{=} \{p_i : p \in \mathbf{Pqr} \ \& \ i \in \mathbf{Num}_W\}$$

Thus, the vocabulary **Vo_{S0}** may be defined as follows:

Definition 3.18 (Vocabulary of S0).

$$\begin{aligned}\mathbf{Aux}_{S0} &\stackrel{\text{DF}}{=} \mathbf{Par}, \\ \mathbf{Vo}_{S0} &\stackrel{\text{DF}}{=} \langle \mathbf{Log}_{S0}, \mathbf{Non}_{S0}, \mathbf{Aux}_{S0} \rangle.\end{aligned}$$

Vocabulary of S1. The vocabulary to the standard first-order logic S1 is similar to that of S0. However, instead of propositional variables, we have individual terms and predicates. Moreover, we include two further logical symbols, the quantifiers '∃' and '∀'.

Given the complexity of this vocabulary, it may be convenient to define its logical and non-logical vocabularies as tuples.

The logical vocabulary $\mathbf{Vo}_{S1}$ may be defined as a pair consisting of the connectives plus the quantifiers.

Task 3.4. Define ' $\mathbf{Log}_{S1}$ '.

Hint: Define ' $\mathbf{Log}_{S1}[1]$ ' similarly as ' $\mathbf{Log}_{S0}$ '.

Definition 3.19 (Non-logical symbols of S1 (answer to task 3.4)).

$$\begin{aligned}\mathbf{Con}_{S1}[1] &\stackrel{\text{DF}}{=} \{\neg\}, \\ \mathbf{Con}_{S1}[2] &\stackrel{\text{DF}}{=} \mathbf{Con} - \{\neg\}, \\ \mathbf{Qua}_{S1} &\stackrel{\text{DF}}{=} \{\exists, \forall\}, \\ \mathbf{Log}_{S1} &\stackrel{\text{DF}}{=} \langle \mathbf{Con}_{S1}, \mathbf{Qua}_{S1} \rangle.\end{aligned}$$

The second member of $\mathbf{Vo}_{S1}$ will also be a pair, consisting of the individual terms $\mathbf{Ind}_{S1}$ plus the predicates $\mathbf{Pre}_{S1}$. However, as we have various types of individual terms and predicates, it may be convenient to codify these also as tuples.

Now, individual terms are either individual constants or individual variables, which are letters and subindexed letters. Thus, we might define a set of individual constants $\mathbf{Icon}_{S1}$ and a set of individual variables $\mathbf{Ivar}_{S1}$ from the sets $\mathbf{Abc}$ and $\mathbf{Xyz}$, respectively.

Task 3.5. Define ' $\mathbf{Icon}_{S1}$ ', ' $\mathbf{Ivar}_{S1}$ ', and ' $\mathbf{Ind}_{S1}$ '.

Definition 3.20 (Individual terms of S1 (answer to task 3.5)).

$$\begin{aligned}\mathbf{Icon}_{S1} &\stackrel{\text{DF}}{=} \{a_i : a \in \mathbf{Abc} \ \& \ i \in \mathbf{Num}_{\mathbb{W}}\}, \\ \mathbf{Ivar}_{S1} &\stackrel{\text{DF}}{=} \{x_i : x \in \mathbf{Xyz} \ \& \ i \in \mathbf{Num}_{\mathbb{W}}\}, \\ \mathbf{Ind}_{S1} &\stackrel{\text{DF}}{=} \langle \mathbf{Icon}_{S1}, \mathbf{Ivar}_{S1} \rangle.\end{aligned}$$

The set of predicates $\mathbf{Pre}_{S1}$ is a bit more complicated, as we need predicates of every arity – that is, one-place, two-place, etc. – and infinitely many of each of those.

Thus, one-place predicates will be the letters in $\mathbf{Pre}_{S1}$ plus the super-index 1 – i.e., by ' P^1 ', ' Q^1 ', ' R^1 ' – plus their subindexed versions – i.e., ' P_1^1 ', ..., ' Q_1^1 ', ..., ' R_1^1 ', etc.

In general, n -place predicates will be of the forms ' P^n ', ' Q^n ', ' R^n ', where n is the numeral that stands for the number n , and of the

forms of the subindexed variants of these expressions – i.e., $\ulcorner P_1^m \urcorner$, ..., $\ulcorner Q_1^m \urcorner$, ..., $\ulcorner R_1^m \urcorner$, etc.

To organise this, we will define a set for each arity. Thus, $\mathbf{Pre}_1$ will contain our one-place predicates, $\mathbf{Pre}_2$ will contain our two-place predicates. In general, $\mathbf{Pre}_n$ will contain our n -place predicates, where n is the numeral corresponding to n .

Task 3.6. Define $\ulcorner \mathbf{Pre}_n \urcorner$ for each $n \in \mathbf{Num}_{\mathbb{W}}$.

Hint: Write a schematic definition $\mathbf{Pre}_n$ for $n \in \mathbf{Num}_{\mathbb{W}}$.

Definition 3.21 (Predicates (answer to task 3.6)).

$$\mathbf{Pre}_n \stackrel{\text{DF}}{=} \{P_i^n : P \in \mathbf{Pre} \ \& \ i \in \mathbf{Num}_{\mathbb{W}}\} : n \in \mathbf{Num}_{\mathbb{N}}.$$

Now we can define $\mathbf{Pre}_{S1}$ as the tuple containing each of the sets defined above in the order expected.

Task 3.7. Define the ' $\mathbf{Pre}_{S1}$ '.

Definition 3.22 (Predicates (answer to task 3.7)).

$$\begin{aligned} \mathbf{Pre}_{S1}[n] &\stackrel{\text{DF}}{=} \mathbf{Pre}_n \\ &: n \in \mathbf{Num}_{\mathbb{N}}. \end{aligned}$$

Now we can define the tuples $\mathbf{Non}_{S1}$ and $\mathbf{Vo}_{S1}$ consisting of the non-logical symbols and full vocabulary of $S1$, respectively.

Task 3.8. Define (a) ' $\mathbf{Non}_{S1}$ ' and (b) ' $\mathbf{Vo}_{S1}$ '.

Definition 3.23 (Non-logical symbols of $S1$ (answer to task 3.8.a)).

$$\mathbf{Non}_{S1} \stackrel{\text{DF}}{=} \langle \mathbf{Ind}_{S1}, \mathbf{Pre}_{S1} \rangle.$$

Definition 3.24 (Vocabulary of $S1$ (answer to task 3.8.b)).

$$\begin{aligned} \mathbf{Aux}_{S1} &\stackrel{\text{DF}}{=} \mathbf{Par}, \\ \mathbf{Vo}_{S1} &\stackrel{\text{DF}}{=} \langle \mathbf{Log}_{S1}, \mathbf{Non}_{S1}, \mathbf{Aux}_{S1} \rangle. \end{aligned}$$

And this is a way we may systematically construct the vocabulary of the standard logical systems.

3.4 Formulae

A formal language is articulated as a set of formulae. We need is a set of formulae of some specific forms, all of which are amenable to a certain logical treatment. The set of formulae $\mathbf{Fo}_S$ of a logical system S is constructed from the symbols of its vocabulary $\mathbf{Vo}_S$.

Formulae of S_0 . The set of formulae $\mathbf{Fo}_{S_0}$ of the standard propositional logic are generated *recursively*. They are built up from simpler expressions according to precise formation rules.

$\mathbf{Fo}_{S_0}$ is based on the set $\mathbf{Non}_{S_0}$ (i.e., $\mathbf{Vo}_{S_0}[2]$) of *propositional variables*, which will be our atomic formulae. More complex formulae are then constructed with the help of the sets $\mathbf{Log}_{S_0}$ (i.e., $\mathbf{Vo}_{S_0}[1]$) of *connectives*, and $\mathbf{Aux}_{S_0}$ of auxiliary signs (i.e., $\mathbf{Vo}_{S_0}[3]$).

Table 3.1 shows the main forms of propositional formulae together with their natural-language analogues. To define this language, we may use a recursive definition like the following:

Preliminary 3.25 (Formulae of propositional logic). The set $\mathbf{Fo}_{S_0}$ is recursively defined as follows:

1. $p \in \mathbf{Non}_{S_0} \Rightarrow p \in \mathbf{X}$.
2. $p \in \mathbf{X} \Rightarrow \neg p \in \mathbf{X}$.
3. $p, q \in \mathbf{X} \Rightarrow (p \vee q), (p \wedge q), (p \rightarrow q), (p \leftrightarrow q) \in \mathbf{X}$.
4. $\mathbf{Fo}_{S_0}$ is the intersection of all sets $\mathbf{X}$ satisfying 1–3.

We might define ' $\mathbf{Fo}_{S_0}$ ' more shortly using ' $\stackrel{\text{DF}}{=}$ '.

Task 3.9. Define $\mathbf{Fo}_{S_0}$ using ' $\stackrel{\text{DF}}{=}$ '.

Definition 3.26 (Formulae of S_0 (answer to task 3.9)).

$$\mathbf{Fo}_{S_0} \stackrel{\text{DF}}{=} \mathbf{Non}_{S_0} \cup \{ \neg p : p \in \mathbf{Fo}_{S_0} \} \cup \{ (p \heartsuit q) : p, q \in \mathbf{Fo}_{S_0} \wedge \heartsuit \in \mathbf{Log}_{S_0}[2] \}.$$

Table 3.1: Propositional formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Atomic	p	Peirce is a logician.
Atomic	q	Quine is a philosopher.
Negation	$\neg p$	It is not the case that Peirce is a teacher.
Disjunction	$(p \vee q)$	Peirce is a logician or Quine is a philosopher.
Conjunction	$(p \wedge q)$	Peirce is a logician and Quine is a philosopher.
Conditional	$(p \rightarrow q)$	If Peirce is a logician, then Quine is a philosopher.
Biconditional	$(p \leftrightarrow q)$	Peirce is a logician iff Quine is a philosopher.

Formulae of S1. $\mathbf{Fo}_{S1}$ shares important features with $\mathbf{Fo}_{S1}$. Their atomic formulae, however, are not propositional variables, but are more complexly built from predicates, which denote properties or relations, and individual terms, which denote specific or unspecified entities. We furthermore have *quantifiers* to specify how many unspecified entities are being referred.

Table 3.2 shows some typical forms of first-order formulae together with their natural-language analogues.

A first useful step towards constructing $\mathbf{Fo}_{S1}$ is to construct a set $\mathbf{ITer}_{S1}$ of individual terms S1, containing the individual constants and individual variables of its language.

Definition 3.27 (Individual terms of S1).

$$\mathbf{ITer}_{S1} \stackrel{\text{DF}}{=} \mathbf{ICon}_{S1} \cup \mathbf{IVar}_{S1}$$

In other words, $\mathbf{ITer}_{S1}$ is the set of individual constants and variables of S1, or:

$$\mathbf{ITer}_{S1} = \{ 'a', 'a_0', \dots, 'c', 'c_0', \dots \} \cup \{ 'x', 'x_0', \dots, 'z', 'z_0', \dots \}.$$

This makes it easy to define ' $\mathbf{Fo}_{S1}$ ' recursively.

Preliminary 3.28 (Formulae of $\mathbf{Fo}_{S1}$). Where $n \in \mathbf{Num}_{\mathbb{W}}$, the set $\mathbf{Fo}_{S1}$ is recursively defined as follows:

1. $P \in \mathbf{Pre}_n \wedge t_1, \dots, t_n \in \mathbf{ITer}_{S1} \Rightarrow P t_1 \dots t_n \in \mathbf{X} : n \in \mathbb{N}$.
2. $F \in \mathbf{X} \Rightarrow \neg F \in \mathbf{X}$.
3. $F, G \in \mathbf{X} \Rightarrow (F \vee G), (F \wedge G), (F \rightarrow G), (F \leftrightarrow G) \in \mathbf{X}$.
4. $F \in \mathbf{X} \wedge x \in \mathbf{IVar}_{S1} \Rightarrow \exists x F, \forall x F \in \mathbf{X}$.
5. $\mathbf{Fo}_{S1}$ is the intersection of all sets $\mathbf{X}$ satisfying 1–4.

As with ' $\mathbf{Fo}_{S0}$ ', we might define ' $\mathbf{Fo}_{S1}$ ' more shortly using ' $\stackrel{\text{DF}}{=}$ '.

Task 3.10. Define ' $\mathbf{Fo}_{S1}$ ' using ' $\stackrel{\text{DF}}{=}$ '.

Table 3.2: First-order formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Atomic	Pa	Aristotle is a philosopher.
Atomic	Px	x is a philosopher.
Atomic	Rab	Aristotle respects Boole.
Atomic	Rxy	x respects y .
Existential	$\exists x Px$	Some object is a philosopher.
Universal	$\forall x Px$	Every object is a philosopher.

Definition 3.29 (Formulae of S_1 (answer to task 3.10)).

$$\begin{aligned} \mathbf{Fo}_{S_1} \stackrel{\text{DF}}{=} & \{P t_1 \dots t_n : P \in \mathbf{Pres}_1[n] \ \& \ t_1 \dots t_n \in \mathbf{ITer}_{S_1}\} \cup \\ & \{\ulcorner \neg F \urcorner : F \in \mathbf{Fo}_{S_1}\} \cup \\ & \{\ulcorner (F \heartsuit G) \urcorner : F, G \in \mathbf{Fo}_{S_1} \ \& \ \heartsuit \in \mathbf{Con}_{S_1}[2]\} \cup \\ & \{\ulcorner \forall x F : \mathbf{Q} \in \mathbf{Qua}_{S_1} \ \& \ x \in \mathbf{IVar}_{S_1} \ \& \ F \in \mathbf{Fo}_{S_1} \urcorner \\ & : n \in \mathbb{N} \end{aligned}$$

3.5 Inference Relations

An inference relation $\mathbf{In}_S$ is a set of inferences $\mathbf{I} = \langle \mathbf{A}, \mathbf{B} \rangle$. An inference is the formal counterpart of an argument, so $\mathbf{A}$ is a collection containing the formulae which represent the premises of the argument that $\mathbf{I}$ represents, while $\mathbf{B}$ contains the formulae representing its conclusions – normally only one.

Within a logical system S , $\mathbf{In}_S$ represents all the forms in which formulae may be combined as premises and conclusions of an argument according to S .

In what follows, we will often use the inference bar notation, which will let us read our inferences with more easy.

Notation 3.30 (Inference bar). The notations $\ulcorner A_1, \dots, A_m / B_1, \dots, B_n \urcorner$ and:

$$\begin{array}{c} A_1 \\ \dots \\ A_m \\ \hline B_1 \\ \dots \\ B_n \end{array}$$

will denote an inference of one of the forms:

$$\begin{aligned} \langle \{A_1, \dots, A_m\}, \{B_1, \dots, B_n\} \rangle, & \quad \langle \{A_1, \dots, A_m\}, \langle B_1, \dots, B_n \rangle \rangle, \\ \langle \langle A_1, \dots, A_m \rangle, \{B_1, \dots, B_n\} \rangle, & \quad \langle \langle A_1, \dots, A_m \rangle, \langle B_1, \dots, B_n \rangle \rangle. \end{aligned}$$

Similarly, the notations $\ulcorner A_1, \dots, A_m // B_1, \dots, B_n \urcorner$ and:

$$\begin{array}{c} A_1 \\ \dots \\ A_m \\ \hline B_1 \\ \dots \\ B_n \end{array}$$

will denote two inferences of the forms: $\ulcorner A_1, \dots, A_m / B_1, \dots, B_n \urcorner$ and $\ulcorner B_1, \dots, B_n / A_1, \dots, A_m \urcorner$.

This notation implicitly encloses each formula in corner marks.

Which of the four forms this notation represents is left to context and may be ambiguous.

It is important to remark here that it is formally possible for inferences to have multiple conclusions. This is because some logical systems allow for this. However, the inferences of most of the systems we will study will have only one formula as a conclusion, which we will represent as a singleton or a one-tuple.

Inference relation of modern logical systems. An inference in S_0 , S_1 , and virtually any modern logical system M is given between any (finite or infinite) collection of formulae representing the premises, and one formula, representing the conclusion.

In particular, in any given modern logical system M , $\langle A, \{F\} \rangle$ is an inference iff $A \subseteq \mathbf{Fo}_M$ and $F \in \mathbf{Fo}_M$.

Task 3.11. Define ' $\mathbf{In}_M$ ', where M is an unspecified modern logical system.

Definition 3.31 (Inference of M (answer to task 3.11)).

$$\mathbf{In}_M \stackrel{\text{DF}}{=} \wp \mathbf{Fo}_M \times \{ \{F\} : F \in \mathbf{Fo}_M \}$$

: M is a standard logical system.

Note that $\{ \{F\} : F \in \mathbf{Fo}_M \}$ is just the set of all singletons of $\mathbf{Fo}_M$. Thus, if we wanted to simplify things, we could have just written:

$$\mathbf{In}_M \stackrel{\text{DF}}{=} \wp \mathbf{Fo}_M \times \mathbf{Fo}_M.$$

However, since we want our definition of inference relation be so general as to include those logical systems which allow for multiple conclusions, we will have to use **definition 3.31**.

3.6 Consequence Relation

A consequence relation $\mathbf{Cn}_S$ is always a subset of the set $\mathbf{In}_S$ of inferences, which contains all and only the S -valid inferences – that is, those inferences which are logically valid in S .

Axiom 3.1 (Consequence relation).

$$\mathbf{Cn}_S \subseteq \mathbf{In}_S : S \text{ is a logical system.}$$

Note that the content of $\mathbf{Cn}_S$, as we understand it here, needs not to depend on a specific semantics or calculus to be determined. We may, e.g., explicitly specify it by saying exactly which inferences are S -valid, as we will do later with A .

However, in most cases, including S_0 and S_1 , it is necessary to specify that content by means of one such calculus or semantics.

Consequence relation of S_0 . As we know, there are infinitely many S_0 -valid inferences, which makes it impossible to list them. We have nevertheless a good idea of what we want $\mathbf{Cn}_{S_0}$ to be.

Among other things, we want all inferences of these forms to be S_0 -valid. In other words, we want all the following inferences to be elements of $\mathbf{Cn}_{S_0}$.

Modus ponendo ponens (MPP): $(A \rightarrow B), A / B$.

Modus tollendo tollens (MTT): $(A \rightarrow B), \neg B / \neg A$.

Reiteration: A / A .

Disjunctive syllogism 1 (DS1): $(A \vee B), \neg A / B$.

Disjunctive syllogism 2 (DS2): $(A \vee B), \neg B / A$.

Law of excluded middle: $/ (A \vee \neg A)$.

Law of non-contradiction: $/ \neg(A \wedge \neg A)$.

Reflexivity of $\rightarrow$: $/ (A \rightarrow A)$.

Exportation: $((A \wedge B) \rightarrow C) / (A \rightarrow (B \rightarrow C))$.

Importation: $(A \rightarrow (B \rightarrow C)) / ((A \wedge B) \rightarrow C)$.

De Morgan 1: $\neg(A \wedge B) // (\neg A \vee \neg B)$.

De Morgan 2: $\neg(A \vee B) // (\neg A \wedge \neg B)$.

Double negation 1: $A / \neg\neg A$.

Double negation 2: $\neg\neg A / A$.

Idempotence of $\vee$: $A // (A \vee A)$.

Idempotence of $\wedge$: $A // (A \wedge A)$.

Simplification 1: $(A \wedge B) / A$.

Simplification 2: $(A \wedge B) / B$.

Addition 1: $A / (A \vee B)$.

Addition 2: $B / (A \vee B)$.

Conjunction: $A, B / (A \wedge B)$.

Explosion (or ECQ): $(A \wedge \neg A) / B$.

Commutativity of $\wedge$: $(A \wedge B) / (B \wedge A)$.

Commutativity of $\vee$: $(B \vee A) / (A \vee B)$.

Simply enumerating these or any other sets of rules instantiated, however, is not the most effective method for establishing $\mathbf{Cn}_{S_0}$. To do so, it may be better to define a semantics or a calculus from which we might define ' $\mathbf{Cn}_{S_0}$ '.

Historical note: Hilbert's positive logic

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Author (XXXX-XXXX)

Consequence relation of S1. Similar to S0, we cannot define the consequence relation of $\mathbf{Cn}_{S1}$ by listing its elements one by one. We have infinitely many S1-valid inferences, which makes it impossible to enumerate them.

We nevertheless also have a good idea of what we want $\mathbf{Cn}_{S1}$ to be. Among other things, where $F, G \in \mathbf{Fo}_{S1} \ \& \ t \in \mathbf{Ind}_{S1} \ \& \ v, w \in \mathbf{IVar}_{S1}$, we want all inferences of the forms below to be S1-valid, that is, to be elements of $\mathbf{Cn}_{S1}$.

Universal instantiation:

$$\forall v F[v] / F[t] : t \text{ is free for } v \text{ in } F.$$

Existential generalisation:

$$F[t] / \exists v F[v] : t \text{ is free for } v \text{ in } F.$$

Quantifier equivalences:

- ▶ $\neg \forall v \neg F[v] // \exists v F[v].$
- ▶ $\neg \exists v \neg F[v] // \forall v F[v].$

Quantifier negation laws:

- ▶ $\neg \forall v F[v] // \exists v \neg F[v].$
- ▶ $\neg \exists v F[v] // \forall v \neg F[v].$

Universal distribution over conjunction:

$$\forall v (F[v] \wedge G[v]) // \forall v F[v] \wedge \forall v G[v].$$

Existential distribution over disjunction:

$$\exists v (F[v] \vee G[v]) // \exists v F[v] \vee \exists v G[v].$$

Universal quantifier commutation:

$$\forall v \forall w F[v][w] // \forall w \forall v F[v][w].$$

Existential quantifier commutation:

$$\exists v \exists w F[v][w] \text{ // } \exists w \exists v F[v][w].$$

As with S_0 , simply enumerating these or any other sets of rules instantiated, however, is not the most effective method for establishing $\mathbf{Cn}_{S_1}$. To do so, it may be better to define a semantics or a calculus from which we might define ' $\mathbf{Cn}_{S_1}$ '.

3.7 Interpretation System

The formulae in a formal language $\mathbf{FoS}$, normally, do not stand for propositions. In the formula ' P^1a ', for instance, ' P^1 ' and ' a ' stand for non-logical symbols without any established meaning. All we know of ' P^1 ' is that it stands for some one-place property, and of ' a ', that it stands for some specific object in the domain.

For this reason, we can say that the formulae in $\mathbf{FoS}$ are normally neither true nor false. However, given some further information about its non-logical symbols is supplied, a formula may be interpreted as being true or false.

Thus, if we interpret ' P^1 ' as standing for the predicate of being a philosopher and ' a ' as standing for Aristotle, then ' P^1a ' would stand for the proposition that Aristotle is a philosopher, making ' P^1 ' true under this interpretation.

In general, an interpretation $\mathfrak{I}$ on a formal language $\mathbf{Fo}_S$ provides a mapping of non-logical symbols $\mathbf{Non}_S$ of S into a domain.

Definition 3.32 (Interpretation).

$$\begin{aligned} \mathfrak{I} \text{ is an interpretation on } \mathbf{Fo}_S &\stackrel{\text{df}}{=} \mathfrak{I} = \langle \mathbf{D}, \iota \rangle \\ &: \iota \subseteq \mathbf{Non}_S \times \mathbf{D}. \end{aligned}$$

The non-logical symbols of a propositional logical system S are propositional variables which an interpretation will map into logical values, e.g. truth and falsity. In a first-order logic, those logical symbols are individual terms and predicates which are mapped into objects and sets in a domain.

In both cases, we then extend those interpretation into *valuations* from all formulae of $\mathbf{Fo}_S$ into logical values.

Definition 3.33 (Valuation).

$$\begin{aligned} \hat{\mathfrak{I}} \text{ is a valuation on } \mathbf{Fo}_S &\stackrel{\text{df}}{=} \hat{\mathfrak{I}} = \langle \mathbf{LV}, \hat{\iota} \rangle \\ &: \hat{\iota} \subseteq \mathbf{Fo}_S \times \mathbf{LV} \text{ } \mathbb{M} \\ &\mathbf{LV} \text{ is a set of logical values.} \end{aligned}$$

An *interpretation system* or a *semantics* $\mathbf{Int}_S$ on a logic system S consists of a set $\mathbf{Int}_S$ of interpretations on $\mathbf{Int}_S$, together with a set $\mathbf{LV}_S$ of logical values, a subset $\mathbf{Des}_S^+$ of $\mathbf{LV}_S$ of designated logical values, a subset $\mathbf{Des}_S^-$ of $\mathbf{LV}_S$ of anti-designated logical values, a set $\mathbf{Val}_S$ of valuations on $\mathbf{Fo}_S$, and a semantic relation of logical consequence $\models_S$.

Definition 3.34 (Interpretation system).

$$\begin{aligned} \mathbf{Sem}_S \text{ is a semantics on } S &\stackrel{\text{df}}{=} \mathbf{Sem}_S = \langle \mathbf{Int}_S, \mathbf{LV}_S, \mathbf{Val}_S, \mathbf{Des}_S^+, \mathbf{Des}_S^-, \models_S \rangle \\ &: \mathbf{Int}_S \text{ is a set of interpretations on } \mathbf{Fo}_S \text{ } \mathbb{M} \\ &\mathbf{LV}_S \text{ is a set of logical values } \mathbb{M} \\ &\mathbf{Des}_S^+ \subseteq \mathbf{LV}_S \text{ } \mathbb{M} \\ &\mathbf{Des}_S^- \subseteq \mathbf{LV}_S \text{ } \mathbb{M} \\ &\mathbf{Val}_S \text{ is a set of valuations on } \mathbf{Fo}_S \text{ } \mathbb{M} \\ &\models_S \subseteq \mathbf{In}_S. \end{aligned}$$

Intuitively, the set $\mathbf{Des}_S^+$ of designated logical values contains those values related to the true or what can be asserted or accepted under an interpretation, while the set $\mathbf{Des}_S^-$ of anti-designated values contains those values related to the false or what can be denied or rejected under an interpretation.

In order to present the semantics of propositional connectives in a more friendly way, we might use truth-tables.

Notation 3.35 (Truth tables). The table

A	$\bullet A$
v_1	w_1
...	...
v_n	w_n

expresses that, under a given valuation $\hat{\mathfrak{S}}$, we have:

$$\iota(A) = v_1 \Rightarrow \hat{\iota}(\ulcorner \bullet A \urcorner) = w_1 \quad \mathbb{M}$$

$$\dots \quad \mathbb{M}$$

$$\iota(A) = v_n \Rightarrow \hat{\iota}(\ulcorner \bullet A \urcorner) = w_n.$$

Moreover, the tables:

A	B	$A \heartsuit B$	$\heartsuit$	v_1	...	v_n
u_1	v_1	w_1	u_1	w_1	...	w_a
...	...	...	...	...	...	...
u_n	v_n	w_n	u_n	w_b	...	w_n

both express that, under a given valuation $\hat{\mathfrak{S}}$, we have:

$$\iota(A) = u_1 \mathbb{M} \iota(B) = v_1 \Rightarrow \hat{\iota}(\ulcorner A \heartsuit B \urcorner) = w_1 \quad \mathbb{M}$$

$$\dots \quad \mathbb{M}$$

$$\iota(A) = u_n \mathbb{M} \iota(B) = v_n \Rightarrow \hat{\iota}(\ulcorner A \heartsuit B \urcorner) = w_n$$

Let us see how the interpretation systems of standard logical systems work.

Semantics for S0. In a standard propositional logical system S0, the non-logical symbols are propositional variables. A classical propositional interpretation, or cpi for short, assigns to each propositional variable a classical truth value, either **1** (truth) or **0** (falsity).

Definition 3.36 (Classical propositional interpretation).

$$\langle \mathbf{D}, \iota \rangle \text{ is a cpi on } \mathbf{Fo}_S \stackrel{\text{df}}{=} \mathbf{D} = \{\mathbf{0}, \mathbf{1}\} \mathbb{M} \iota : \mathbf{Pro}_S \rightarrow \{\mathbf{0}, \mathbf{1}\}.$$

In other words, $\langle \mathbf{D}, \iota \rangle$ is a cpi on $\mathbf{Fo}_S$ iff its domain is the set of logical values $\{\mathbf{0}, \mathbf{1}\}$ and its interpretation map ι is a function from the set of propositional variables to $\{\mathbf{0}, \mathbf{1}\}$ – that is, each propositional variable is assigned either **0** or **1**, but not both.

A cpi $\mathfrak{I}$ is extended to all formulae of $\mathbf{Fo}_{S0}$ by means of a valuation $\hat{\mathfrak{I}}$, which is its classical propositional extension (cpe).

Definition 3.37 (Classical propositional extension). A valuation $\hat{\mathfrak{I}}$ is a **cpe** of the cpi $\mathfrak{I}$ on $\mathbf{Fo}_S$

$$\begin{aligned}
 &\stackrel{\text{DF}}{=} \hat{\iota}(\mathbf{A}) = \iota(\mathbf{A}) && \mathbb{M} \\
 &\hat{\iota}(\neg \mathbf{F}) = \mathbf{1} \Leftrightarrow \hat{\iota}(\mathbf{F}) = \mathbf{0} && \mathbb{M} \\
 &\hat{\iota}(\neg(\mathbf{F} \wedge \mathbf{G})) = \mathbf{1} \Leftrightarrow \hat{\iota}(\mathbf{F}) = \mathbf{1} \mathbb{M} \hat{\iota}(\mathbf{G}) = \mathbf{1} && \mathbb{M} \\
 &\hat{\iota}(\neg(\mathbf{F} \vee \mathbf{G})) = \mathbf{1} \Leftrightarrow \hat{\iota}(\mathbf{F}) = \mathbf{1} \mathbb{W} \hat{\iota}(\mathbf{G}) = \mathbf{1} && \mathbb{M} \\
 &\hat{\iota}(\neg(\mathbf{F} \rightarrow \mathbf{G})) = \mathbf{1} \Leftrightarrow \hat{\iota}(\mathbf{F}) = \mathbf{0} \mathbb{W} \hat{\iota}(\mathbf{G}) = \mathbf{1} && \mathbb{M} \\
 &\hat{\iota}(\neg(\mathbf{F} \leftrightarrow \mathbf{G})) = \mathbf{1} \Leftrightarrow \hat{\iota}(\mathbf{F}) = \hat{\iota}(\mathbf{G}) && \mathbb{M} \\
 &: \mathbf{A} \in \mathbf{Pro}_S \mathbb{M} \mathbf{F}, \mathbf{G} \in \mathbf{Fo}_S.
 \end{aligned}$$

Notation 3.35 makes it possible to rephrase **definition 3.37** as follows:

Definition 3.38 (Classical propositional valuation). A valuation $\hat{\mathfrak{I}}$ is a **classical extension** of the cpi $\mathfrak{I}$ on $\mathbf{Fo}_S \stackrel{\text{DF}}{=} \hat{\iota}(\mathbf{A}) = \iota(\mathbf{A}) \mathbb{M}$

F	$\neg F$	$\wedge$	0	1	$\vee$	0	1
0	1	0	0	0	0	0	1
1	0	1	0	1	1	1	1

$\rightarrow$	0	1	$\leftrightarrow$	0	1
0	1	1	0	1	0
1	0	1	1	0	1

: $\mathbf{A} \in \mathbf{Pro}_S \mathbb{M} \mathbf{F} \in \mathbf{Fo}_S$.

Now, as you might have guessed the one designated value in a semantics for $S0$ is **1**, that is, truth; while the only anti-designated value is **0**, that is, falsity. The standard notion of semantic logical consequence says that, $\mathbf{B}$ is a semantic logical consequence of $\mathbf{A}$ iff there is no interpretation $\mathfrak{I}$ such that $\hat{\iota}(\mathbf{A}) = \mathbf{1}$ but $\hat{\iota}(\mathbf{B}) = \mathbf{0}$.

We can generalise this standard notion of semantic logical consequence using the notions of designated and anti-designated value.

Definition 3.39 (Semantic consequence). $\models_S$ is a standard semantic consequence relation $\stackrel{\text{DF}}{=}$

$$\begin{aligned}
 \langle \mathbf{A}, \{\mathbf{B}\} \rangle \in \models_S &\Leftrightarrow \exists \mathfrak{I} (\mathfrak{I} \text{ is a cpi } \mathbb{M} \\
 &\forall \mathbf{A} \in \mathbf{A} (\hat{\iota}(\mathbf{A}) \in \mathbf{Des}_S^+ \mathbb{M} \hat{\iota}(\mathbf{B}) \in \mathbf{Des}_S^-).
 \end{aligned}$$

We will adopt the following convention:

Notation 3.40.

$$A \models_S B \stackrel{\text{DF}}{=} \ulcorner \langle A, B \rangle \in \models_S \urcorner$$

$$A \not\models_S B \stackrel{\text{DF}}{=} \ulcorner \langle A, B \rangle \notin \models_S \urcorner$$

Now we can define an interpretation system for S_0 , which we will call $\mathbf{Sem}_{S_0}$.

Definition 3.41 ($\mathbf{Sem}_{S_0}$).

$\mathbf{Sem}_{S_0}$ is a semantics on S_0

$$\stackrel{\text{DF}}{=} \mathbf{Int}_{S_0} = \{ \mathfrak{I} : \mathfrak{I} \text{ is a cpi on } \mathbf{Fo}_{S_0} \} \quad \mathbb{M}$$

$$\mathbf{LV}_{S_0} = \{ \mathbf{0}, \mathbf{1} \} \quad \mathbb{M}$$

$$\mathbf{Des}_{S_0}^+ = \{ \mathbf{1} \} \quad \mathbb{M}$$

$$\mathbf{Des}_{S_0}^- = \{ \mathbf{0} \} \quad \mathbb{M}$$

$$\mathbf{Val}_{S_0} = \{ \hat{\mathfrak{I}} : \exists \mathfrak{I} \in \mathbf{Int}_{S_0} (\hat{\mathfrak{I}} \text{ is a cpe of } \mathfrak{I}) \} \quad \mathbb{M}$$

$$\models_{S_0} \subseteq \mathbf{Int}_{S_0} \quad \mathbb{M} \quad \models_{S_0} \text{ is classical.}$$

Task 3.12. Show that all inferences of the forms listed on p. 90 are members of $\models_{S_0}$.

As a previous remark, note $\models_{S_0}$ is classical. Hence, by [definition 3.39](#), $A_1, \dots, A_n \models_{S_0} B$ iff there is no cpi $\mathfrak{I}$ such that $\hat{l}(A_1) = \dots = \hat{l}(A_n) = \mathbf{1}$, but $\hat{l}(B) = \mathbf{0}$.

We will show this for two of the cases required by [task 3.12](#).

Theorem 3.42. $(A \rightarrow B), A \models_{S_0} B$ – i.e., $(A \rightarrow B), A / B \in \models_{S_0}$.

Proof. Let $A, B \in \mathbf{Fo}_{S_0}$ and assume, for reductio, that there is a cpi $\langle \{ \mathbf{0}, \mathbf{1} \}, \iota \rangle$ such that $\hat{l}(\ulcorner (A \rightarrow B) \urcorner) = \hat{l}(A) = \mathbf{1}$, but $\hat{l}(B) = \mathbf{0}$. However, by [definition 3.37](#), $\hat{l}(A) = \mathbf{1}$ and $\hat{l}(B) = \mathbf{0}$ entail that $\hat{l}(\ulcorner (A \Rightarrow B) \urcorner) = \mathbf{1}$. As this contradicts our assumption that $\hat{l}(\ulcorner (A \rightarrow B) \urcorner) = \mathbf{1}$, we must conclude that there is no cpi $\mathfrak{I}$ such that $\hat{l}(\ulcorner (A \rightarrow B) \urcorner) = \hat{l}(A) = \mathbf{1}$, but $\hat{l}(B) = \mathbf{0}$.

Moreover, since $\mathbf{1} \in \mathbf{Des}_{S_0}^+$ and $\mathbf{0} \in \mathbf{Des}_{S_0}^-$, this entails, by [definitions 3.39](#) and [3.41](#), that $(A \rightarrow B), A \models_{S_0} B$. $\square$

An alternative proof may be made via a schematic truth table.

Proof. Let $A, B \in \mathbf{Fo}_{S_0}$ and consider the schematic truth table:

A	B	$(A \rightarrow B)$	A	B
0	0	1	0	0
0	1	1	0	1
1	0	0	1	0
1	1	1	1	1

Note that the rows in this table correspond to all possible cpi in relation to A and B. However, there is no cpi $\langle \{0, 1\}, \iota \rangle$ such that where $\hat{\iota}(\ulcorner(A \rightarrow B)\urcorner) = \hat{\iota}(A) = 1$, but $\hat{\iota}(B) = 0$. Hence, as $1 \in \mathbf{Des}_{S_0}^+$ and $0 \in \mathbf{Des}_{S_0}^-$, this entails, by definitions 3.39 and 3.41, that $(A \rightarrow B), A \models_{S_0} B$. $\square$

Theorem 3.43. $\neg(A \vee B) \models_{S_0} (\neg A \wedge \neg B) \wedge (\neg A \wedge \neg B) \models_{S_0} \neg(A \vee B)$
 – i.e., $\neg(A \vee B) \parallel (\neg A \wedge \neg B) \in \mathbf{F}_{S_0}$.

Proof. Let $A, B \in \mathbf{Fo}_{S_0}$ and consider the schematic truth table:

A	B	$\neg(A \vee B)$	$(\neg A \wedge \neg B)$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Note that the rows in this table correspond to all possible cpi in relation to A and B. Hence, we see that all cpi $\langle \{0, 1\}, \iota \rangle$ are such that $\hat{\iota}(\ulcorner\neg(A \vee B)\urcorner) = \hat{\iota}(\ulcorner\neg A \wedge \neg B\urcorner)$. Consequently, since $1 \in \mathbf{Des}_{S_0}^+$ and $0 \in \mathbf{Des}_{S_0}^-$, this entails, by definitions 3.39 and 3.41, that $\neg(A \vee B) \models_{S_0} (\neg A \wedge \neg B)$ and $(\neg A \wedge \neg B) \models_{S_0} \neg(A \vee B)$. $\square$

Task 3.13. Show that the following inferences are not in $\mathbf{F}_{S_0}$:

1. $/ \neg(\neg p \rightarrow p)$
2. $/ \neg(p \rightarrow \neg p)$
3. $(p \rightarrow q) / \neg(p \rightarrow \neg q)$
4. $(p \rightarrow \neg q) / \neg(p \rightarrow q)$

We show just one case.

Theorem 3.44. $(p \rightarrow q) \not\models_{S0} \neg(p \rightarrow \neg q)$. – i.e., $(p \rightarrow q) / \neg(p \rightarrow \neg q) \notin \models_{S0}$.

Proof. Let the cpi $\langle \{0, 1\}, \iota \rangle$ be such that $\iota('p') = 0$. By **definition 3.36**, this entails $\iota('(p \rightarrow q)') = 1$ and $\iota('(p \rightarrow \neg q)') = 1$. The latter implies, also by **definition 3.36**, that $\iota('(p \rightarrow \neg q)') = 0$. Now, since $\iota('(p \rightarrow q)') = 1$ and $\iota('(p \rightarrow \neg q)') = 0$, we can infer by **definition 3.39**, that $(p \rightarrow q) \not\models_{S0} \neg(p \rightarrow \neg q)$. $\square$

Semantics for S1. See chapter 10 of Hannes Leitgeb's script to the course *Logik 1*.

3.8 Derivation System

While an interpretation system provides a semantic account of logical consequence, a *derivation system* or *calculus* provides a *syntactic* account. Instead of assigning meanings to formulae, we specify how formulae may be obtained from others by means of *derivation rules*.

Notion 3.45 (Derivation rule). A **derivation rule** is a rule enabling an expression E to be written provided other expressions $E_1, \dots, E_m$ are written somewhere else.

A set of derivation rules is *deductive base*.

Definition 3.46 (Deductive base). $\mathbf{DBas}_S$ is a **deductive base** $\stackrel{\text{df}}{=} \mathbf{DBas}_S \subseteq \{R : R \text{ is a derivation rule}\}$.

Deductive bases allow the construction of derivations. A **derivation** may be understood as series of steps some of which are assumptions, and others of which can be obtained from previous steps via a derivation rule.

Notion 3.47 (Derivation). A derivation from a deductive base $\mathbf{DBas}_S$ is a series of expressions:

$$E_1, \dots, E_n.$$

Such that each E_i is either an assumption or may be obtained from other expressions $E_{j_1}, \dots, E_{j_m}$, where $j_1, \dots, j_m < n$.

We usually write a derivation as an enumerated list of formulae, and the right of each of which we write within parentheses whether it is an assumption, or whether it follows from other formulae, and its justification.

We will use a special kind of notation for derivations in this style allowing us not only to represent derivations, but also inference rules for a derivation system written in that notation.

Notation 3.48 (Fitch notation). The notation

$$\begin{array}{l|l} a_1 & A_1 \\ & \vdots \\ a_l & A_l \\ \hline b_1 & B_1 \quad R_1, b_{1_1}, \dots, b_{1_{k_1}} \\ & \vdots \\ b_n & B_n \quad R_n, b_{n_1}, \dots, b_{n_{k_n}} \end{array}$$

describes a derivation on $\mathbf{Fo}_S$ from a deductive base $\mathbf{DBas}_S$ iff:

- $A_1, \dots, A_m, B_n \in \mathbf{Fo}_{S0}$,
- $A_1, \dots, A_m$ are the premises and B_n is the conclusion, and
- each B_i is obtained from steps $b_{i_1}, \dots, b_{i_{k_i}}$ by the rule R_i , where $b_{i_1}, \dots, b_{i_{k_i}} < b_i$.

This notation allows us to outline the notion of a Fitch derivation.

Notion 3.49 (Fitch derivation). A derivation is a Fitch derivation on $\mathbf{Fo}_S$ from a deductive base $\mathbf{DBas}_S$ iff it is a derivation of the form specified in [notation 3.48](#) or it is of the form:

a_1	A_1	
	$\vdots$	
a_l	A_l	
b_1	B_1	$R_1, b_{1_1}, \dots, b_{1_{k_1}}$
	$\vdots$	
b_m	B_m	$R_m, b_{m_1}, \dots, b_{m_{k_m}}$
	D	
c_1	C_1	$S_1, c_{1_1}, \dots, c_{1_{k_1}}$
	$\vdots$	
c_n	C_n	$S_n, c_{n_1}, \dots, c_{n_{k_n}}$

where:

- D is a Fitch derivation,
- $\{A_1, \dots, A_l, C_n\} \subseteq \mathbf{Fo}_{S0}$,
- $A_1, \dots, A_m$ are the premises and C_n is the conclusion,
- Each B_i is obtained from steps $b_{i_1}, \dots, b_{i_{k_i}}$ by the rule R_i , where $b_{i_1}, \dots, b_{i_{k_i}} < b_i$,
- Each C_j is obtained from steps $c_{j_1}, \dots, c_{j_{k_j}}$ by the rule S_j , where $c_{j_1}, \dots, c_{j_{k_j}} < c_j$.

Thus, Fitch derivations allow for embedding a derivation within another (see next page).

Thus, a *derivation system* or (*logical*) *calculus* $\mathbf{Cal}_S$ on a logic system S consists of a deductive base $\mathbf{DBas}_S$, a set of derivations $\mathbf{Der}_S$, a function $\mathbf{Prem}_S$ establishing the set of premises of $\mathbf{Der}_S$, a function $\mathbf{Conc}_S$ establishing the set of conclusions of $\mathbf{Der}_S$, and a syntactic relation of logical consequence $\vdash_S$.

Definition 3.50 (Derivation system).

$\mathbf{Cal}_S$ is a calculus on S

$$\stackrel{\text{DF}}{=} \mathbf{Cal}_S = \langle \mathbf{DBas}_S, \mathbf{Der}_S, \mathbf{Prem}_S, \mathbf{Conc}_S, \vdash_S \rangle$$

: $\mathbf{DBas}_S$ is a deductive base on $\mathbf{Fo}_S$ $\mathbb{M}$

$\mathbf{Der}_S = \{D : D \text{ is a derivation on } \mathbf{Fo}_S \text{ from } \mathbf{DBas}_S\}$ $\mathbb{M}$

$\mathbf{Prem}_S : \mathbf{Der}_S \rightarrow \wp \mathbf{Fo}_S$ $\mathbb{M}$

$\mathbf{Conc}_S : \mathbf{Der}_S \rightarrow \wp \mathbf{Fo}_S$ $\mathbb{M}$

$\vdash_S = \{ \langle A, B \rangle : \exists X \in \mathbf{Der}_S (\mathbf{Prem}_S(X) = A \mathbb{M} \mathbf{Conc}_S(X) = B) \}.$

Calculus for S_0 . The deductive base of S_0 may be presented as a set of rules governing the use of logical connectives. Each connective is associated with characteristic ways in which statements containing it may be established or used within an argument.

The rules of the system are traditionally divided into introduction and elimination rules. Introduction rules specify the conditions under which a formula containing a given connective may be inferred. Elimination rules, by contrast, describe how information may be extracted from such formulas once they have been established.

The best way to express those rules is in the style of [notation 3.48](#).

Reiteration.

a		A	
		$\vdots$	
		$\dots$	A Reit, a

Negation.

a		$\neg A$	
		$\vdots$	
		B	
		$\vdots$	
b		$\neg B$	
		$\vdots$	
		A	$\neg^-, a-b$

a		A	
		$\vdots$	
		B	
		$\vdots$	
b		$\neg B$	
		$\vdots$	
		$\neg A$	$\neg^+, a-b$

Conjunction.

a		$(A \wedge B)$	
		$\vdots$	
		A	$\wedge_1^-, a$

a		$(A \wedge B)$	
		$\vdots$	
		B	$\wedge_2^-, a$

a		A	
		$\vdots$	
b		B	
		$\vdots$	
		$(A \wedge B)$	$\wedge^+, a, b$

Disjunction.

a		$(A \vee B)$	
		$\vdots$	
b		A	
		$\vdots$	
c		C	
		$\vdots$	
d		B	
		$\vdots$	
e		C	
		$\vdots$	
		C	$\vee^-, a, b-c, d-e$

a		A	
		$\vdots$	
		$(A \vee B)$	$\vee_1^+, a$

a		B	
		$\vdots$	
		$(A \vee B)$	$\vee_2^+, a$

Conditional.

$$\begin{array}{c|c}
 a & (A \rightarrow B) \\
 & \vdots \\
 b & A \\
 & \vdots \\
 & B \quad \rightarrow^-, a, b
 \end{array}
 \qquad
 \begin{array}{c|c}
 a & \frac{A}{\vdots} \\
 & B \\
 & \vdots \\
 & (A \rightarrow B) \quad \rightarrow^+, a-b
 \end{array}$$

Biconditional.

$$\begin{array}{c|c}
 a & (A \leftrightarrow B) \\
 & \vdots \\
 b & A \\
 & \vdots \\
 & B \quad \leftrightarrow_1^-, a, b
 \end{array}
 \qquad
 \begin{array}{c|c}
 a & \frac{A}{\vdots} \\
 & B \\
 & \vdots \\
 c & \frac{B}{\vdots} \\
 & A \\
 & \vdots \\
 & (A \leftrightarrow B) \quad \leftrightarrow^+, a-b, c-d
 \end{array}$$

Thus, we might define $\mathbf{DBas}_{S_0}$ as follows:

Definition 3.51 (Deductive base for S_0).

$$\mathbf{DBas}_{S_0} \stackrel{\text{DF}}{=} \{\text{Reit}, \neg^+, \neg^-, \wedge^+, \wedge_1^-, \wedge_2^-, \vee_1^+, \vee_2^+, \vee^-, \rightarrow^+, \rightarrow^-, \leftrightarrow^+, \leftrightarrow^-\}.$$

As for the derivation system $\mathbf{Cal}_{S_0}$, we will assume that the functions $\mathbf{Prem}_{S_0}$ and $\mathbf{Conc}_{S_0}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 3.52 ($\mathbf{Cal}_{S_0}$).

$\mathbf{Cal}_{S_0}$ is a calculus on S_0

$$\stackrel{\text{DF}}{=} \mathbf{Cal}_{S_0} = \langle \mathbf{DBas}_{S_0}, \mathbf{Der}_{S_0}, \mathbf{Prem}_{S_0}, \mathbf{Conc}_{S_0}, \vdash_{S_0} \rangle$$

: $\mathbf{DBas}_{S_0}$ is a deductive base on $\mathbf{Fo}_{S_0} \mathrel{\mathbb{M}}$

$$\mathbf{Der}_{S_0} = \{D : D \text{ is a Fitch derivation on } \mathbf{Fo}_{S_0} \text{ from } \mathbf{DBas}_{S_0}\} \mathrel{\mathbb{M}}$$

$$\mathbf{Prem}_{S_0} : \mathbf{Der}_{S_0} \rightarrow \wp \mathbf{Fo}_{S_0} \mathrel{\mathbb{M}}$$

$$\mathbf{Conc}_{S_0} : \mathbf{Der}_{S_0} \rightarrow \wp \mathbf{Fo}_{S_0} \mathrel{\mathbb{M}}$$

$$\vdash_S = \{ \langle A, B \rangle : \exists X \in \mathbf{Der}_{S_0} (\mathbf{Prem}_{S_0}(X) = A \mathrel{\mathbb{M}} \mathbf{Conc}_{S_0}(X) = B) \}.$$

Task 3.14. Show that all inferences of the forms listed on p. 90 are members of $\vdash_{S0}$.

Theorem 3.53. $(A \rightarrow B), \neg B \vdash_{S0} \neg A$ – i.e., $(A \rightarrow B), \neg B / \neg A \in \vdash_{S0}$.

Proof. Let $A, B \in \mathbf{Fo}_{S0}$ and consider the schematic derivation:

1		$(A \rightarrow B)$	
2		$\neg B$	
3			A
4			B $\rightarrow^-, 1, 3$
5			$\neg B$ Reit, 2
6		$\neg A$	$\rightarrow^+, 3-5$

Call this schematic derivation ‘D’. It is clear, for all $A, B \in \mathbf{Fo}_{S0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{S0}$ from $\mathbf{DBas}_{S0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{S0}$ with $\{\ulcorner(A \rightarrow B)\urcorner, \ulcorner\neg B\urcorner\} \subseteq \mathbf{Prem}_{S0}(\mathbf{D})$ and $\ulcorner\neg A\urcorner \in \mathbf{Conc}_{S0}(\mathbf{D})$. Therefore, by definition 3.52, we have that $(A \rightarrow B), \neg B / \neg A \in \vdash_{S0}$. $\square$

Theorem 3.54 (Importation). $((A \wedge B) \rightarrow C) \vdash_{S0} (A \rightarrow (B \rightarrow C))$ – i.e., $((A \wedge B) \rightarrow C) / (A \rightarrow (B \rightarrow C)) \in \vdash_{S0}$.

Proof. Let $A, B, C \in \mathbf{Fo}_{S0}$ and consider the schematic derivation:

1		$((A \wedge B) \rightarrow C)$		
2			A	
3				
3				B
4				$(A \wedge B)$ $\wedge^+, 2, 3$
5				C $\rightarrow^-, 1, 4$
6			$(B \rightarrow C)$ $\rightarrow^+, 3-5$	
7		$(A \rightarrow (B \rightarrow C))$ $\rightarrow^+, 2, 6$		

Call this schematic derivation ‘D’. It is clear, for all $A, B, C \in \mathbf{Fo}_{S0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{S0}$ from $\mathbf{DBas}_{S0}$. This means that there is a derivation in $\mathbf{Der}_{S0}$ with $\{\ulcorner((A \wedge B) \rightarrow C)\urcorner\} \subseteq \mathbf{Prem}_{S0}(\mathbf{D})$ and $\ulcorner(A \rightarrow (B \rightarrow C))\urcorner \in \mathbf{Conc}_{S0}(\mathbf{D})$. Therefore, by definition 3.52, we have that $((A \wedge B) \rightarrow C) / (A \rightarrow (B \rightarrow C)) \in \vdash_{S0}$. $\square$

Note that many of these rules cannot be translated into rules in the sense of **notion 3.45**. However, they can be recaptured by a logical system on S_0 with appropriate definitions of $\mathbf{Der}_{S_0}$, $\mathbf{Prem}_{S_0}$, and $\mathbf{Conc}_{S_0}$.

Calculus for S_1 . See chapter 11 of Hannes Leitgeb's script to the course *Logik 1*.

Part II

LOGICAL SYSTEMS

Aristotle's **canonical syllogistic** is probably the most influential logical system of all times – modern classical logic still has a long wait before getting there.

I will formalise this system representing all inferences admitted by the 192 moods admitted by the 3 Aristotelian figures, among which only 14 are the canonical syllogisms admitted by Aristotle.

4.1 Vocabulary of A.

The vocabulary of the canonical syllogistic can be formalised with: four logical symbols, quantifiers 'A', 'E', 'I', 'O'; plus three non-logical ones, the terms 'a', 'b', 'c'. We will not need auxiliary symbols.

Task 4.1. Define (a) ' Log_A ', (b) ' Non_A ', and (c) ' Vo_A '.

Definition 4.1 (Vocabulary of A (answer to task 4.1)).

$$\begin{aligned}\text{Non}_A &\stackrel{\text{DF}}{=} \{ 'A', 'E', 'I', 'O' \}, \\ \text{Log}_A &\stackrel{\text{DF}}{=} \text{A}bc, \\ \text{Vo}_A &\stackrel{\text{DF}}{=} \langle \text{Log}_A, \text{Non}_A \rangle.\end{aligned}$$

Note that, since $\text{A}bc = \{ 'a', 'b', 'c' \}$, **definition 4.1** entails that $\text{Log}_A = \{ 'a', 'b', 'c' \}$.

4.2 Formulae of A.

Aristotelian syllogistic deals with *categorical propositions*, which are basic forms of propositions that express a relation between two terms. They are composed of a *subject term*, a *copula*, and a *predicate term*.

Where 'x' stands for the predicate term and 'y' for the subject term, the basic form of Aristotelian categorical sentences is:

x is belongs to y.

For his logical treatment of these expressions, Aristotle considers four kinds of categorical propositions: universal affirmative or *A-type*, universal negative or *E-type*, particular affirmative or *I-type*, and particular negative or *O-type*.

In a formal language, categorical proposition are represented by combining one logical symbol, indicating the kind of proposition,

Table 4.1: Categorical formulae and their natural-language interpretations.

	Affirmative	Negative
Universal	x belongs to all y. Axy	x belongs to no y. Exy
Particular	x belongs to some y. Ixy	x does not belong to all y. Oxy

and two different non-logical symbols, indicating the terms. Thus, we obtain formulae like 'Aab' or 'Ocb'. Table 4.1 summarises how we represent such sentences..

However, there is no categorical formula such as 'Abb', since the system does not permit the same non-logical symbol to occupy both term positions in a formula.

Task 4.2. Define $\mathbf{Fo}_A$ using $\mathbf{Vo}_A$.

Hint: Make sure no formula such as 'Abb' is included.

Definition 4.2 (Formulae of A (answer to task 4.2)).

$$\mathbf{Fo}_A \stackrel{\text{DF}}{=} \{\bullet xy : \bullet \in \mathbf{Log}_A \wedge x, y \in \mathbf{Non}_A \wedge x \neq y\}.$$

Note that, by definitions , the following holds:

$$\begin{aligned} \mathbf{Vo}_A &= \langle \mathbf{Log}_A, \mathbf{Non}_A \rangle, \\ \mathbf{Log}_A &= \{ 'A', 'E', 'I', 'O' \}, \\ \mathbf{Non}_A &= \mathbf{A}bc = \{ 'a', 'b', 'c' \}. \end{aligned}$$

Hence, the definition above entails that:

$$\mathbf{Fo}_A = \{\bullet xy : \bullet \in \{ 'A', 'E', 'I', 'O' \} \wedge x, y \in \{ 'a', 'b', 'c' \} \wedge x \neq y\}.$$

It is easy to see that $\mathbf{Fo}_A$ excludes all formulae of the form Qtt , that is, such that it has a repeated term.

Corollary 4.3. For all $\bullet \in \mathbf{Log}_A$ and $t \in \mathbf{Non}_A$, $\bullet tt \notin \mathbf{Fo}_A$.

Proof. Assume, $Qtt \in \mathbf{Fo}_A$. However, by definition 4.2, it would follow that $t \neq t$, which is absurd. Thus, $Qtt \notin \mathbf{Fo}_A$. $\square$

It is also easy to show that $\mathbf{Fo}_A$ is finite.

Corollary 4.4. $|\mathbf{Fo}_A| = 24$.

Proof. We count each instance of the schema which defines $\mathbf{Fo}_A$, which are the following:

'Aab', 'Aac', 'Aba', 'Abc', 'Aca', 'Acb',
 'Eab', 'Eac', 'Eba', 'Ebc', 'Eca', 'Ecb',
 'Iab', 'Iac', 'Iba', 'Ibc', 'Ica', 'Icb',
 'Oab', 'Oac', 'Oba', 'Obc', 'Oca', 'Ocb'.

These are 24 in total. Hence, $|\mathbf{Fo}_A| = 24$. $\square$

Syllogistic formal languages are very special, since they are the only ones which may be formulated as finite sets of formulae, and with explicit definitions rather than with recursive ones. Neither of these situations will hold for the formal languages of modern logical systems, including the following two.

4.3 Inference relation of A.

Categorical formulae such as 'Aab', 'Eab', and 'Obc' are all well-formed on their own. However, they cannot be combined to form inferences in A (whether valid or not).

Inferences in A must be constructed according to the three *Aristotelian figures**, all of which require exactly two premises and one conclusion. Moreover, the individual term shared by the premises, the *middle term*, should not occur in the conclusion.

*: There is a fourth figure, but which was not explicitly formulated by Aristotle.

Thus, an inference in A has the form:

$$\langle \langle A, B \rangle, \{C\} \rangle,$$

where A is the major premise, B is the minor premise, and C is the conclusion. A, B, C are all categorical formulae of $\mathbf{Fo}_A$.

Thus, where $\mathbf{0}_1, \mathbf{0}_2, \mathbf{0}_3 \in \mathbf{Log}_A$, the three Aristotelian figures may be explained as follows:

- In the *first figure*, the middle term m occurs as the predicate of the major premise and as the subject of the minor premise:

$$\mathbf{0}_1xm, \mathbf{0}_2my / \mathbf{0}_3xy.$$

- In the **second figure**, the middle term m occurs as the predicate in both premises:

$$\mathbf{0}_1mx, \mathbf{0}_2my / \mathbf{0}_3xy.$$

- Finally, in the **third figure**, the middle term m occurs as the subject in both premises:

$$\mathbf{0}_1xm, \mathbf{0}_2ym / \mathbf{0}_3xy.$$

In order to formalise, we will construct inferences so that 'c' always stands for the middle term, 'a' for the subject term of the conclusion and 'b' for the predicate term of the conclusion.

Task 4.3. Define 'In_A'.

Definition 4.5 (Inference in A (answer to task 4.3)).

$$\begin{aligned} \text{In}_A \stackrel{\text{DF}}{=} & \{ \mathbf{0}_1 ac, \mathbf{0}_2 cb / \mathbf{0}_3 ab : \mathbf{0}_1, \mathbf{0}_2, \mathbf{0}_3 \in \text{Log}_A \} \cup \\ & \{ \mathbf{0}_1 ca, \mathbf{0}_2 cb / \mathbf{0}_3 ab : \mathbf{0}_1, \mathbf{0}_2, \mathbf{0}_3 \in \text{Log}_A \} \cup \\ & \{ \mathbf{0}_1 ac, \mathbf{0}_2 bc / \mathbf{0}_3 ab : \mathbf{0}_1, \mathbf{0}_2, \mathbf{0}_3 \in \text{Log}_A \}. \end{aligned}$$

As with **Fo_A** the cardinality of **In_A** is finite and can be easily established.

Corollary 4.6. $|\text{In}_A| = 192$.

Proof. Easily (though tediously) shown by counting the instances. Alternatively, to each figure corresponds $4 \times 4 \times 4 = 64$ inferences. Since there three such figures, we have $64 \times 3 = 192$. $\square$

4.4 Consequence relation of A.

Out of the 192 inferential combinations included in **In_A**, Aristotle accepts exactly 14 as syllogisms – in our terms, as A-valid inferences.

It must be noted that the sense of a categorical expression is similar to the sense in which it would be interpreted by modern standard logic. The main important difference is that, in expressions of the forms $\lceil \text{Axy} \rceil$ (x belongs to all y) and $\lceil \text{Ixy} \rceil$ (i.e., x belongs to no y), we assume that there exists at least one y.

Definition 4.7 (Consequence in A).

$$\begin{aligned} \text{Con}_A \stackrel{\text{DF}}{=} & \{ \langle \langle \text{Aac}, \text{Ac b} \rangle, \{ \text{Aab} \} \rangle, \langle \langle \text{Eac}, \text{Ac b} \rangle, \{ \text{Eab} \} \rangle, \\ & \langle \langle \text{Aac}, \text{Ic b} \rangle, \{ \text{Iab} \} \rangle, \langle \langle \text{Eac}, \text{Ic b} \rangle, \{ \text{Oab} \} \rangle, \\ & \langle \langle \text{Eca}, \text{Ac b} \rangle, \{ \text{Eab} \} \rangle, \langle \langle \text{Aca}, \text{Ec b} \rangle, \{ \text{Eab} \} \rangle, \\ & \langle \langle \text{Eca}, \text{Ic b} \rangle, \{ \text{Oab} \} \rangle, \langle \langle \text{Aca}, \text{Oc b} \rangle, \{ \text{Oab} \} \rangle, \\ & \langle \langle \text{Aac}, \text{Abc} \rangle, \{ \text{Iab} \} \rangle, \langle \langle \text{Eac}, \text{Abc} \rangle, \{ \text{Oab} \} \rangle, \\ & \langle \langle \text{Iac}, \text{Abc} \rangle, \{ \text{Iab} \} \rangle, \langle \langle \text{Aac}, \text{Ibc} \rangle, \{ \text{Iab} \} \rangle, \\ & \langle \langle \text{Oac}, \text{Abc} \rangle, \{ \text{Oab} \} \rangle, \langle \langle \text{Eac}, \text{Ibc} \rangle, \{ \text{Oab} \} \rangle \}. \end{aligned}$$

4.5 Interpretation and derivation systems for A

Since we have a complete definition of $\mathbf{Con}_A$, it may seem useless to construct an interpretation or a derivation system for A.

If the aim is just to formalise A, it may be useless. However, if we want to extend A into a system admitting inferences of more than two premises. Although we do not know that Aristotle had develop systematically a syllogistic more than two premises, this development is consistent with his interests. In fact, he makes some syllogistic chains in his works, articulating various syllogisms to produce a conclusion from more than to premises.

We will not develop such semantics and calculus, though.

Łukasiewicz's and Kleene's 3-Valued Logics

5

5.1 Philosophical Motivations

Standard logic is built on a simple and powerful idea: every proposition is either true or false. Thus, a logical system is constructed in which every formula can be interpreted as being either true or false.

Yet many situations motivate deviating from this duality. What should we say about a statement whose truth is unknown? What about propositions concerning future contingents, incomplete information, or computations that fail to terminate?

Questions like these motivated the development of three-valued logics, systems in which a third logical value supplements the classical pair of truth and falsity.

In the early twentieth century, Jan Łukasiewicz developed one of the first systematic three-valued logics. Łukasiewicz was motivated partly by philosophical problems concerning future contingent statements, inspired by Aristotle's discussion of the issue.

In *De Interpretatione* (Book 9), Aristotle considers propositions about future contingent events, such as that there will be a sea battle tomorrow. If every proposition must already be either true or false, then it seems that either the sea battle must occur with necessity or it must fail to occur with necessity. This appears to lead to a form of fatalism: the future would already be fixed independently of human action or chance.

Aristotle and Łukasiewicz, therefore, questioned whether propositions about contingent future events should be assigned a definite truth value before the events themselves occur. Łukasiewicz regarded this problem as evidence that classical two-valued logic is not always adequate for reasoning about the future.

Moreover, he proposed introducing a third truth value, often interpreted as *possibility* or *indeterminacy*. Under this interpretation, the statement that there will be a sea battle tomorrow is neither true nor false today, but would be assigned this third value.

A different motivation emerged in the work of Stephen Cole Kleene. In connection with partial recursive functions and computations that may fail to terminate, Kleene introduced influential three-valued semantics in which the third value represents *undefinedness* (Kleene 1971, §64).

Kleene proposed two different understanding for the logical operators in this three-valued framework. We know these systems as *weak Kleene* WK and *strong Kleene* SK.

5.2 Vocabularies and Languages

The vocabularies and formal languages $\mathbf{Fo}_{WK,SK}$, and $\mathbf{Fo}_{L3}$ will be the same as those of the standard system.

Definition 5.1.

$$\begin{array}{lll} \mathbf{Vo}_{SK} \stackrel{\text{DF}}{=} \mathbf{Vo}_{S0} & \mathbf{Vo}_{WK} \stackrel{\text{DF}}{=} \mathbf{Vo}_{S0} & \mathbf{Vo}_{L3} \stackrel{\text{DF}}{=} \mathbf{Vo}_{S0} \\ \mathbf{Fo}_{SK} \stackrel{\text{DF}}{=} \mathbf{Fo}_{S0} & \mathbf{Fo}_{WK} \stackrel{\text{DF}}{=} \mathbf{Fo}_{S0} & \mathbf{Fo}_{L3} \stackrel{\text{DF}}{=} \mathbf{Fo}_{S0} \end{array}$$

An alternative way would be to define a language including additional logical operators which behave differently to the classical ones. However, we will here take the usual approach of giving an alternative account of the usual logical operators.

5.3 Inference and Consequence Relations

WK, SK, and L3 fall among modern systems, so their inference relations are just cases of [definition 3.31](#).

As for the contents of their consequence relations, we want the law of excluded third to be not valid in general. In other words, we do not want that all inferences of the following form be members of their consequence relations:

1. $/ (A \vee \neg A)$

5.4 Interpretation Systems

In both WK and SK, the non-logical symbols are propositional variables. A three-valued propositional interpretation, or 3pi for short, assigns to each propositional variable a logical truth value among **1** (truth), $\frac{1}{2}$ (the third value), **0** (falsity).

Definition 5.2 (Three-valued propositional interpretation).

$$\langle \mathbf{D}, \iota \rangle \text{ is a 3pi on } \mathbf{FoS} \\ \stackrel{\text{DF}}{=} \mathbf{D} = \{\mathbf{0}, \frac{1}{2}, \mathbf{1}\} \models \iota : \mathbf{ProS} \rightarrow \{\mathbf{0}, \frac{1}{2}, \mathbf{1}\}.$$

In other words, $\langle \mathbf{D}, \iota \rangle$ is a 3pi on $\mathbf{FoS}$ iff its domain is the set of logical values $\{\mathbf{0}, \frac{1}{2}, \mathbf{1}\}$ and its interpretation map ι is a function from the set of propositional variables to $\{\mathbf{0}, \frac{1}{2}, \mathbf{1}\}$ – that is, each propositional variable is assigned either just **0** or just $\frac{1}{2}$ or just **1**.

A 3pi $\mathfrak{I}$ is extended to all formulae of $\mathbf{FoS}$ by means of a valuation.

We have two kinds of extensions of a 3pi, related to the two systems we are studying: weak 3-valued extensions (w3e) for WK, strong 3-valued extensions (s3e) for SK, and Łukasiewicz 3-valued extensions (Ł3e) for L3.

In the following, note that the notion of semantic consequence we need for all these logics is standard, thus, it satisfies [definition 3.39](#). Moreover, the designated value of all these logics will be **1**, while the others will be anti-designated.

Weak Kleene WK. A weak 3-valued extensions (w3e) on a 3pi is defined as follows.

Definition 5.3 (Weak 3-valued propositional valuation). A valuation $\hat{\mathfrak{I}}$ is a w3e of the 3pi $\mathfrak{I}$ on $\mathbf{FoS} \stackrel{\text{DF}}{=} \hat{\iota}(\mathbf{A}) = \iota(\mathbf{A}) \models$

F	$\neg F$	$\wedge$	0	$\frac{1}{2}$	1	$\vee$	0	$\frac{1}{2}$	1
0	1	0	0	$\frac{1}{2}$	0	0	0	$\frac{1}{2}$	1
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1	0	1	0	$\frac{1}{2}$	1	1	1	$\frac{1}{2}$	1

$\rightarrow$	0	$\frac{1}{2}$	1	$\leftrightarrow$	0	$\frac{1}{2}$	1
0	1	$\frac{1}{2}$	1	0	1	$\frac{1}{2}$	0
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1	0	$\frac{1}{2}$	1	1	0	$\frac{1}{2}$	1

: $\mathbf{A} \in \mathbf{ProS} \models \mathbf{F} \in \mathbf{FoS}$.

Note that weak extensions are obtained by filling the additional cells in the resulting truth tables with ' $\frac{1}{2}$ ', which stands for the third value.

$\mathbf{Sem}_{\mathbf{WK}}$ is defined based on [definition 5.3](#) as follows:

Definition 5.4 ($\mathbf{Sem}_{\mathbf{WK}}$).

$\mathbf{Sem}_{\mathbf{WK}}$ is a semantics on $\mathbf{Fo}_{\mathbf{WK}}$

$$\begin{aligned} &\stackrel{\text{DF}}{=} \mathbf{Int}_{\mathbf{WK}} = \{\mathfrak{I} : \mathfrak{I} \text{ is a 3pi on } \mathbf{Fo}_{\mathbf{WK}}\} && \mathbb{M} \\ &\mathbf{LV}_{\mathbf{WK}} = \{0, \frac{1}{2}, 1\} && \mathbb{M} \\ &\mathbf{Des}_{\mathbf{WK}}^+ = \{1\} && \mathbb{M} \\ &\mathbf{Des}_{\mathbf{WK}}^- = \{\frac{1}{2}, 0\} && \mathbb{M} \\ &\mathbf{Val}_{\mathbf{WK}} = \{\hat{\mathfrak{I}} : \exists \mathfrak{I} \in \mathbf{Int}_{\mathbf{WK}} (\hat{\mathfrak{I}} \text{ is a w3e of } \mathfrak{I})\} && \mathbb{M} \\ &\models_{\mathbf{WK}} \subseteq \mathbf{In}_{\mathbf{WK}} \text{ } \mathbb{M} \models_{\mathbf{WK}} \text{ is classical.} \end{aligned}$$

Strong Kleene SK. A strong 3-valued extensions (s3e) on a 3pi is defined as follows.

Definition 5.5 (Strong 3-valued propositional valuation). A valuation $\hat{\mathfrak{I}}$ is a s3e of the 3pi $\mathfrak{I}$ on $\mathbf{Fo}_{\mathbf{S}} \stackrel{\text{DF}}{=} \hat{\iota}(\mathbf{A}) = \iota(\mathbf{A}) \mathbb{M}$

F	$\neg F$	$\wedge$	0	$\frac{1}{2}$	1	$\vee$	0	$\frac{1}{2}$	1
0	1	0	0	0	0	0	0	$\frac{1}{2}$	1
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
1	0	1	0	$\frac{1}{2}$	1	1	1	1	1

$\rightarrow$	0	$\frac{1}{2}$	1	$\leftrightarrow$	0	$\frac{1}{2}$	1
0	1	1	1	0	1	$\frac{1}{2}$	0
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1	0	$\frac{1}{2}$	1	1	0	$\frac{1}{2}$	1

: $\mathbf{A} \in \mathbf{Pro}_{\mathbf{S}} \mathbb{M} \mathbf{F} \in \mathbf{Fo}_{\mathbf{S}}$.

Strong extensions differ from weak extensions with regard to the value assigned to formulae with ' $\wedge$ ', ' $\vee$ ', ' $\rightarrow$ '. The truth-tables of such formulae are obtained by filling the additional cells in the resulting truth tables according to the criteria:

- There is an ordering $0 < \frac{1}{2} < 1$.
- A disjunction preserves the higher logical value of its disjuncts.
- A conjunction preserves the lower logical value of its conjuncts.
- A conditional is assigned 1 when the antecedent is assigned 0 or its consequent is assigned 0; it is assigned 0 when the antecedent is assigned 1 and the consequent is assigned 0; otherwise, it is assigned $\frac{1}{2}$.

Sem_{SK} is defined based on **definition 5.5** as follows:

Definition 5.6 (Sem_{SK}).

Sem_{SK} is a semantics on Fo_{SK}

$$\stackrel{\text{DF}}{=} \text{Int}_{\text{SK}} = \{\mathfrak{I} : \mathfrak{I} \text{ is a 3pi on } \text{Fo}_{\text{SK}}\} \quad \mathbb{M}$$

$$\text{LV}_{\text{SK}} = \{0, \frac{1}{2}, 1\} \quad \mathbb{M}$$

$$\text{Des}_{\text{SK}}^+ = \{1\} \quad \mathbb{M}$$

$$\text{Des}_{\text{SK}}^- = \{\frac{1}{2}, 0\} \quad \mathbb{M}$$

$$\text{Val}_{\text{SK}} = \{\hat{\mathfrak{I}} : \exists \mathfrak{I} \in \text{Int}_{\text{SK}} (\hat{\mathfrak{I}} \text{ is a s3e of } \mathfrak{I})\} \quad \mathbb{M}$$

$$\models_{\text{SK}} \subseteq \text{In}_{\text{SK}} \quad \mathbb{M} \quad \models_{\text{SK}} \text{ is classical.}$$

Łukasiewicz 3-valued logic L3. A Łukasiewicz 3-valued extensions (Ł3e) on a 3pi is defined as follows.

Definition 5.7 (Łukasiewicz 3-valued propositional valuation).

A valuation $\hat{\mathfrak{I}}$ is a **Ł3e** of the 3pi $\mathfrak{I}$ on $\text{Fo}_{\text{S}} \stackrel{\text{DF}}{=} \hat{\iota}(\mathbf{A}) = \iota(\mathbf{A}) \quad \mathbb{M}$

F	$\neg F$	$\wedge$	0	$\frac{1}{2}$	1	$\vee$	0	$\frac{1}{2}$	1
0	1	0	0	0	0	0	0	$\frac{1}{2}$	1
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
1	0	1	0	$\frac{1}{2}$	1	1	1	1	1

$\rightarrow$	0	$\frac{1}{2}$	1	$\leftrightarrow$	0	$\frac{1}{2}$	1
0	1	1	1	0	1	$\frac{1}{2}$	0
$\frac{1}{2}$	$\frac{1}{2}$	1	1	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$
1	0	$\frac{1}{2}$	1	1	0	$\frac{1}{2}$	1

: $\mathbf{A} \in \text{Pro}_{\text{S}} \quad \mathbb{M} \quad \mathbf{F} \in \text{Fo}_{\text{S}}.$

Łukasiewicz extensions differ from weak extensions with regard to the value assigned to formulae with ' $\rightarrow$ ', ' $\leftrightarrow$ '. In these truth tables, we replace ' $\frac{1}{2}$ ' in the cell $(\frac{1}{2}, \frac{1}{2})$ with '**1**'.*

The logic behind this decision is, again, to assume an ordering $0 < \frac{1}{2} < 1$. Hence, we assign the conditional **1** in an interpretation iff the antecedent and the consequent are assigned values $\mathbf{v}$ and $\mathbf{w}$, respectively, such that $\mathbf{v} \leq \mathbf{w}$.

*: Another version of Łukasiewicz's system includes additional operators to represent modal notions such as possibility, necessity, and contingency, with semantics which are very alien to the treatment of these notions in modern modal logics.

$\mathbf{Sem}_{L3}$ is defined based on [definition 5.7](#) as follows:

Definition 5.8 ($\mathbf{Sem}_{L3}$).

$\mathbf{Sem}_{L3}$ is a semantics on $\mathbf{Fo}_{L3}$

$$\stackrel{\text{DF}}{=} \mathbf{Int}_{L3} = \{ \mathfrak{I} : \mathfrak{I} \text{ is a 3pi on } \mathbf{Fo}_{L3} \} \quad \mathbb{M}$$

$$\mathbf{LV}_{L3} = \{ \mathbf{0}, \frac{1}{2}, \mathbf{1} \} \quad \mathbb{M}$$

$$\mathbf{Des}_{L3}^+ = \{ \mathbf{1} \} \quad \mathbb{M}$$

$$\mathbf{Des}_{L3}^- = \{ \frac{1}{2}, \mathbf{0} \} \quad \mathbb{M}$$

$$\mathbf{Val}_{L3} = \{ \hat{\mathfrak{I}} : \exists \mathfrak{I} \in \mathbf{Int}_{L3} (\hat{\mathfrak{I}} \text{ is a L3e of } \mathfrak{I}) \} \quad \mathbb{M}$$

$$\models_{L3} \subseteq \mathbf{In}_{L3} \quad \mathbb{M} \models_{L3} \text{ is classical.}$$

Results As desired, ' $(p \vee \neg p)$ ' is not a member of any of these semantic consequence relations.

Theorem 5.9. $\not\models_{WK} (p \vee \neg p) \quad \not\models_{SK} (p \vee \neg p) \quad \not\models_{L3} (p \vee \neg p).$

Proof. Consider the following truth table:

p	$(p \vee \neg p)$
$\mathbf{0}$	$\mathbf{1} \quad \mathbf{1}$
$\frac{1}{2}$	$\frac{1}{2} \quad \frac{1}{2}$
$\mathbf{1}$	$\mathbf{1} \quad \mathbf{0}$

Note that the second row of this table corresponds to a 3pi $\langle \{ \mathbf{0}, \frac{1}{2}, \mathbf{1} \}, \iota \rangle$ with a 3pe $\hat{\mathfrak{I}}$ such that $\hat{\iota}((p \vee \neg p)) = \frac{1}{2}$. But since $\frac{1}{2}$ is in $\mathbf{Des}_{WK}^-$, $\mathbf{Des}_{SK}^-$, and $\mathbf{Des}_{L3}^-$, we have that $\not\models_{WK} (p \vee \neg p)$, $\not\models_{SK} (p \vee \neg p)$, and $\not\models_{L3} (p \vee \neg p)$. $\square$

Task 5.1. Show whether the following inferences are members of $\models_{WK}$, $\models_{SK}$, and $\models_{L3}$.

1. $q / (p \rightarrow q)$
2. $\neg p / (p \rightarrow q)$
3. $((p \wedge q) \rightarrow r) / ((p \rightarrow r) \vee (q \rightarrow r))$
4. $((p \rightarrow q) \wedge (r \rightarrow s)) / ((p \rightarrow s) \vee (r \rightarrow q))$
5. $\neg(p \rightarrow q) / p$
6. $(p \rightarrow r) / ((p \wedge q) \rightarrow r)$
7. $(p \rightarrow q), (q \rightarrow r) / (p \rightarrow r)$
8. $(p \rightarrow q) / (\neg p \rightarrow \neg q)$
9. $p / (q \vee \neg q)$
10. $/ (p \rightarrow (q \vee \neg q))$
11. $p, \neg p / q$
12. $(p \wedge \neg p) / q$
13. $/ ((p \wedge \neg p) \rightarrow q)$
14. $(p \rightarrow q), p / q$
15. $(p \vee q), \neg p / q$
16. $/ (p \vee \neg p)$
17. $/ \neg\neg(p \vee \neg p)$
18. $\neg\neg p / p$
19. $p / \neg\neg p$
20. $\neg\neg\neg p / \neg p$

5.5 Derivation Systems

It is perfectly possible to provide derivation systems for three-valued logics. We will not do it here, as their semantics provide the essence of these logics.

6.1 Philosophical Motivations

Intuitionistic logic emerged in the early twentieth century as an alternative to classical logic. Its development was closely tied to foundational debates in mathematics, especially concerning the nature of mathematical truth, existence, and proof. Unlike classical logic, but like most three-valued logics, intuitionistic logic rejects the unrestricted validity of the law of excluded middle.

The philosophical roots of intuitionism lie primarily in the work of the Dutch mathematician and philosopher Luitzen Egbertus Jan Brouwer, whose programme sought to reconstruct mathematics on a purely constructive basis. Later, Arend Heyting (among others) formalised and extended these ideas into a system of logic usually regarded as intuitionistic logic.

Brouwer regarded mathematics not as a discovery of objective truths existing independently of human cognition, but as an activity of mental construction. According to him, mathematics originates in the basic intuition of time: the awareness of the passage from one moment to another. From this primordial intuition, the natural numbers and all mathematical constructions arise.

For Brouwer, mathematical objects do not exist independently of our capacity to construct them. Existence in mathematics therefore means constructibility. To claim that an object exists is to possess a method for explicitly constructing it. Moreover, to assert something, means that we have the means to prove it.

The most famous consequence of Brouwer's philosophy is the rejection of the law of excluded middle: $\neg(A \vee \neg A)$. Classical logic treats this principle as universally valid. Brouwer argued, however, that such a principle is justified only when one possesses a method either to prove A or to refute it. In infinite domains, many propositions may remain undecided.

Now the *principium tertii exclusi*: this demands that every supposition is either correct or incorrect, mathematically: that of every supposed fitting in a certain way of systems in one another, either the termination or the blockage by impossibility, can be constructed. The question of the validity of the *principium tertii exclusi* is thus equivalent to the question concerning the possibility of unsolvable mathematical problems. For the already proclaimed conviction that unsolvable mathematical problems do not exist, no indication of a demonstration is present.

As long as only certain finite discrete systems are posited, the investigation into the possibility or impossibility of a fitting can always be terminated and leads to an answer whence the *principium tertii exclusi* is a reliable principle of reasoning. (Brouwer 2017, p. 42)

Brouwer therefore distinguished between:

- ▶ proving that a proposition is true,
- ▶ proving that it is false,
- ▶ and merely lacking either proof.

A proposition can be regarded as true only if there exists a construction proving it. As examples of statements about mathematics which Brouwer regarded as not certain to have a, he included (cf. Brouwer 2017, p. 44):

1. In the decimal expansion of π , there is ‘a digit that occurs enduringly more often than all others’.
2. In the decimal expansion of π , ‘there occur ... infinitely many pairs of equal consecutive digits’.

In sum, while in classical logic, truth is generally understood semantically: a statement is true if it corresponds to reality or satisfies a model; intuitionism instead identifies truth with provability aka demonstrability.

In this section we will present three systems related to intuitionistic logic: Hilbert’s positive propositional logic P0, Ingebrigt Johansson’s minimal propositional logic M0, and Heyting’s intuitionistic propositional logic I0. The first of these is a system of logic without negation, on the basis of which many other logics can be built. The second adds negation, but

In this section we will present three systems related to intuitionistic logic: Hilbert’s positive propositional logic P0, Ingebrigt Johansson’s minimal propositional logic M0, and Heyting’s intuitionistic propositional logic I0.

The first of these is a system of logic without negation, on the basis of which many other logic systems can be constructed. The second adds negation, but without validating neither the *ex contradictione quodlibet* nor the *tertium non datur*. Finally, Heyting’s intuitionistic logic extends minimal logic by incorporating this principle, thereby yielding the standard propositional system associated with Brouwer’s intuitionistic conception of mathematics.

6.2 Vocabularies and Languages

The vocabularies and formal languages $\mathbf{Fo}_{M0}$ and $\mathbf{Fo}_{I0}$ will be defined as those of $S0$.

Definition 6.1.

$$\begin{array}{ll} \mathbf{Vo}_{M0} \stackrel{\text{DF}}{=} \mathbf{Vo}_{S0} & \mathbf{Vo}_{I0} \stackrel{\text{DF}}{=} \mathbf{Vo}_{S0} \\ \mathbf{Fo}_{M0} \stackrel{\text{DF}}{=} \mathbf{Fo}_{S0} & \mathbf{Fo}_{I0} \stackrel{\text{DF}}{=} \mathbf{Fo}_{S0} \end{array}$$

The vocabulary and language of $P0$, however, must be different, as there is no negation in this system.

Task 6.1. Define (a) ' $\mathbf{Vo}_{P0}$ ' and (b) ' $\mathbf{Fo}_{P0}$ '.

Definition 6.2 (Vocabulary of $P0$ (answer to task 6.1.a)).

$$\begin{array}{l} \mathbf{Log}_{P0} \stackrel{\text{DF}}{=} \mathbf{Con} - \{ '\neg' \}, \\ \mathbf{Non}_{P0} \stackrel{\text{DF}}{=} \mathbf{Non}_{S0}, \\ \mathbf{Aux}_{P0} \stackrel{\text{DF}}{=} \mathbf{Par}, \\ \mathbf{Vo}_{P0} \stackrel{\text{DF}}{=} \langle \mathbf{Log}_{P0}, \mathbf{Non}_{P0}, \mathbf{Aux}_{P0} \rangle. \end{array}$$

Definition 6.3 (Formulae of $P0$ (solution to task 6.1.b)).

$$\mathbf{Fo}_{P0} \stackrel{\text{DF}}{=} \mathbf{Non}_{P0} \cup \{ \ulcorner (p \heartsuit q) \urcorner : p, q \in \mathbf{Fo}_{P0} \wedge \heartsuit \in \mathbf{Log}_{P0} \}.$$

6.3 Inference and Consequence Relations

$P0, M0$, and $I0$ fall among modern systems, so their inference relations are just special cases of [definition 3.31](#).

As for the contents of their consequence relations, we want the law of excluded third to be not valid in general, as with the three-valued logics we have seen before. In other words, we do not want that all inferences of the following form be members of their consequence relations:

$$1. \ / (A \vee \neg A)$$

This will be necessarily the case for $P0$, since it has no negation symbol. For the other systems, we will need to construct their consequence relation so that this does law does not hold in general.

However, we do want, following Brouwer, that:

$$2. \ / \neg(A \wedge \neg A)$$

$$3. (A \rightarrow B), (B \rightarrow C) / (A \rightarrow C)$$

Moreover, contrary to what happens in three-valued logics, we want that:

$$4. \quad / \neg\neg(A \vee \neg A)$$

This law makes sense in the context of intuitionistic logic, as it may be read as stating that there is no construction that turns a proof of $\ulcorner(A \vee \neg A)\urcorner$ into a contradiction, or it is impossible to refute $\ulcorner(A \vee \neg A)\urcorner$.

This is not necessarily against the principles of intuitionistic logic. Thus, intuitionism asserts that no proof of $\ulcorner(A \vee \neg A)\urcorner$ exists for some cases: those cases in which no proof of neither A nor $\ulcorner\neg A\urcorner$ is given. However, it does not assert that $\ulcorner\neg(A \vee \neg A)\urcorner$. Moreover, it states that

Something else we want is all these to be *proper sublogics* of $S0$. In other words, we want that:

$$\mathbf{Con}_{P0}, \mathbf{Con}_{M0}, \mathbf{Con}_{I0} \subset \mathbf{Con}_{S0}.$$

Moreover, we want that:

$$\mathbf{Con}_{P0} \subset \mathbf{Con}_{M0} \subset \mathbf{Con}_{I0} \subset \mathbf{Con}_{S0}.$$

We can show this if we show that all inferences valid in any of $\mathbf{Con}_{P0}, \mathbf{Con}_{M0}, \mathbf{Con}_{I0}$ are also valid in $\mathbf{Con}_{S0}$, and that, for each $\mathbf{Con}_{P0}, \mathbf{Con}_{M0}, \mathbf{Con}_{I0}$, there is at least one inference that is $\mathbf{Con}_{S0}$ but not in $\mathbf{Con}_{P0}, \mathbf{Con}_{M0}, \mathbf{Con}_{I0}$, respectively.

6.4 Interpretation Systems

There are many interpretation systems for intuitionistic logics. Many of these require a possible worlds semantics, for which we would need at least to lessons to introduce. So we will focus on the derivation systems.

6.5 Derivation Systems

There are many derivation systems proposed for intuitionistic logics. It is a good idea to start by showing the axiomatic derivation systems which have been proposed for these systems, which are as follows.

Consider the following schematic inference forms:

$(A \rightarrow B), A$	$/ B$	(MPP)
$/ (A \rightarrow (B \rightarrow A))$		(P1)
$/ ((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$		(P2)
$/ (A \rightarrow (B \rightarrow (A \wedge B)))$		(P3)
$/ ((A \wedge B) \rightarrow A)$		(P4)
$/ ((A \wedge B) \rightarrow B)$		(P5)
$/ (A \rightarrow (A \vee B))$		(P6)
$/ (B \rightarrow (A \vee B))$		(P7)
$/ ((A \rightarrow C) \rightarrow ((B \rightarrow C) \rightarrow ((A \vee B) \rightarrow C)))$		(P8)
$/ ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A))$		(M1)
$/ (\neg A \rightarrow (A \rightarrow B))$		(I1)
$/ (\neg \neg A \rightarrow A)$		(S1)

In the context of these rules, we assume that $\ulcorner(A \leftrightarrow B)\urcorner$ is an abbreviation of $\ulcorner((A \rightarrow B) \wedge (B \rightarrow A))\urcorner$.

From certain combinations these inference forms, it is possible to construct axiomatic deductive bases for P0, M0, I0, and even S0.

The axiomatic deductive base for P0 would consist of (MPP) plus (P1)–(P8). This is the basis for many deductive bases.

Thus, the axiomatic deductive base for M0 consists of P0's plus (M1). The axiomatic deductive base for I0 consists of M0's plus (I1). Finally, the axiomatic deductive base for S0 consists of I0's plus (S1).

This is sufficient definition of the deductive bases of these systems. However, we want to construct Fitch derivation systems, as they are more adequate representations of how reasoning works in the framework of these systems.

Derivation System for P0. The deductive base of the propositional positive logic P0 may be presented in the style of [notation 3.48](#) as follows.

Reiteration.

$$\begin{array}{c|c} a & A \\ & \vdots \\ & | \dots | A \text{ Reit, } a \end{array}$$

Conjunction.

$$\begin{array}{c|c} a & (A \wedge B) \\ & \vdots \\ & A \quad \wedge_1^-, a \end{array} \quad \begin{array}{c|c} a & (A \wedge B) \\ & \vdots \\ & B \quad \wedge_2^-, a \end{array} \quad \begin{array}{c|c} a & A \\ & \vdots \\ b & B \\ & \vdots \\ & (A \wedge B) \quad \wedge^+, a, b \end{array}$$

Disjunction.

$$\begin{array}{c|c} a & (A \vee B) \\ & \vdots \\ b & \begin{array}{c|c} A \\ \hline \vdots \end{array} \\ c & \begin{array}{c|c} C \\ \hline \vdots \end{array} \\ d & \begin{array}{c|c} B \\ \hline \vdots \end{array} \\ e & \begin{array}{c|c} C \\ \hline \vdots \end{array} \\ & C \quad \vee^-, a, b-c, d-e \end{array} \quad \begin{array}{c|c} a & A \\ & \vdots \\ & (A \vee B) \quad \vee_1^+, a \end{array} \quad \begin{array}{c|c} a & B \\ & \vdots \\ & (A \vee B) \quad \vee_2^+, a \end{array}$$

Conditional.

$$\begin{array}{c|c} a & (A \rightarrow B) \\ & \vdots \\ b & A \\ & \vdots \\ & B \quad \rightarrow^-, a, b \end{array} \quad \begin{array}{c|c} a & \begin{array}{c|c} A \\ \hline \vdots \end{array} \\ b & B \\ & \vdots \\ & (A \rightarrow B) \quad \rightarrow^+, a-b \end{array}$$

Biconditional.

$ \begin{array}{c l} a & (A \leftrightarrow B) \\ & \vdots \\ b & A \\ & \vdots \\ & B \quad \leftrightarrow_1^-, a, b \end{array} $	$ \begin{array}{c l} a & A \\ & \vdots \\ b & B \\ & \vdots \\ c & B \\ & \vdots \\ d & A \\ & \vdots \\ & (A \leftrightarrow B) \quad \leftrightarrow^+, a-b, c-d \end{array} $
--	--

Thus, we might define $\mathbf{DBas}_{P0}$ as follows:

Definition 6.4 (Deductive base for $P0$).

$\mathbf{DBas}_{P0} \stackrel{\text{DF}}{=} \{\text{Reit}, \wedge^+, \wedge_1^-, \wedge_2^-, \vee_1^+, \vee_2^+, \vee^-, \rightarrow^+, \rightarrow^-, \leftrightarrow^+, \leftrightarrow_1^-, \leftrightarrow_2^-\}.$

As for the derivation system $\mathbf{Cal}_{P0}$, we will assume that the functions $\mathbf{Prem}_{P0}$ and $\mathbf{Conc}_{P0}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 6.5 ($\mathbf{Cal}_{P0}$).

$\mathbf{Cal}_{P0}$ is a calculus on $P0$

$\stackrel{\text{DF}}{=} \mathbf{Cal}_{P0} = \langle \mathbf{DBas}_{P0}, \mathbf{Der}_{P0}, \mathbf{Prem}_{P0}, \mathbf{Conc}_{P0}, \vdash_{P0} \rangle$

: $\mathbf{DBas}_{P0}$ is a deductive base on $\mathbf{Fo}_{P0} \mathrel{\mathbb{M}}$

$\mathbf{Der}_{P0} = \{D : D \text{ is a Fitch derivation on } \mathbf{Fo}_{P0} \text{ from } \mathbf{DBas}_{P0}\} \mathrel{\mathbb{M}}$

$\mathbf{Prem}_{P0} : \mathbf{Der}_{P0} \rightarrow \wp \mathbf{Fo}_{P0} \mathrel{\mathbb{M}}$

$\mathbf{Conc}_{P0} : \mathbf{Der}_{P0} \rightarrow \wp \mathbf{Fo}_{P0} \mathrel{\mathbb{M}}$

$\vdash_S = \{ \langle A, B \rangle : \exists X \in \mathbf{Der}_{P0} (\mathbf{Prem}_{P0}(X) = A \mathrel{\mathbb{M}} \mathbf{Conc}_{P0}(X) = B) \}.$

Task 6.2. Show that all inferences of the forms (MPP), (P1)–(P8) (on p. 127) are members of $\vdash_{P0}$.

Theorem 6.6. $\vdash_{\mathbf{P0}} ((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$.

Proof. Let $A, B, C \in \mathbf{Fo}_{\mathbf{P0}}$ and consider the schematic derivation:

1		(A $\rightarrow$ B)	
2		(A $\rightarrow$ (B $\rightarrow$ C))	
3		A	
4		(B $\rightarrow$ C)	$\rightarrow^-$, 2, 3
5		B	$\rightarrow^-$, 1, 3
6		C	$\rightarrow^-$, 4, 5
7		(A $\rightarrow$ C)	$\rightarrow^+$, 3–6
8		((A $\rightarrow$ (B $\rightarrow$ C)) $\rightarrow$ (A $\rightarrow$ C))	$\rightarrow^+$, 2–7
9		((A $\rightarrow$ B) $\rightarrow$ ((A $\rightarrow$ (B $\rightarrow$ C)) $\rightarrow$ (A $\rightarrow$ C)))	$\rightarrow^+$, 1–8

Call this schematic derivation ‘**D**’. It is clear, for all $A, B, C \in \mathbf{Fo}_{\mathbf{P0}}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{\mathbf{P0}}$ from $\mathbf{DBas}_{\mathbf{P0}}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathbf{P0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathbf{P0}}(\mathbf{D})$ and $\ulcorner ((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C))) \urcorner \in \mathbf{Conc}_{\mathbf{P0}}(\mathbf{D})$. Therefore, $\vdash_{\mathbf{P0}} ((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$ has to hold according to [definition 6.5](#). $\square$

Derivation System for M0. We want to present the deductive base for the minimal propositional logic M0 in the style of [notation 3.48](#).

To do so, it is convenient (though not strictly necessary), to introduce the symbol ‘ $\perp$ ’, called ‘the falsum’. This is not a symbol of M0’s language (though it could have been introduced as such). However, we will have rules associated to this symbol which will allow us to obtain formulae of $\mathbf{Fo}_{\mathbf{M0}}$. Those rules are as follows:

Falsum.

a		$\perp$		a		A
		$\vdots$				$\vdots$
		B	$\perp^-, a$	b		$\neg A$
						$\vdots$
						$\perp$
						$\perp^+, a, b$

We moreover rewrite the rules for negation as follows:

Negation.

$$\begin{array}{c|c}
 a & \neg A \\
 \hline
 & \vdots \\
 b & \perp \\
 & \vdots \\
 & A \quad \neg^-, a-b
 \end{array}
 \qquad
 \begin{array}{c|c}
 a & A \\
 \hline
 & \vdots \\
 b & \perp \\
 & \vdots \\
 & \neg A \quad \neg^+, a-b
 \end{array}$$

We will not use all these rules for neither M1 nor I1, though. We want to have negated sentences in both M1 and I1, so $\neg^+$ seems to be acceptable in both systems. In order to obtain a negated formula in these derivation systems, however, we also need a way to introduce the falsum, so $\perp^+$ will also be acceptable in both systems.

We nevertheless do not want to do *reductio ad absurdum* aka indirect proofs, so we will have to dispense with $\neg^-$ in both systems. In the particular case of minimal logic, we also do not want to have anything be derivable from contradictions, so we also want to dispense with $\perp^-$.

Thus, we might define $\mathbf{DBas}_{M0}$ as follows:

Definition 6.7 (Deductive base for M0).

$$\mathbf{DBas}_{M0} \stackrel{\text{DF}}{=} \mathbf{DBas}_{P0} \cup \{\neg^+, \perp^+\}.$$

As for the derivation system $\mathbf{Cal}_{M0}$, we will once again assume that the functions $\mathbf{Prem}_{M0}$ and $\mathbf{Conc}_{M0}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 6.8 ($\mathbf{Cal}_{M0}$).

$\mathbf{Cal}_{M0}$ is a calculus on M0

$$\stackrel{\text{DF}}{=} \mathbf{Cal}_{M0} = \langle \mathbf{DBas}_{M0}, \mathbf{Der}_{M0}, \mathbf{Prem}_{M0}, \mathbf{Conc}_{M0}, \vdash_{M0} \rangle$$

: $\mathbf{DBas}_{M0}$ is a deductive base on $\mathbf{Fo}_{M0} \curvearrowright$

$$\mathbf{Der}_{M0} = \{D : D \text{ is a Fitch derivation on } \mathbf{Fo}_{M0} \text{ from } \mathbf{DBas}_{M0}\} \curvearrowright$$

$$\mathbf{Prem}_{M0} : \mathbf{Der}_{M0} \rightarrow \wp \mathbf{Fo}_{M0} \curvearrowright$$

$$\mathbf{Conc}_{M0} : \mathbf{Der}_{M0} \rightarrow \wp \mathbf{Fo}_{M0} \curvearrowright$$

$$\vdash_S = \{ \langle A, B \rangle : \exists X \in \mathbf{Der}_{M0} (\mathbf{Prem}_{M0}(X) = A \curvearrowright \mathbf{Conc}_{M0}(X) = B) \}.$$

Task 6.3. Show that all inferences of the forms (MPP), (P1)–(P8), (M1) (on p. 127) are members of $\vdash_{M0}$.

It is clear that all inferences of the forms (P1)–(P8) are members of $\vdash_{M0}$, since $\vdash_{M0}$ is an extension of $\vdash_{P0}$. Let us, then, prove the case of the inference form distinctive of $M0$, i.e., (M1).

Theorem 6.9. $\vdash_{M0} ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A))$.

Proof. Let $A, B \in \mathbf{Fo}_{M0}$ and consider the schematic derivation:

1		(A $\rightarrow$ B)	
2		(A $\rightarrow$ $\neg$ B)	
3		A	
4		B	$\rightarrow^-$, 1, 3
5		$\neg$ B	$\rightarrow^-$, 2, 3
6		$\perp$	$\perp^+$, 4, 5
7		$\neg$ A	$\neg^+$, 3–6
8		((A $\rightarrow$ $\neg$ B) $\rightarrow$ $\neg$ A)	$\rightarrow^+$, 2–7
9		((A $\rightarrow$ B) $\rightarrow$ ((A $\rightarrow$ $\neg$ B) $\rightarrow$ $\neg$ A))	$\rightarrow^+$, 1–8

Call this schematic derivation ‘D’. It is clear, for all $A, B \in \mathbf{Fo}_{M0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{M0}$ from $\mathbf{DBas}_{M0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{M0}$ with $\{\} \subseteq \mathbf{Prem}_{M0}(\mathbf{D})$ and $\ulcorner ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)) \urcorner \in \mathbf{Conc}_{M0}(\mathbf{D})$. Therefore, by [definition 6.8](#), $\vdash_{M0} ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A))$. $\square$

Derivation System for $I0$. To present the deductive base $\mathbf{Cal}_{I0}$ for the intuitionistic propositional logic $I0$ in the style of [notation 3.48](#), we work from $\mathbf{Cal}_{M0}$ and just add $\perp^-$. However, as we argued above, we should not add $\neg^-$ in order to avoid *reductio ad absurdum* proofs. Thus, we might define $\mathbf{DBas}_{I0}$ as follows:

Definition 6.10 (Deductive base for $I0$).

$$\mathbf{DBas}_{I0} \stackrel{\text{DF}}{=} \mathbf{DBas}_{M0} \cup \{\perp^-\}.$$

As for the derivation system $\mathbf{Cal}_{I0}$, we will once again assume that the functions $\mathbf{Prem}_{I0}$ and $\mathbf{Conc}_{I0}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 6.11 (Cal_{I_0}).

Cal_{I_0} is a calculus on I_0

$$\stackrel{\text{DF}}{=} \text{Cal}_{I_0} = \langle \text{DBas}_{I_0}, \text{Der}_{I_0}, \text{Prem}_{I_0}, \text{Conc}_{I_0}, \vdash_{I_0} \rangle$$

: DBas_{I_0} is a deductive base on $\text{Fo}_{I_0} \mathrel{\Vdash}$

$\text{Der}_{I_0} = \{D : D \text{ is a Fitch derivation on } \text{Fo}_{I_0} \text{ from } \text{DBas}_{I_0}\} \mathrel{\Vdash}$

$\text{Prem}_{I_0} : \text{Der}_{I_0} \rightarrow \wp \text{Fo}_{I_0} \mathrel{\Vdash}$

$\text{Conc}_{I_0} : \text{Der}_{I_0} \rightarrow \wp \text{Fo}_{I_0} \mathrel{\Vdash}$

$\vdash_S = \{ \langle A, B \rangle : \exists X \in \text{Der}_{I_0} (\text{Prem}_{I_0}(X) = A \mathrel{\Vdash} \text{Conc}_{I_0}(X) = B) \}.$

Task 6.4. Show that all inferences of the forms (MPP), (P1)–(P8), (M1), (I1) (on p. 127) are members of $\vdash_{I_0}$.

It is clear that all inferences of the forms (P1)–(P8), (M1) are members of $\vdash_{I_0}$, since $\vdash_{I_0}$ is an extension of $\vdash_{M_0}$. Let us, then, prove the case of the inference form distinctive of I_0 , i.e., (I1).

Theorem 6.12. $\vdash_{I_0} (\neg A \rightarrow (A \rightarrow B)).$

Proof. Let $A, B \in \text{Fo}_{I_0}$ and consider the schematic derivation:

1			$\neg A$	
2				A
3				$\perp$ $\perp^+, 2, 1$
4				B $\perp^-, 3$
5				$(A \rightarrow B)$ $\rightarrow^+, 2-4$
6				$(\neg A \rightarrow (A \rightarrow B))$ $\rightarrow^+, 1-5$

Call this schematic derivation ‘D’. It is clear, for all $A, B \in \text{Fo}_{I_0}$, that D is a Fitch derivation on Fo_{I_0} from DBas_{I_0} . This means that there is a derivation $D \in \text{Der}_{I_0}$ with $\{\} \subseteq \text{Prem}_{I_0}(D)$ and $\ulcorner (\neg A \rightarrow (A \rightarrow B)) \urcorner \in \text{Conc}_{I_0}(D)$. Therefore, by definition 6.11, $\vdash_{I_0} (\neg A \rightarrow (A \rightarrow B)).$ $\square$

An interesting fact about Heyting’s intuitionistic logic is that, although we cannot have $A \mathrel{\Vdash} \neg A$ in general, we do have $\neg A \mathrel{\Vdash} \neg \neg \neg A$.

Task 6.5. Show that all inferences of the forms $\neg A \mathrel{\Vdash} \neg \neg \neg A$ are members of $\vdash_{I_0}$.

In order to do task 6.5 we may prove each side independently.

Theorem 6.13. $\neg A \vdash_{I0} \neg\neg\neg A$

Proof. Let $A \in \mathbf{Fo}_{I0}$ and consider the schematic derivation:

1	$\neg A$	
2	$\neg\neg A$	
3	$\perp$	$\perp^+, 1, 2$
4	$\neg\neg\neg A$	$\neg^+, 2-3$

Call this schematic derivation ‘D’. It is clear, for all $A \in \mathbf{Fo}_{I0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{I0}$ from $\mathbf{DBas}_{I0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{I0}$ with $\{\ulcorner \neg A \urcorner\} \subseteq \mathbf{Prem}_{I0}(\mathbf{D})$ and $\ulcorner \neg\neg\neg A \urcorner \in \mathbf{Conc}_{I0}(\mathbf{D})$. Therefore, by [definition 6.11](#), $\neg A \vdash_{I0} \neg\neg\neg A$. $\square$

Theorem 6.14. $\neg\neg\neg A \vdash_{I0} \neg A$

Proof. Let $A \in \mathbf{Fo}_{I0}$ and consider the schematic derivation:

1	$\neg\neg\neg A$	
2	A	
3	$\neg A$	
4	$\perp$	$\perp^+, 2, 3$
5	$\neg\neg A$	$\neg^+, 3-4$
6	$\perp$	$\perp^+, 5, 1$
7	$\neg A$	$\neg^+, 2-6$

Call this schematic derivation ‘D’. It is clear, for all $A \in \mathbf{Fo}_{I0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{I0}$ from $\mathbf{DBas}_{I0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{I0}$ with $\{\ulcorner \neg\neg\neg A \urcorner\} \subseteq \mathbf{Prem}_{I0}(\mathbf{D})$ and $\ulcorner \neg A \urcorner \in \mathbf{Conc}_{I0}(\mathbf{D})$. Therefore, by [definition 6.11](#), $\neg\neg\neg A \vdash_{I0} \neg A$. $\square$

We can also prove, as desired, the double-negated variant of the *tertium non datur*.

Task 6.6. Show that all inferences of the form $\Vdash \neg\neg(A \vee \neg A)$ are members of $\vdash_{\text{I0}}$.

Theorem 6.15. $\vdash_{\text{I0}} \neg\neg(A \vee \neg A)$

Proof. Let $A \in \mathbf{Fo}_{\text{I0}}$ and consider the schematic derivation:

1		$\neg(A \vee \neg A)$	
2			A
3			$(A \vee \neg A) \quad \vee_1^+, 2$
4			$\perp \quad \perp^+, 3, 1$
5		$\neg A \quad \neg^+, 2-4$	
6		$(A \vee \neg A) \quad \vee_2^+, 5$	
7		$\perp \quad \perp^+, 1, 6$	
8		$\neg\neg(A \vee \neg A) \quad \neg^+, 1-7$	

Call this schematic derivation ‘**D**’. It is clear, for all $A \in \mathbf{Fo}_{\text{I0}}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{\text{I0}}$ from $\mathbf{DBas}_{\text{I0}}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\text{I0}}$ with $\{\} \subseteq \mathbf{Prem}_{\text{I0}}(\mathbf{D})$ and $\ulcorner \neg\neg(A \vee \neg A) \urcorner \in \mathbf{Conc}_{\text{I0}}(\mathbf{D})$. Therefore, by **definition 6.11**, $\vdash_{\text{I0}} \neg\neg(A \vee \neg A)$. $\square$

7.1 Philosophical Motivations

One of its most distinctive features of S_0 is its treatment of contradictions: once a contradiction is admitted, any formula follows. This consequence is expressed by the principle *ex contradictione quodlibet* (ECQ), also known as the principle of explosion:

$$A, \neg A \text{ / } B. \quad (\text{ECQ})$$

As a result, any inconsistent set of or sequence of formulae becomes trivial, in the sense that every formula follows from it by (ECQ).

The idea that contradictions leads to triviality has been used by many authors (e.g., Popper) to discard inconsistencies from science and knowledge in general. This idea has nevertheless been challenged on both philosophical and practical grounds.

A source of motivation comes from the study of semantic paradoxes. An example is the liar's paradox, generated by a sentence like:

$$(\neg) \text{ is false.} \quad (\neg)$$

This sentence leads to a contradiction and, hence, to a triviality if reasoned by means of classical logic. However, there does not seem to be anything ungrammatical about it. So it has the form of a propositional sentence – although it is arguably not.

Similar difficulties arise in set theory. Naïve set theory, based on unrestricted comprehension, gives rise to Russell's paradox and other antinomies. Such paradoxes suggest that contradictions may arise naturally within formal systems and that their presence need not imply the collapse of all reasoning.

Another important motivation is the existence of inconsistent information. Databases, legal systems, regulations, and large bodies of knowledge often contain conflicting statements. Such inconsistencies may result from errors, temporal changes, or the integration of data from multiple sources. Under classical logic, it seems impossible to reason about those databases, as they would be trivial. In practice, however, people are able to draw sensible conclusions while avoiding arbitrary ones.

The history of science provides further examples. Scientific theories have often contained tensions and even outright contradictions. Yet scientists have continued to employ such theories successfully, extracting valuable predictions and explanations despite these inconsistencies. The development of science therefore suggests that inconsistency and rationality are not incompatible. This also

suggests a non-classical approach where not every contradiction leads to triviality.

These considerations motivate the development of *paraconsistent logics*. A logic is paraconsistent (with respect to the negation operator ‘ $\neg$ ’) when the (ECQ) does not hold in general. Such logics make possible the formal study of inconsistent but non-trivial systems (e.g., theories, databases, etc.).

Among the most influential systems of paraconsistent logic are those introduced by Newton da Costa (1963). His calculi, which we will call $\mathsf{C0}_n$ (where $n \in \mathbf{Num}_{\mathbb{N}}$)*, constituted the first systematic family of paraconsistent propositional logics explicitly intended to solve semantic paradoxes by accepting contradictions.

*: Originally called C_n , where $n \in \mathbf{Num}_{\mathbb{N}} \cup \{\omega\}$.

7.2 Vocabularies and Languages

The vocabularies and formal languages $\mathbf{Vo}_{\mathsf{C0}_1}, \dots$ and $\mathbf{Fo}_{\mathsf{C0}_1}, \dots$ will be defined as those of $\mathsf{S0}$.

Definition 7.1.

$$\mathbf{Fo}_{\mathsf{C0}_n} \stackrel{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}.$$

$$\mathbf{Vo}_{\mathsf{C0}_n} \stackrel{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}.$$

In order to better express some principles of these systems, we will use the following abbreviations:

Definition 7.2 (Abbreviations for $\mathbf{Fo}_{\mathsf{C0}_1}, \dots$).

$$(A)^1 \stackrel{\text{DF}}{=} \neg(A \wedge \neg A)$$

$$(A)^{N(m)} \stackrel{\text{DF}}{=} ((A)^m)^1$$

$$A^{(1)} \stackrel{\text{DF}}{=} A^1$$

$$(A)^{N(m)} \stackrel{\text{DF}}{=} ((A)^{(m)} \wedge (A)^{N(m)})$$

$$\neg^{(m)} A \stackrel{\text{DF}}{=} (\neg A \wedge (A)^{(m)})$$

The expressions defined above are not formulae of $\mathbf{Fo}_{\mathsf{C0}_1}, \dots$, but rather abbreviations of some formulae of them. We will use these abbreviations in our meta-language in order to more easily describe the properties of the systems $\mathsf{C0}_1, \dots$ by da Costa. We will dispense with the parentheses when this does not result in any ambiguity.

Thus, the expression ‘ p^1 ’, which abbreviates the formula ‘ $\neg(p \wedge \neg p)$ ’, is understood by da Costa to intuitively express that ‘ p ’ is *well-behaved*. One might also think of it as stating that ‘ p ’ is consistent, that is, that it is not the case that both ‘ p ’ and ‘ $\neg p$ ’. A formula is well-behaved when it is not simultaneously true or asserted together with its negation.

The expression ' p^2 ', which abbreviates the expression ' $\neg(p^1 \wedge \neg p^1)$ ', which in turn abbreviates the formula ' $\neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p))$ ', has the intended sense that ' $\neg(p \wedge \neg p)$ ' is well-behaved (though not necessarily also ' p '). That is, not only ' p ' is well-behaved, but also the formulae expressing that it is well-behaved and that is ill-behaved – that is, the formula $(p \wedge \neg p)$.

An expression of the form ' $p^{(m)}$ ' abbreviates a formula of the form ' $p^1 \wedge \dots \wedge p^m$ ' (if we remove parentheses). The intuitive sense is that ' p ' is well-behaved up to level m . In other words, not only is ' p ' well-behaved, but so are ' p^1 ' and ' p^2 ' and so forth until ' p^m '.

Finally, the expression ' $\neg^{(1)} p$ ', which abbreviates ' $(\neg p \wedge p^{(1)})$ ', has the intended sense of strong negation in the system $C0_1$. It is strong because, as we will see, although ' $(\neg p \wedge p^{(1)})$ ' does not lead to explosion in $C0_1$, the formula abbreviated by ' $(p \wedge \neg^{(1)} p)$ ' does. In general, for all $n \in \mathbf{Num}$, the symbol ' $\neg^{(n)}$ ' expresses the operation of strong negation in the system called ' $C0_n$ '.

Task 7.1. Determine the formulae abbreviated by the following expressions:

1. p^2
2. $(p \wedge q)^1$
3. $(p \wedge q)^2$
4. $p^{(1)}$
5. $p^{(2)}$
6. $\neg^{(1)} p$
7. $\neg^{(2)} p$
8. $\neg^{(1)}(p \wedge \neg^{(1)} p)$

Solution to task 7.1.

1. $p^2 = (p^1)^1$
 $= \neg(p^1 \wedge \neg p^1)$
 $= \neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p))$
2. $(p \wedge q)^1 = \neg((p \wedge q) \wedge \neg(p \wedge q))$
3. $(p \wedge q)^2 = ((p \wedge q)^1)^1$
 $= \neg((p \wedge q)^1 \wedge \neg(p \wedge q)^1)$
 $= \neg(\neg(\neg((p \wedge q) \wedge \neg(p \wedge q))) \wedge \neg\neg((p \wedge q) \wedge \neg(p \wedge q)))$
4. $p^{(1)} = p^1$
 $= \neg(p \wedge \neg p)$
5. $p^{(2)} = (p^{(1)} \wedge p^2)$
 $= (p^1 \wedge p^2)$
 $= (\neg(p \wedge \neg p) \wedge \neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p)))$
6. $\stackrel{(1)}{\neg} p = (\neg p \wedge p^{(1)})$
 $= (\neg p \wedge p^1)$
 $= (\neg p \wedge \neg(p \wedge \neg p))$
7. $\stackrel{(2)}{\neg} p = (\neg p \wedge p^{(2)})$
 $= (\neg p \wedge (p^1 \wedge p^2))$
 $= (\neg p \wedge (\neg(p \wedge \neg p) \wedge \neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p))))$
8. $\stackrel{(1)}{\neg}(p \wedge \stackrel{(1)}{\neg} p) = (\neg(p \wedge \stackrel{(1)}{\neg} p) \wedge (p \wedge \stackrel{(1)}{\neg} p))$
 $= (\neg(p \wedge \stackrel{(1)}{\neg} p) \wedge (p \wedge \stackrel{(1)}{\neg} p))$
 $= (\neg(p \wedge (\neg p \wedge p^{(1)})) \wedge (p \wedge (\neg p \wedge p^{(1)})))$
 $= (\neg(p \wedge (\neg p \wedge p^1)) \wedge (p \wedge (\neg p \wedge p^1)))$
 $= (\neg(p \wedge (\neg p \wedge \neg(p \wedge \neg p))) \wedge (p \wedge (\neg p \wedge \neg(p \wedge \neg p))))$

7.3 Inference and Consequence Relations

All systems $C0_1, \dots$ fall among modern systems, so their inference relations are just special cases of **definition 3.31**.

As for the contents of their consequence relations, there are two principles in question: the law of non contradiction

ion (LNC) and *ex contradictione sequitur quodlibet* (ECQ):

$$/ \neg(A \wedge \neg A) \quad (\text{LNC})$$

$$A, \neg A / B \quad (\text{ECQ})$$

While ECQ is restricted in all systems of paraconsistent logic, LNC will not be restricted in some of them.

It is worth noting that, in Graham Priest's LP system, LNC is valid in general. The underlying philosophy of LP is *dialtheism*, the thesis that there are true contradictions. That is, some expressions will be regarded as true, which are of the form:

$$(A \wedge \neg A) \quad (\neg)$$

Some of those will be precisely expressions of the form above. Thus, say that ' p ' stands for an expression such that ' $(p \wedge \neg p)$ ' is true. Thus, we see that $(\neg)$ holds for ' p '. Moreover, there would be no problem, under this framework, with saying that $(\neg)$ also holds for ' $(p \wedge \neg p)$ '. Hence, we would have that:

$$((p \wedge \neg p) \wedge \neg(p \wedge \neg p)). \quad (\forall)$$

From this, both ' $(p \wedge \neg p)$ ' and ' $\neg(p \wedge \neg p)$ ' would obtain. Thus, under this framework, LNC can safely be said to hold. ' p ' even when it is such that ' $(p \wedge \neg p)$ ' is the case.

For the rest, we want to have as many of the principles of standard logic as we can have, without having to include these principles.

A couple of problematic principles, in this respect, are the principle of addition ($\vee_1^+$ and $\vee_2^+$) and disjunctive syllogism (DS1 and DS2).

a	$(A \vee B)$	
	$\vdots$	
b	$\neg A$	
	$\vdots$	
	B	DS1, a, b

a	$(A \vee B)$	
	$\vdots$	
b	$\neg B$	
	$\vdots$	
	A	DS2, a, b

These principles are derivable in S0, and have a long-standing tradition in Western logic. The reason these principles are problematic in paraconsistent logics is that they easily allow to obtain ECQ as a derived rule.

Theorem 7.3. $\mathbf{DBas}_S = \{\vee_1^+, \vee_2^+, \text{DS1}, \text{DS2}\} \Rightarrow \exists \mathbf{D} : \mathbf{D} \text{ is a Fitch derivation on } \mathbf{Fo}_S \text{ from } \mathbf{DBas}_S \text{ with } A, \ulcorner \neg A \urcorner \in \mathbf{Prem}_S(\mathbf{D}) \text{ and } B \in \mathbf{Conc}_S(\mathbf{D}).$

Proof. Let $A, B \in \mathbf{Fo}_S$ and $\mathbf{DBas}_S = \{\vee_1^+, \vee_2^+, \text{DS1}, \text{DS2}\}$. Consider the schematic derivation:

1	A	
2	$\neg A$	
3	$(A \vee B)$	$\vee_1^+, 1$
4	B	DS2, 3, 2

Call this schematic derivation ‘**D**’. It is clear, for all $A, B \in \mathbf{Fo}_S$, that **D** is a Fitch derivation on $\mathbf{Fo}_S$ from $\mathbf{DBas}_S$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_S$ with $A, \neg A \in \mathbf{Prem}_S(\mathbf{D})$ and $B \in \mathbf{Conc}_S(\mathbf{D})$. $\square$

We have only used $\vee_1^+$ and DS2, but we could have alternatively used $\vee_2^+$ and DS1. Hence, a paraconsistent system will normally have to restrict some of the principles $\vee_1^+, \vee_2^+, \text{DS1}, \text{DS2}$.

7.4 Interpretation System for $\mathbf{C0}_n$

The semantics for da Costa’s $\mathbf{C0}_1$ systems uses cpi (classical propositional interpretations), thus assigning to each propositional variable a classical truth value, either **1** (truth) or **0** (falsity).

A cpi $\mathfrak{I}$ is extended to all formulae of $\mathbf{Fo}_{\mathbf{C0}_1}$ by means of a valuation $\hat{\mathfrak{I}}$, which is its $\mathbf{C0}_1$ -extension, or $\mathbf{C0}_1\text{e}$, for short.

Definition 7.4 ($\mathbf{C0}_1$ -valuation). A valuation $\hat{\mathfrak{I}} = \langle \{\mathbf{0}, \mathbf{1}\}, \hat{\iota} \rangle$ is a **dCe** of the cpi $\mathfrak{I}$ on $\mathbf{Fo}_S$

$$\begin{aligned}
 &\stackrel{\text{DE}}{=} \hat{\iota}(\neg A) = \mathbf{0} \Rightarrow \hat{\iota}(A) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(\neg \neg A) = \mathbf{1} \Rightarrow \hat{\iota}(A) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(\neg B) = \hat{\iota}(\neg(A \rightarrow B)) = \hat{\iota}(\neg(A \rightarrow \neg B)) = \mathbf{1} \Rightarrow \hat{\iota}(A) = \mathbf{0} && \mathfrak{M} \\
 &\hat{\iota}(A \wedge B) = \mathbf{1} \Leftrightarrow \hat{\iota}(A) = \mathbf{1} \wedge \hat{\iota}(B) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(A \vee B) = \mathbf{1} \Leftrightarrow \hat{\iota}(A) = \mathbf{1} \vee \hat{\iota}(B) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(A \rightarrow B) = \mathbf{1} \Leftrightarrow \hat{\iota}(A) = \mathbf{0} \vee \hat{\iota}(B) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(A \heartsuit B) = \mathbf{0} \Rightarrow \hat{\iota}(A) = \mathbf{0} \vee \hat{\iota}(B) = \mathbf{0} \\
 &: \heartsuit \in \{\wedge, \vee, \rightarrow\} \mathfrak{M} A, B \in \mathbf{Fo}_S.
 \end{aligned}$$

7.5 Derivation Systems for $\mathbf{C0}_n$

As with intuitionistic logics, it is a good idea to start with axiomatic systems. In particular, an axiomatic derivation system for $\mathbf{C0}_1$ can be defined using the following inference forms as the rules of its deductive base:

$(A \rightarrow B), A \ / \ B$	(MPP)
$\ / \ (A \rightarrow (B \rightarrow A))$	(P1)
$\ / \ (((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$	(P2)
$\ / \ (A \rightarrow (B \rightarrow (A \wedge B)))$	(P3)
$\ / \ ((A \wedge B) \rightarrow A)$	(P4)
$\ / \ ((A \wedge B) \rightarrow B)$	(P5)
$\ / \ (A \rightarrow (A \vee B))$	(P6)
$\ / \ (B \rightarrow (A \vee B))$	(P7)
$\ / \ (((A \rightarrow C) \rightarrow ((B \rightarrow C) \rightarrow ((A \vee B) \rightarrow C)))$	(P8)
$\ / \ (A \vee \neg A)$	(Tertium)
$\ / \ (\neg \neg A \rightarrow A)$	(DN2)
$\ / \ (B^1 \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$	(C0 ₁ 1)
$\ / \ (A^1 \rightarrow (\neg A)^1)$	(C0 ₁ 2)
$\ / \ ((A^1 \wedge B^1) \rightarrow (A \wedge B)^1)$	(C0 ₁ 3)
$\ / \ ((A^1 \wedge B^1) \rightarrow (A \vee B)^1)$	(C0 ₁ 4)
$\ / \ ((A^1 \wedge B^1) \rightarrow (A \rightarrow B)^1)$	(C0 ₁ 5)

As before, we assume, in the context of these rules, that $\ulcorner(A \leftrightarrow B)\urcorner$ is an abbreviation of $\ulcorner((A \rightarrow B) \wedge (B \rightarrow A))\urcorner$.

Note that we have all axioms of positive logic, plus two axioms satisfied by S0 which are not satisfied by I0: (Tertium) and (DN2).

As for the rest of $C0_n$ systems, their axioms are very similar. We just change the last four with their corresponding $C0_n$ -versions, which are as follows:

$\ / \ (B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$	(C0 _n 1)
$\ / \ (A^n \rightarrow (\neg A)^n)$	(C0 _n 2)
$\ / \ ((A^n \wedge B^n) \rightarrow (A \wedge B)^n)$	(C0 _n 3)
$\ / \ ((A^n \wedge B^n) \rightarrow (A \vee B)^n)$	(C0 _n 4)
$\ / \ ((A^n \wedge B^n) \rightarrow (A \rightarrow B)^n)$	(C0 _n 5)

It is important to note that the larger the n , the weaker is the system. In other words, if we define the consequence relations of $C0_n$ based on the axiom schemata above, we will have:

$$\mathbf{Cn}_{C0_1} \supset \mathbf{Cn}_{C0_2} \supset \dots \supset \mathbf{Cn}_{C0_\omega}.$$

The system $C0_\omega$ – which we will not study here – results from applying the axioms above to expressions of the form $\ulcorner A^\omega \urcorner$, which results from applying the superindex ‘1’ infinitely many times. That is, we are not dealing with a system describable with the language S0, but with an infinitary language, i.e., a language containing formulae of infinite length.

Constructing natural derivation systems for C0_n systems is a bit more complicated than with the previous systems. To do so, we work, again, from the rules in $\mathbf{DBas}_{\text{P0}}$, and we add some which will allow us to obtain the additional results.

As the axioms show, both *tertium non datur* and double negation work, which means that we do not need many of the restrictions of intuitionistic logic. In particular, $\neg^+$ and $\neg^-$ will be rules in all C0_n systems. We also want to admit $\perp^-$, as we will want to apply the ECQ to some restricted set of formulae.

We nevertheless want to restrict $\perp^+$. In particular, ' $\perp$ ' will not be obtainable from any contradiction, but just from contradictions obtained of well-behaved formulae. Thus, we want the following rule to hold for a system named $\ulcorner \text{C0}_n \urcorner$.

Falsum-introduction.

$$\begin{array}{c|l} a & A^n \\ & \vdots \\ b & A \\ & \vdots \\ c & \neg A \\ & \vdots \\ & \perp \quad \perp^{n+}, a, b, c \end{array}$$

Furthermore, we introduce the following *propagation rules*, which allow us to infer that formulae composed from well-behaved formulae are themselves well-behaved.

Propagation rules.

$$\begin{array}{c|l} a & A^n \\ & \vdots \\ b & B^n \\ & \vdots \\ & (A \heartsuit B)^n \quad \heartsuit^{n+}, a, b \\ & : \heartsuit \in \{ \wedge, \vee, \rightarrow, \leftrightarrow \} \end{array} \qquad \begin{array}{c|l} a & A^n \\ & \vdots \\ & (\neg A)^n \quad \neg^{n+}, a \end{array}$$

The rules $\wedge^{n+}, \vee^{n+}, \rightarrow^{n+}, \leftrightarrow^{n+}$ defined here are to be intuitively understood as follows: if A is well-behaved and B is well-behaved, then so are $\ulcorner (A \wedge B) \urcorner, \ulcorner (A \vee B) \urcorner, \ulcorner (A \rightarrow B) \urcorner, \ulcorner (A \leftrightarrow B) \urcorner$. Note, however, that $A, B, \ulcorner (A \wedge B) \urcorner, \ulcorner (A \vee B) \urcorner, \ulcorner (A \rightarrow B) \urcorner, \ulcorner (A \leftrightarrow B) \urcorner$ are not themselves derivable by this rule.

Finally, we need to add the proof by cases rules, which corresponds to the axiom (**Tertium**).

Proof by cases rule

a		A
		$\vdots$
b		B
		$\vdots$
c		$\neg A$
		$\vdots$
d		B
		$\vdots$
		$B \text{ PC, } a-b, c-d$

The system above suffices for characterising all systems $\mathbf{Cn}_{C0_n}$. Thus, we might define $\mathbf{DBas}_{C0_n}$ as follows:

Definition 7.5 (Deductive base for $C0_n$).

$$\mathbf{DBas}_{C0_n} \stackrel{\text{DF}}{=} \mathbf{DBas}_{P0} \cup \{\neg^-, \neg^+, \perp^-, \perp^{n+}, \neg^{n+}, \wedge^{n+}, \vee^{n+}, \rightarrow^{n+}, \leftrightarrow^{n+}, \text{PC}\}.$$

As for the derivation system $\mathbf{Cal}_{C0_n}$, we will assume that the functions $\mathbf{Prem}_{C0_n}$ and $\mathbf{Conc}_{C0_n}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 7.6 ($\mathbf{Cal}_{C0_n}$).

$\mathbf{Cal}_{C0_n}$ is a calculus on $C0_n$

$$\begin{aligned} &\stackrel{\text{DF}}{=} \mathbf{Cal}_{C0_n} = \langle \mathbf{DBas}_{C0_n}, \mathbf{Der}_{C0_n}, \mathbf{Prem}_{C0_n}, \mathbf{Conc}_{C0_n}, \vdash_{C0_n} \rangle \\ &: \mathbf{DBas}_{C0_n} \text{ is a deductive base on } \mathbf{Fo}_{C0_n} \text{ } \mathbb{M} \\ &\mathbf{Der}_{C0_n} = \{D : D \text{ is a Fitch derivation on } \mathbf{Fo}_{C0_n} \text{ from } \mathbf{DBas}_{C0_n}\} \text{ } \mathbb{M} \\ &\mathbf{Prem}_{C0_n} : \mathbf{Der}_{C0_n} \rightarrow \wp \mathbf{Fo}_{C0_n} \text{ } \mathbb{M} \\ &\mathbf{Conc}_{C0_n} : \mathbf{Der}_{C0_n} \rightarrow \wp \mathbf{Fo}_{C0_n} \text{ } \mathbb{M} \\ &\vdash_{C0_n} = \{\langle A, B \rangle : \exists X \in \mathbf{Der}_{C0_n} (\mathbf{Prem}_{C0_n}(X) = A \text{ } \mathbb{M} \text{ } \mathbf{Conc}_{C0_n}(X) = B)\}. \end{aligned}$$

Task 7.2. Show that all inferences of the forms (**Tertium**), (**DN2**), (**C0_n1**)–(**C0_n4**) are members of $\vdash_{C0_n}$.

Theorem 7.7. $\vdash_{\mathbf{Co}_n} (\neg\neg A \rightarrow A)$.

Proof. Let $A \in \mathbf{Fo}_{\mathbf{Co}_n}$ and consider the schematic derivation:

1			$\neg\neg A$	
2			$(A \wedge \neg A)$	
3			A	$\wedge_1^-, 2$
4			$\neg(A \wedge \neg A)$	
			A^n	Abbr., 4
5			$(\neg A)^n$	$\neg^{n+}, 4$
6				
7				
8				
9				
10				
1			$\neg A$	
2			$\perp$	$\perp^{n+}, 5, 6, 1$
3			A	$\neg^-, 6-7$
4			A	PC, 2-3, 4-8
5			$(\neg\neg A \rightarrow A)$	$\rightarrow^+, 1-9$

Call this schematic derivation ‘**D**’. It is clear, for all $A \in \mathbf{Fo}_{\mathbf{Co}_n}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{\mathbf{Co}_n}$ from $\mathbf{DBas}_{\mathbf{Co}_n}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathbf{Co}_n}$ with $\{\} \subseteq \mathbf{Prem}_{\mathbf{Co}_n}(\mathbf{D})$ and $\ulcorner (\neg\neg A \rightarrow A) \urcorner \in \mathbf{Conc}_{\mathbf{Co}_n}(\mathbf{D})$. Therefore, $\vdash_{\mathbf{Co}_n} (\neg\neg A \rightarrow A)$ has to hold according to [definition 7.6](#). $\square$

Theorem 7.8. $\vdash_{\mathbf{C0}_n} (B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$.

Proof. Let $A, B \in \mathbf{Fo}_{\mathbf{C0}_n}$ and consider the schematic derivation:

1			B^n	
2			$(A \rightarrow B)$	
3			$(A \rightarrow \neg B)$	
4			A	
5			B	$\rightarrow^-, 2, 4$
6			$\neg B$	$\rightarrow^-, 3, 4$
7			$\perp$	$\perp^{n+}, 1, 5, 6$
8			$\neg A$	$\neg^+, 4-7$
9			$((A \rightarrow \neg B) \rightarrow \neg A)$	$\rightarrow^+, 3-8$
10			$((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A))$	$\rightarrow^+, 2-9$
11			$(B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$	$\rightarrow^+, 1-11$

Call this schematic derivation 'D'. It is clear, for all $A, B \in \mathbf{Fo}_{\mathbf{C0}_n}$, that D is a Fitch derivation on $\mathbf{Fo}_{\mathbf{C0}_n}$ from $\mathbf{DBas}_{\mathbf{C0}_n}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathbf{C0}_n}$ with $\{\} \subseteq \mathbf{Prem}_{\mathbf{C0}_n}(\mathbf{D})$ and $\ulcorner (B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A))) \urcorner \in \mathbf{Conc}_{\mathbf{C0}_n}(\mathbf{D})$. Therefore, $\vdash_{\mathbf{C0}_n} ((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$ has to hold according to [definition 7.6](#). $\square$

Note also that, in $\mathbf{C}_n$, the strong negation ' $\neg^{(n)}$ ' has the properties of classical negation.

Task 7.3. Show that all inferences of the following forms are members of $\vdash_{\mathbf{C0}_n}$.

1. $/ (A \vee \neg^{(n)} A)$
2. $/ \neg^{(n)}(A \wedge \neg^{(n)} A)$
3. $(A \rightarrow B), (A \rightarrow \neg^{(n)} B) / \neg^{(n)} A$
4. $A, \neg^{(n)} A / B$
5. $A / \neg^{(n)} \neg^{(n)} A$
6. $(A \leftrightarrow \neg^{(n)} A) // B$

Theorem 7.9. $A, \overset{(n)}{\neg} A \vdash_{C0_n} B$.

*: In step 5, remember that $p^{(n)} = p^n \wedge p^{n-1} \wedge \dots \wedge p^1$.

Proof. Let $A \in \mathbf{Fo}_{C0_n}$ and consider the schematic derivation*:

1	A	
2	$\overset{(n)}{\neg} A$	
	$(\neg A \wedge A^{(n)})$	Abbr., 2
3	$\neg A$	$\wedge_1^-, 2$
4	$A^{(n)}$	$\wedge_2^-, 2$
5	A^n	$\wedge_2^-, 4$
6	$\perp$	$\perp^{n+}, 5, 1, 3$
7	B	$\perp^-, 6$

Call this schematic derivation 'D'. It is clear, for all $A, B, C \in \mathbf{Fo}_{C0_n}$, that D is a Fitch derivation on $\mathbf{Fo}_{C0_n}$ from $\mathbf{DBas}_{C0_n}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{C0_n}$ with $A, \overset{(n)}{\neg} A \in \mathbf{Prem}_{C0_n}(\mathbf{D})$ and $B \in \mathbf{Conc}_{C0_n}(\mathbf{D})$. Therefore, $A, \overset{(n)}{\neg} A \vdash_{C0_n} B$ has to hold according to [definition 7.6](#). $\square$

8.1 Philosophical Motivations

Connexive logics are motivated by some dissatisfaction with standard accounts of implication. A central intuition is that a proposition should not imply its own negation, nor should its negation imply the proposition itself.

In other words, we want that inferences of the following form be logically valid:

$$/ \neg(\neg A \rightarrow A) \quad (\text{AT1})$$

Connexive logics are logical systems in which this and other similar inference forms are logically valid.

A second important motivation is historical. Connexive theses are commonly associated with the theses about the conditional by several authors such as Aristotle, Chrysippus, and Boethius. Many researchers regard connexive logic as a way of recovering and systematising ideas that can be traced back to ancient and medieval traditions of logical thought.

More generally, connexive logics are motivated by the desire to provide a more demanding notion of implication. According to this perspective, a genuine implication should involve some form of coherence or compatibility between antecedent and consequent.

8.2 Vocabularies and Languages

The vocabulary and formal language $\mathbf{Fo}_{W0}$ and $\mathbf{Fo}_{W0}$ will be defined as those of $\mathbf{S0}$.

Definition 8.1.

$$\mathbf{Vo}_{W0} \stackrel{\text{DF}}{=} \mathbf{Vo}_{S0}. \quad \mathbf{Fo}_{W0} \stackrel{\text{DF}}{=} \mathbf{Fo}_{S0}.$$

8.3 Inference and Consequence Relations

$W0$ falls among modern systems, so their inference relations are just special cases of [definition 3.31](#).

As for the contents of their consequence relations, we want that all inferences of these forms be valid:

$$/ \neg(\neg A \rightarrow A) \quad (\text{AT1})$$

$$/ \neg(A \rightarrow \neg A) \quad (\text{AT2})$$

$$(A \rightarrow B) / \neg(A \rightarrow \neg B) \quad (\text{BT1})$$

$$(A \rightarrow \neg B) / \neg(A \rightarrow B) \quad (\text{BT2})$$

Since none of these inference forms are valid in S_0 , the classical logic, we say that W_0 is a *contra-classical logic*. A contra-classical logic is a non-classical logic which not reject or restrict some of its theses, but also propose theses which are not part of classical logic.

Moreover, following McCall (1966, p. 416), we do not want all inferences of the following form to be valid in a connexive logic:

$$/ ((A \rightarrow B) \rightarrow (B \rightarrow A)).$$

This inference form is already non-valid in S_0 , so we must only be careful not to define W_0 so that it becomes valid.

For the rest, we want W_0 to be very much like S_0 , as this system is, after all, inspired by many of the intuitions on which classical logic is also inspired.

8.4 Interpretation Systems

The semantics of this system involves possible worlds, and we will not develop it in full.

It is important to remark, however, that this semantics treats truth and falsity conditions independently.

8.5 Derivation Systems

The derivation system originally proposed for $W0$ is an axiomatic one, which is also based on the positive logic. Such system is defined by the following inference forms:

$(A \rightarrow B), A / B$	(MPP)
$/ (A \rightarrow (B \rightarrow A))$	(P1)
$/ ((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$	(P2)
$/ (A \rightarrow (B \rightarrow (A \wedge B)))$	(P3)
$/ ((A \wedge B) \rightarrow A)$	(P4)
$/ ((A \wedge B) \rightarrow B)$	(P5)
$/ (A \rightarrow (A \vee B))$	(P6)
$/ (B \rightarrow (A \vee B))$	(P7)
$/ ((A \rightarrow C) \rightarrow ((B \rightarrow C) \rightarrow ((A \vee B) \rightarrow C)))$	(P8)
$/ (\neg\neg A \leftrightarrow A)$	(DN)
$/ (\neg(A \vee B) \leftrightarrow (\neg A \wedge \neg B))$	(DM1)
$/ (\neg(A \wedge B) \leftrightarrow (\neg A \vee \neg B))$	(DM2)
$/ (\neg(A \rightarrow B) \leftrightarrow (A \rightarrow \neg B))$	(W0)

As usual, $\ulcorner(A \leftrightarrow B)\urcorner$ is an abbreviation of $\ulcorner(A \rightarrow B) \wedge (B \rightarrow A)\urcorner$ in the context of these axioms.

Note that the axioms added to positive logic in this system are satisfied by $S0$, except for (W0), which $W0$'s characteristic axiom. Thus, an axiomatic deductive base for $W0$ would consist of $P0$'s plus (DN), (DM1), (DM2), and (W0).

8.5.1 Fitch-style derivations for $W0$

To present the deductive base $\mathbf{Cal}_{W0}$ for the connexive propositional logic $W0$ in the style of notation 3.48, we will, again, extend that of $\mathbf{Cal}_{P0}$. First, we will add rules corresponding to De Morgan rules and double negation:

Double negation rules.

$a \mid A$	$a \mid \neg\neg A$
$\vdots$	$\vdots$
$\neg\neg A \quad \neg\neg^+, a$	$A \quad \neg\neg^-, a$

De Morgan rules.

$$\begin{array}{c|c}
 a & \neg(A \vee B) \\
 & \vdots \\
 & (\neg A \wedge \neg B) \quad \text{DM}_1^-, a
 \end{array}
 \qquad
 \begin{array}{c|c}
 a & (\neg A \wedge \neg B) \\
 & \vdots \\
 & \neg(A \vee B) \quad \text{DM}_1^+, a
 \end{array}$$

$$\begin{array}{c|c}
 a & \neg(A \wedge B) \\
 & \vdots \\
 & (\neg A \vee \neg B) \quad \text{DM}_2^-, a
 \end{array}
 \qquad
 \begin{array}{c|c}
 a & (\neg A \vee \neg B) \\
 & \vdots \\
 & \neg(A \wedge B) \quad \text{DM}_2^+, a
 \end{array}$$

These rules easily allow us to derive axioms (DN), (DM1), and (DM2). As for axiom (W0), we need to add the following rules:

Connexive rules.

$$\begin{array}{c|c}
 a & A \\
 & \vdots \\
 b & \neg B \\
 & \vdots \\
 & \neg(A \rightarrow B) \quad \rightarrow^+, a-b
 \end{array}
 \qquad
 \begin{array}{c|c}
 a & \neg(A \rightarrow B) \\
 & \vdots \\
 b & A \\
 & \vdots \\
 & \neg B \quad \rightarrow^-, a, b
 \end{array}$$

Thus, we might define $\mathbf{DBas}_{W0}$ as follows:

Definition 8.2 (Deductive base for W0).

$$\mathbf{DBas}_{W0} \stackrel{\text{DF}}{=} \mathbf{DBas}_{P0} \cup \{ \neg\neg^-, \neg\neg^+, \text{DM}_1^-, \text{DM}_2^-, \text{DM}_1^+, \text{DM}_2^+, \rightarrow^-, \rightarrow^+ \}.$$

Note that no specific rules for negation or the falsum were added.

As for the derivation system $\mathbf{Cal}_{W0}$, we will once again assume that the functions $\mathbf{Prem}_{W0}$ and $\mathbf{Conc}_{W0}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 8.3 (Cal_{W0}).

Cal_{W0} is a calculus on $W0$

$$\stackrel{\text{DF}}{=} \text{Cal}_{W0} = \langle \text{DBas}_{W0}, \text{Der}_{W0}, \text{Prem}_{W0}, \text{Conc}_{W0}, \vdash_{W0} \rangle$$

: DBas_{W0} is a deductive base on $\text{Fo}_{W0} \mathrel{\mathcal{M}}$

$\text{Der}_{W0} = \{D : D \text{ is a Fitch derivation on } \text{Fo}_{W0} \text{ from } \text{DBas}_{W0}\} \mathrel{\mathcal{M}}$

$\text{Prem}_{W0} : \text{Der}_{W0} \rightarrow \wp \text{Fo}_{W0} \mathrel{\mathcal{M}}$

$\text{Conc}_{W0} : \text{Der}_{W0} \rightarrow \wp \text{Fo}_{W0} \mathrel{\mathcal{M}}$

$\vdash_S = \{ \langle A, B \rangle : \exists X \in \text{Der}_{W0} (\text{Prem}_{W0}(X) = A \mathrel{\mathcal{M}} \text{Conc}_{W0}(X) = B) \}.$

Task 8.1. Show that all inferences of the forms (DN), (DM1), (DM2), and (W0) are members of $\vdash_{W0}$.

The cases (DN), (DM1), and (DM2) are easy. Let us, then, prove the case of the axiom schema distinctive of $W0$, i.e., (W0).

Theorem 8.4. $\vdash_{W0} (\neg(A \rightarrow B) \leftrightarrow (A \rightarrow \neg B)).$

Proof. Let $A, B \in \text{Fo}_{W0}$ and consider the schematic derivation:

1		$\neg(A \rightarrow B)$	
2			A
3			$\neg B$ $\rightarrow^-, 1, 2$
4		$(A \rightarrow \neg B)$ $\rightarrow^+, 2-3$	
5		$(\neg(A \rightarrow B) \rightarrow (A \rightarrow \neg B))$ $\rightarrow^+, 2-4$	
6			$(A \rightarrow \neg B)$
7			A
8			$\neg B$ $\rightarrow^-, 6, 7$
9		$\neg(A \rightarrow B)$ $\rightarrow^+, 7-8$	
10		$((A \rightarrow \neg B) \rightarrow \neg(A \rightarrow B))$ $\rightarrow^+, 6-9$	
11		$(\neg(A \rightarrow B) \leftrightarrow (A \rightarrow \neg B))$ $\wedge^+, 5, 10$	

Call this schematic derivation 'D'. It is clear, for all $A, B \in \text{Fo}_{W0}$, that D is a Fitch derivation on Fo_{W0} from DBas_{W0} . This means that there is a derivation $D \in \text{Der}_{W0}$ with $\{\} \subseteq \text{Prem}_{W0}(D)$ and $\ulcorner (\neg(A \rightarrow B) \leftrightarrow (A \rightarrow \neg B)) \urcorner \in \text{Conc}_{W0}(D)$. Therefore, by **definition 8.3**, $\vdash_{W0} (\neg(A \rightarrow B) \leftrightarrow (A \rightarrow \neg B)).$ $\square$

Similarly, all instances of (AT1), (AT2), (BT1), and (BT2) are derivable in this Der_{W0} .

Task 8.2. Show that all inferences of the forms (AT1), (AT2), (BT1), and (BT2) are members of $\vdash_{W0}$.

We will prove the cases of (AT1) and (BT1).

Theorem 8.5. $\vdash_{W0} \neg(\neg A \rightarrow A)$

Proof. Let $A \in \mathbf{Fo}_{W0}$ and consider the schematic derivation:

1			$\neg A$	
2			$\neg A$	Reit, 1
3			$\neg(\neg A \rightarrow A)$	$\rightarrow^+, 1-2$

Call this schematic derivation 'D'. It is clear, for all $A \in \mathbf{Fo}_{W0}$, that D is a Fitch derivation on $\mathbf{Fo}_{W0}$ from $\mathbf{DBas}_{W0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{W0}$ with $\{\} \subseteq \mathbf{Prem}_{W0}(\mathbf{D})$ and $\ulcorner \neg(\neg A \rightarrow A) \urcorner \in \mathbf{Conc}_{W0}(\mathbf{D})$. Therefore, by [definition 8.3](#), $\vdash_{W0} \neg(\neg A \rightarrow A)$. $\square$

Theorem 8.6. $(A \rightarrow B) \vdash_{W0} \neg(A \rightarrow \neg B)$

Proof. Let $A, B \in \mathbf{Fo}_{W0}$ and consider the schematic derivation:

1			$(A \rightarrow B)$	
2			A	
3			B	$\rightarrow^-, 1, 2$
4			$\neg\neg B$	$\neg\neg^+, 3$
5			$\neg(A \rightarrow \neg B)$	$\rightarrow^+, 2-4$

Call this schematic derivation 'D'. It is clear, for all $A, B \in \mathbf{Fo}_{W0}$, that D is a Fitch derivation on $\mathbf{Fo}_{W0}$ from $\mathbf{DBas}_{W0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{W0}$ with $\{\ulcorner A \rightarrow B \urcorner\} \subseteq \mathbf{Prem}_{W0}(\mathbf{D})$ and $\ulcorner \neg(A \rightarrow \neg B) \urcorner \in \mathbf{Conc}_{W0}(\mathbf{D})$. Therefore, by [definition 8.3](#), we have that $(A \rightarrow B) \vdash_{W0} \neg(A \rightarrow \neg B)$. $\square$

A curious fact about W_0 is that it is a *contradictory logic*, i.e., it is a logic which has at least one formula A such that both A and $\neg A$ are theorems of W_0 .

Task 8.3. Show that $\vdash_{W_0}$ has contradictory theorems, i.e., show that, for some $A \in \mathbf{Fo}_{W_0}$, the inferences $\vdash A$ and $\vdash \neg A$ are members of $\vdash_{W_0}$.

Theorem 8.7. $\vdash_{W_0}$ has contradictory theorems.

Proof. Let $A \in \mathbf{Fo}_{W_0}$. We show that the formula $\neg(A \wedge \neg A)$ and its negation $\neg\neg(A \wedge \neg A)$ are derivable in $\vdash_{W_0}$.

For the first, consider the schematic derivation:

1			$(A \wedge \neg A)$	
2			A	$\wedge_1^-, 1$
3			$((A \wedge \neg A) \rightarrow A)$	$\rightarrow^+, 1-2$

Call this schematic derivation 'D1'. It is clear, for all $A \in \mathbf{Fo}_{W_0}$, that **D1** is a Fitch derivation on $\mathbf{Fo}_{W_0}$ from $\mathbf{DBas}_{W_0}$. This means that there is a derivation $\mathbf{D1} \in \mathbf{Der}_{W_0}$ with $\{\}$ $\subseteq \mathbf{Prem}_{W_0}(\mathbf{D1})$ and $\neg((A \wedge \neg A) \rightarrow A) \in \mathbf{Conc}_{W_0}(\mathbf{D1})$. Hence, by [definition 8.3](#), we have that $\vdash_{W_0} ((A \wedge \neg A) \rightarrow A)$.

For the contradictory, consider this other schematic derivation:

1			$(A \wedge \neg A)$	
2			$\neg A$	$\wedge_2^-, 1$
3			$\neg((A \wedge \neg A) \rightarrow A)$	$\neg\rightarrow^+, 1-2$

Call this schematic derivation 'D2'. It is clear, for all $A \in \mathbf{Fo}_{W_0}$, that **D2** is a Fitch derivation on $\mathbf{Fo}_{W_0}$ from $\mathbf{DBas}_{W_0}$. This means that there is a derivation $\mathbf{D2} \in \mathbf{Der}_{W_0}$ with $\{\}$ $\subseteq \mathbf{Prem}_{W_0}(\mathbf{D2})$ and $\neg((A \wedge \neg A) \rightarrow A) \in \mathbf{Conc}_{W_0}(\mathbf{D2})$. Hence, by [definition 8.3](#), we have that $\vdash_{W_0} \neg((A \wedge \neg A) \rightarrow A)$.

Note that $\neg((A \wedge \neg A) \rightarrow A)$ and $\neg\neg((A \wedge \neg A) \rightarrow A)$, both members of W_0 , are mutually contradictory. Therefore, $\vdash_{W_0}$ has contradictory theorems. $\square$

Despite this inconsistency, the system is non-trivial.

Part III

NON-STANDARD LANGUAGES

9.1 Kind of definition

First, we may distinguish between systems whose languages (i.e. sets of wffs) are **defined recursively** and those whose languages are **defined explicitly**.

This distinction is not common, since formal languages used in logic are normally defined recursively.

However, as we previously saw, the formal language of Aristotelian syllogistic can be defined as a finite set of categorical propositions. Thus, it is basically syllogistic languages vs modern logical languages.

Let us consider some non-standard logical symbols. That is, symbols featured in formal languages used in logic, which are not part of standard languages.

10.1 Logical value constants

Some languages include constant for logical values. For instance, it is not uncommon to have languages with '**1**' as a propositional constant for representing *truth* and '**0**' for *falsity*.

We may define one such enhancement S of S_0 as follows:

Preliminary 10.1 (Formulae with logical value constants). The set $\mathbf{Fo}_S$ is recursively defined as follows:

1. $p \in \mathbf{Non}_S \cup \{\mathbf{0}, \mathbf{1}\} \Rightarrow p \in \mathbf{X}$.
2. $p \in \mathbf{X} \Rightarrow \ulcorner \neg p \urcorner \in \mathbf{X}$.
3. $p, q \in \mathbf{X} \Rightarrow \ulcorner (p \vee q) \urcorner, \ulcorner (p \wedge q) \urcorner, \ulcorner (p \rightarrow q) \urcorner, \ulcorner (p \leftrightarrow q) \urcorner \in \mathbf{X}$.
4. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–3.

We might define ' $\mathbf{Fo}_S$ ' more shortly using ' $\stackrel{\text{DF}}{=}$ '.

Definition 10.2 (Formulae with logical value constants).

$$\begin{aligned} \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \mathbf{Non}_S \cup \{\mathbf{0}, \mathbf{1}\} \cup \\ &\quad \{\ulcorner \bullet p \urcorner : p \in \mathbf{Fo}_S \ \& \ \bullet \in \{\ulcorner \neg \urcorner, \ulcorner \square \urcorner, \ulcorner \diamond \urcorner\}\} \cup \\ &\quad \{\ulcorner (p \heartsuit q) \urcorner : p, q \in \mathbf{Fo}_S \ \& \ \heartsuit \in \mathbf{Con}\}. \end{aligned}$$

The sense of an expression featuring these kinds of logical symbols is difficult to translate into natural language. The following are examples, two of which are expressed in a language different to that defined in [definition 10.2](#).

1. If p , then the true.
Representation: ' $(p \rightarrow \mathbf{1})$ '.
2. If the false, then I sing like Freddy Mercury.
Representation: ' $(\mathbf{0} \rightarrow q)$ '.
3. If every proposition is true, then every proposition is false.
Representation 1: ' $((p \leftrightarrow \mathbf{0}) \rightarrow (p \leftrightarrow \mathbf{1}))$ '.
Representation 2: ' $\forall p (p = \mathbf{0} \rightarrow p = \mathbf{1})$ '.
Representation 3: ' $(\forall p (p = \mathbf{0}) \rightarrow \forall p (p = \mathbf{1}))$ '.

This may also be done of any other logical value used for constructing interpretation in for a logical system.

10.2 Modal operators

Modal logic is nowadays a standard extension of classical logic, providing formal tools to reason about notions such as necessity, possibility, knowledge, time, obligation, and location.

There is a wide variety of modal systems, each motivated by different applications, and often equipped with distinct symbols. Below we present a non-exhaustive overview of some of the most commonly used modal operators and their intended readings.

10.2.1 Alethic Modality

We may speak about possibilities, contingencies, necessities, impossibilities, and the like, in modal languages. We commonly use ‘ $\Box$ ’ for expressing necessity and ‘ $\Diamond$ ’ for possibility. A modal 0th-order formal language $\mathbf{Fo}_S$ may be constructed as follows.

Preliminary 10.3 (Formulae of propositional modal logic). The set $\mathbf{Fo}_S$ is recursively defined as follows:

1. $p \in \mathbf{Non}_S \Rightarrow p \in \mathbf{X}$.
2. $p \in \mathbf{X} \Rightarrow \ulcorner \neg p \urcorner, \ulcorner \Box p \urcorner, \ulcorner \Diamond p \urcorner \in \mathbf{X}$.
3. $p, q \in \mathbf{X} \Rightarrow \ulcorner (p \vee q) \urcorner, \ulcorner (p \wedge q) \urcorner, \ulcorner (p \rightarrow q) \urcorner, \ulcorner (p \leftrightarrow q) \urcorner \in \mathbf{X}$.
4. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–3.

We might define ‘ $\mathbf{Fo}_S$ ’ more shortly using ‘ $\stackrel{\text{DF}}{=}$ ’.

Definition 10.4 (Formulae of propositional modal logic).

$$\begin{aligned} \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \mathbf{Non}_S \cup \\ &\quad \{ \ulcorner \bullet p \urcorner : p \in \mathbf{Fo}_S \ \& \ \bullet \in \{ \neg, \Box, \Diamond \} \} \cup \\ &\quad \{ \ulcorner (p \heartsuit q) \urcorner : p, q \in \mathbf{Fo}_S \ \& \ \heartsuit \in \mathbf{Con} \}. \end{aligned}$$

Table 10.1: Alethic modal formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Necessity	$\Box p$	It is necessary that Peirce is a logician. It is necessarily the case that Peirce is a logician. In all possible worlds, Peirce is a logician.
Possibility	$\Diamond p$	It is possible that Peirce is a logician. It is possibly the case that Peirce is a logician. In some possible world, Peirce is a logician.
Impossibility	$\neg \Diamond p$	It is not possible that Peirce is a logician. It is impossible that Peirce is a logician.
Contingency	$(\Diamond p \wedge \Diamond \neg p)$	It is contingent that Peirce is a logician.

The informal sense of usual modal expressions is summarised in [Table 10.1](#).

Task 10.1. Formalise the following modal expressions.

1. Necessarily, either it rains now or it does not rain now.
2. Either it necessarily rains now or it necessarily does not rain now.
3. It is necessary that, if you are a bachelor, then you are unmarried.
4. If you are necessarily human, then you are possibly an animal.

Solution to task 10.1.

1. Necessarily, either it rains now or it does not rain now.

Representation: ' $\Box(r \vee \neg r)$ '.

2. Either it necessarily rains now or it necessarily does not rain now.

Representation: ' $(\Box r \vee \Box \neg r)$ '.

3. It is necessary that, if you are a bachelor, then you are unmarried.

Representation: ' $\Box(p \rightarrow q)$ '.

4. If you are necessarily an animal, then you are possibly human.

Representation: ' $(\Box p \rightarrow \Diamond q)$ '.

It is important to note that ' $\Box$ ' and ' $\Diamond$ ' may also be used in non-alethic modal senses, which include epistemic, deontic, and temporal senses, among others.

10.2.2 Epistemic modality

Epistemic expressions serve to express that a given agent knows, believes, or has some epistemic attitude towards some statement.

An epistemic language may use a modal epistemic operator 'K', so that a modal 0th-order epistemic language $\mathbf{Fo}_S$ may be constructed as follows.

Preliminary 10.5 (Formulae of propositional epistemic logic). The set $\mathbf{Fo}_S$ is recursively defined as follows:

1. $p \in \mathbf{Non}_S \Rightarrow p \in \mathbf{X}$.
2. $p \in \mathbf{X} \Rightarrow \ulcorner \neg p \urcorner, \ulcorner Kp \urcorner \in \mathbf{X}$.
3. $p, q \in \mathbf{X} \Rightarrow \ulcorner (p \vee q) \urcorner, \ulcorner (p \wedge q) \urcorner, \ulcorner (p \rightarrow q) \urcorner, \ulcorner (p \leftrightarrow q) \urcorner \in \mathbf{X}$.
4. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–3.

We might define ' $\mathbf{Fo}_S$ ' more shortly using ' $\stackrel{\text{DF}}{=}$ '.

Definition 10.6 (Formulae of propositional epistemic logic).

$$\begin{aligned} \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \mathbf{Non}_S \cup \\ &\quad \{ \ulcorner \bullet p \urcorner : p \in \mathbf{Fo}_S \wedge \bullet \in \{ \ulcorner \neg \urcorner, \ulcorner K \urcorner \} \} \cup \\ &\quad \{ \ulcorner (p \heartsuit q) \urcorner : p, q \in \mathbf{Fo}_S \wedge \heartsuit \in \mathbf{Con} \}. \end{aligned}$$

The informal sense of usual modal expressions is summarised in [Table 10.2](#).

Task 10.2. Formalise the following epistemic expressions.

1. Either we know that it rains now or we do not know that it rains now.
2. Either we know that it rains now or we do know that it does not rains now.
3. We know that, if it God is known to exist, then life has a divine origin.
4. We know that, if it God is known to exist, then we know that life has a divine origin.

Table 10.2: Epistemic formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Positive knowledge	Kp	It is known that Peirce is a logician. We know that Peirce is a logician.
Negative knowledge	$K\neg p$	It is known that Peirce is not a logician. We know that Peirce is not a logician.
Ignorance	$\neg Kp$	It is not known that Peirce is a logician. We ignore if Peirce is a logician (he may be or may not be).

Solution to task 10.2.

1. Either we know that it rains now or we do not know that it rains now.

Representation: ' $(Kr \vee \neg Kr)$ '.

2. Either we know that it rains now or we do know that it does not rains now.

Representation: ' $(Kr \vee K\neg r)$ '.

3. We know that, if it God is known to exist, then life has a divine origin.

Representation: ' $K(Kp \rightarrow q)$ '.

4. We know that, if it God is known to exist, then we know that life has a divine origin.

Representation: ' $K(Kp \rightarrow Kq)$ '.

Other kinds of epistemic attitudes, like belief or entertainment, may expressed with similar languages.

Moreover, we may index an epistemic operator, so that ' $K_a p$ ' means that agent a knows that p .

10.2.3 Deontic Modality

Deontic expressions serve to express obligations or permission in relation to the state of affairs denoted by some statement.

An deontic language may use a modal deontic operators 'O', 'P', so that a modal 0th-order deontic language $\mathbf{Fo}_S$ may be constructed as follows.

Preliminary 10.7 (Formulae of propositional deontic logic). The set $\mathbf{Fo}_S$ is recursively defined as follows:

1. $p \in \mathbf{Non}_S \Rightarrow p \in \mathbf{X}$.
2. $p \in \mathbf{X} \Rightarrow \ulcorner \neg p \urcorner, \ulcorner Op \urcorner, \ulcorner Pp \urcorner \in \mathbf{X}$.
3. $p, q \in \mathbf{X} \Rightarrow \ulcorner (p \vee q) \urcorner, \ulcorner (p \wedge q) \urcorner, \ulcorner (p \rightarrow q) \urcorner, \ulcorner (p \leftrightarrow q) \urcorner \in \mathbf{X}$.
4. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–3.

We might define ' $\mathbf{Fo}_S$ ' more shortly using ' $\stackrel{\text{DF}}{=}$ '.

Definition 10.8 (Formulae of propositional deontic logic).

$$\begin{aligned} \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \mathbf{Non}_S \cup \\ &\quad \{ \ulcorner \bullet p \urcorner : p \in \mathbf{Fo}_S \ \& \ \bullet \in \{ \ulcorner \neg \urcorner, \ulcorner O \urcorner, \ulcorner P \urcorner \} \} \cup \\ &\quad \{ \ulcorner (p \heartsuit q) \urcorner : p, q \in \mathbf{Fo}_S \ \& \ \heartsuit \in \mathbf{Con} \}. \end{aligned}$$

The informal sense of usual modal expressions is summarised in [Table 10.3](#).

Task 10.3. Formalise the following deontic expressions.

1. It is permitted both to stay and not to stay.
2. It is obligatory both to stay and not to stay.
3. If it is not obligatory to stay, then it is permitted not to stay.
4. It is mandatory that it be permitted to stay.

Table 10.3: Epistemic formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Obligation	Op	It is obligatory that the door be open.
		It is mandatory that the door be open.
	$O\neg p$	It is obligatory that the door be not open.
	$\neg Op$	It is not obligatory that the door be open.
Permission	Pp	It is permitted that the door be open.
	$P\neg p$	It is permitted that the door be not open.
	$\neg Pp$	It is not permitted that the door be open.

Solution to task 10.3.

1. It is permitted to stay and it is permitted not to stay.

Representation: ' $(Pp \wedge P\neg p)$ '.

2. It is obligatory both to stay and not to stay.

Representation: ' $O(p \wedge \neg p)$ '.

3. If it is not obligatory to stay, then it is permitted not to stay.

Representation: ' $(\neg Op \rightarrow P\neg p)$ '.

4. It is mandatory that it be permitted to stay and not to stay.

Representation: ' $OP(p \wedge \neg p)$ '.

10.2.4 Temporal Modality

Temporal or tense-relative expressions serve to express statements in relation to time.

A temporal or tense language may use a modal tense operators 'G', 'H', so that a modal 0th-order temporal language $\mathbf{Fo}_S$ may be constructed as follows.

Preliminary 10.9 (Formulae of propositional temporal logic). The set $\mathbf{Fo}_S$ is recursively defined as follows:

1. $p \in \mathbf{Non}_S \Rightarrow p \in \mathbf{X}$.
2. $p \in \mathbf{X} \Rightarrow \neg p, Gp, Hp \in \mathbf{X}$.
3. $p, q \in \mathbf{X} \Rightarrow (p \vee q), (p \wedge q), (p \rightarrow q), (p \leftrightarrow q) \in \mathbf{X}$.
4. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–3.

We might define ' $\mathbf{Fo}_S$ ' more shortly using ' $\stackrel{\text{DF}}{=}$ '.

Definition 10.10 (Formulae of propositional temporal logic).

$$\begin{aligned} \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \mathbf{Non}_S \cup \\ &\quad \{ \neg p : p \in \mathbf{Fo}_S \} \cup \{ Gp, Hp : p \in \mathbf{Fo}_S \} \cup \\ &\quad \{ (p \vee q), (p \wedge q), (p \rightarrow q), (p \leftrightarrow q) : p, q \in \mathbf{Fo}_S \}. \end{aligned}$$

The informal sense of usual modal expressions is summarised in [Table 10.4](#).

Task 10.4. Formalise the following temporal expressions.

1. Peter was German and will always be German.
2. Peter was French, but will no longer be.
3. If you were a comedian, you will always be a comedian..
4. If George is French, then he will always be.

Table 10.4: Temporal formulae and their natural-language interpretations.

Type	Formula	Natural language sentence
Future	Gp	It always will be the case, that the door is open.
	$\neg G\neg p$	Not always will be the case, that the door is not open.
		It will be the case that the door is open.
Past	Hp	It always was the case, that the the door is open.
	$\neg H\neg p$	Not always was the case, that the door is not open.
		It was the case that the door is open.

Solution to task 10.4.

1. Peter was German and will always be German.

Representation: $(\neg H\neg p \wedge Gp)$.

2. Peter was French, but will no longer be.

Representation: $(\neg H\neg p \wedge G\neg p)$.

3. If you were always a comedian, you will be a comedian.

Representation: $(Hp \rightarrow \neg G\neg p)$.

4. If George is French, then he will always be.

Representation: $(p \rightarrow Gp)$.

10.2.5 Remark

A given modal operator may receive different semantic interpretations depending on the underlying logical framework (e.g. Kripke semantics, neighbourhood semantics, topological semantics), and different modal logics may validate different principles. Consequently, modal operators should always be understood relative to the specific system and application in which they are employed.

10.3 Term-forming symbols

In this category we have several symbols. The following are the most usual examples.

10.3.1 The definite descriptor

A definite description is an expression whose intended referent is the unique entity satisfying some property. We do so in the context of a first-order language, and construct singular terms with ‘ l ’ of the form ‘ lxF ’.

For our vocabularies, we will use those of $S1$. Thus, a first-order language $\mathbf{Fo}_S$ with a definite descriptor with ‘ l ’ may be constructed as follows.

Preliminary 10.11 (Formulae with definite descriptors). Where $n \in \mathbf{Num}_{\mathbb{W}}$, the set $\mathbf{Fo}_S$ is recursively defined as follows:

1. $t \in \mathbf{ICon}_{S1} \cup \mathbf{IVar}_{S1} \Rightarrow t \in \mathbf{Ter}_S$.
2. $F \in \mathbf{X} \Rightarrow \ulcorner \neg F \urcorner \in \mathbf{X}$.
3. $F, G \in \mathbf{X} \Rightarrow \ulcorner (F \vee G) \urcorner, \ulcorner (F \wedge G) \urcorner, \ulcorner (F \rightarrow G) \urcorner, \ulcorner (F \leftrightarrow G) \urcorner \in \mathbf{X}$.
4. $F \in \mathbf{X} \ \& \ x \in \mathbf{IVar}_{S1} \Rightarrow \ulcorner lxF \urcorner \in \mathbf{Ter}_S$.
5. $P \in \mathbf{Pre}_n \ \& \ t_1, \dots, t_n \in \mathbf{Ter}_S \Rightarrow Pt_1 \dots t_n \in \mathbf{X} : n \in \mathbb{N}$.
6. $F \in \mathbf{X} \ \& \ x \in \mathbf{IVar}_{S1} \Rightarrow \ulcorner \exists x F \urcorner, \ulcorner \forall x F \urcorner \in \mathbf{X}$.
7. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–6.

As with ‘ $\mathbf{Fo}_{S0}$ ’, we might define ‘ $\mathbf{Fo}_S$ ’ more shortly using ‘ $\stackrel{\text{DF}}{=}$ ’.

Definition 10.12 (Formulae with definite descriptors).

$$\begin{aligned}
 \mathbf{Ter}_S &\stackrel{\text{DF}}{=} \mathbf{ICon}_{S1} \cup \mathbf{IVar}_{S1} \cup \{ \ulcorner lxF \urcorner : F \in \mathbf{Fo}_S \ \& \ x \in \mathbf{IVar}_{S1} \} \\
 \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \{ Pt_1 \dots t_n : P \in \mathbf{Pre}_S[n] \ \& \ t_1 \dots t_n \in \mathbf{Ter}_S \} \cup \\
 &\quad \{ \ulcorner \neg F \urcorner : F \in \mathbf{Fo}_S \} \cup \\
 &\quad \{ \ulcorner (F \heartsuit G) \urcorner : F, G \in \mathbf{Fo}_S \ \& \ \heartsuit \in \mathbf{Con}_{S1}[2] \} \cup \\
 &\quad \{ \ulcorner \mathbf{O}xF \urcorner : \mathbf{O} \in \mathbf{Qua}_{S1} \ \& \ x \in \mathbf{IVar}_{S1} \ \& \ F \in \mathbf{Fo}_S \} \\
 &\quad : n \in \mathbb{N}
 \end{aligned}$$

An expression like ' $\iota x A(x)$ ' denotes the unique x such that $A(x)$. However, this expression is not a formula: it cannot be interpreted as true or false.

Task 10.5. Provide natural language expressions with the structure of the following formulae with definite descriptors:

- | | |
|------------------------------------|---|
| 1. ' $Q^1 \iota x P x$ '. | Regimented: ' $Q^1(\iota x P(x))$ '. |
| 2. ' $R^2 \iota x P^2 x a b$ '. | Regimented: ' $R^2(\iota x P^2(x, a), b)$ '. |
| 3. ' $\neg P^1 \iota x S^2 x b$ '. | Regimented: ' $\neg P^1(\iota x S^2(x, b))$ '. |
| 4. ' $R^2 a \iota x P^1 x$ '. | Regimented: ' $R^2(a, \iota x P^1(x))$ '. |

Solution to task 10.5.

- | | |
|---|---|
| 1. ' $Q^1 \iota x P x$ '. | Regimented: ' $Q^1(\iota x P(x))$ '. |
| Natural language expression: | |
| The king of France is bald. | |
| 2. ' $R^2 \iota x P^2 x a b$ '. | Regimented: ' $R^2(\iota x P^2(x, a), b)$ '. |
| Natural language expression: | |
| The editor of <i>Mind</i> rejected the paper. | |
| 3. ' $\neg P^1 \iota x S^2 x b$ '. | Regimented: ' $\neg P^1(\iota x S^2(x, b))$ '. |
| Natural language expression: | |
| Bertha's unique sister is not Prussian. | |
| 4. ' $R^2 a \iota x P^1 x$ '. | Regimented: ' $R^2(a, \iota x P^1(x))$ '. |
| Natural language expression: | |
| Ana loves the author of <i>Hamlet</i> . | |

10.3.2 The indefinite descriptor

An indefinite description is an expression whose intended referent is some (not necessarily unique) entity satisfying a given property, if there is one. We introduce such expressions in a first-order language by means of Hilbert's ε -operator, which forms singular terms with ' ε ' of the form $\ulcorner \varepsilon x F \urcorner$.

As before, we work with the vocabulary of $S1$. Thus, a first-order language $\mathbf{Fo}_S$ with an indefinite descriptor using ε may be constructed as follows.

Preliminary 10.13 (Formulae with indefinite descriptors). Where $n \in \mathbf{Num}_{\mathbb{W}}$, the set $\mathbf{Fo}_S$ is recursively defined as follows:

1. $t \in \mathbf{ICon}_{S1} \cup \mathbf{IVar}_{S1} \Rightarrow t \in \mathbf{Ter}_S$.
2. $F \in \mathbf{X} \Rightarrow \ulcorner \neg F \urcorner \in \mathbf{X}$.
3. $F, G \in \mathbf{X} \Rightarrow \ulcorner (F \vee G) \urcorner, \ulcorner (F \wedge G) \urcorner, \ulcorner (F \rightarrow G) \urcorner, \ulcorner (F \leftrightarrow G) \urcorner \in \mathbf{X}$.
4. $F \in \mathbf{X} \ \& \ x \in \mathbf{IVar}_{S1} \Rightarrow \ulcorner \varepsilon x F \urcorner \in \mathbf{Ter}_S$.
5. $P \in \mathbf{Pre}_n \ \& \ t_1, \dots, t_n \in \mathbf{Ter}_S \Rightarrow Pt_1 \dots t_n \in \mathbf{X} : n \in \mathbb{N}$.
6. $F \in \mathbf{X} \ \& \ x \in \mathbf{IVar}_{S1} \Rightarrow \ulcorner \exists x F \urcorner, \ulcorner \forall x F \urcorner \in \mathbf{X}$.
7. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–6.

As with $\mathbf{Fo}_{S0}$, we may also present the formation rules more succinctly using $\stackrel{\text{DF}}{=}$.

Definition 10.14 (Formulae with indefinite descriptors).

$$\begin{aligned} \mathbf{Ter}_S &\stackrel{\text{DF}}{=} \mathbf{ICon}_{S1} \cup \mathbf{IVar}_{S1} \cup \{ \ulcorner \varepsilon x F \urcorner : F \in \mathbf{Fo}_S \ \& \ x \in \mathbf{IVar}_{S1} \} \\ \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \{ Pt_1 \dots t_n : P \in \mathbf{Pres}[n] \ \& \ t_1 \dots t_n \in \mathbf{Ter}_S \} \cup \\ &\quad \{ \ulcorner \neg F \urcorner : F \in \mathbf{Fo}_S \} \cup \\ &\quad \{ \ulcorner (F \heartsuit G) \urcorner : F, G \in \mathbf{Fo}_S \ \& \ \heartsuit \in \mathbf{Con}_{S1}[2] \} \cup \\ &\quad \{ \ulcorner \bullet x F \urcorner : \bullet \in \mathbf{Qua}_{S1} \ \& \ x \in \mathbf{IVar}_{S1} \ \& \ F \in \mathbf{Fo}_S \} \\ &\quad : n \in \mathbb{N} \end{aligned}$$

An expression like ' $\epsilon x A(x)$ ' denotes some (not necessarily unique) x such that $A(x)$, if such an x exists; otherwise it denotes an arbitrary object in the domain.

Task 10.6. Provide natural language expressions with the structure of the following formulae with definite descriptors:

- | | |
|---------------------------------------|--|
| 1. ' $S^1 \epsilon x P^1 x$ '. | Regimented: ' $S^1(\epsilon x P^1(x))$ '. |
| 2. ' $S^1 \epsilon x P^1 x$ '. | Regimented: ' $S^1(\epsilon x P^1(x))$ '. |
| 3. ' $\neg P^1 \epsilon x Q^2 x c$ '. | Regimented: ' $\neg P^1(\epsilon x Q^2(x, c))$ '. |
| 4. ' $P^2 \epsilon x Q^2 x a a$ '. | Regimented: ' $P^2(\epsilon x Q^2(x, a), a)$ '. |

Solution to task 10.6.

- | | |
|--|--|
| 1. ' $S^1 \epsilon x P^1 x$ '. | Regimented: ' $S^1(\epsilon x P^1(x))$ '. |
| Natural language expression: | |
| Whoever is president is sly. | |
| 2. ' $\neg Q^1 \epsilon x S x$ '. | Regimented: ' $\neg Q^1(\epsilon x S(x))$ '. |
| Natural language expression: | |
| Whoever stole this is not the Queen. | |
| 3. ' $\neg P^1 \epsilon x Q^2 x c$ '. | Regimented: ' $\neg P^1(\epsilon x Q^2(x, c))$ '. |
| Natural language expression: | |
| Whoever witnessed the crime is not around. | |
| 4. ' $P^2 \epsilon x Q^2 x a a$ '. | Regimented: ' $P^2(\epsilon x Q^2(x, a), a)$ '. |
| Natural language expression: | |
| The person who called Andrew is after him. | |

There is a remarkable logical system that uses ' ϵ ' and is commonly referred to as ϵ -calculus. In the ϵ -calculus, we usually do not have quantifiers as primitive symbols. We rather define the existential quantifier as follows:

$$\begin{aligned} \exists x P x &\stackrel{\text{df}}{=} P \epsilon x P x \\ &P(\epsilon x P(x)) && \text{(regimented)} \\ &: P \in \mathbf{Pre}_1 \end{aligned}$$

Can you make intuitive sense of that?

10.3.3 The lambda abstractor

The lambda abstractor ' λ ' is an operator for forming complex predicate-terms from formulae. Intuitively, while the operators ' ι ' and ' ε ' form *singular terms* intended to denote objects, the lambda abstractor ' λ ' forms *predicate-terms* intended to denote properties or relations.

We introduce it in the form of expressions ' $\ulcorner \lambda x F \urcorner$ '. Such expressions are treated as predicate-expressions which can be applied to terms in the usual way.

As before, we work in the vocabulary of $S1$. Thus, we extend the language $\mathbf{Fo}_{S1}$ to a language $\mathbf{Fo}_S$ with the lambda-abstractor as follows.

Preliminary 10.15 (Formulae with the lambda abstractor).
Where $n \in \mathbf{Num}_{\mathbb{W}}$, the set $\mathbf{Fo}_{S1}$ is recursively defined as follows:

1. $P \in (\mathbf{Pre}_n \cup \mathbf{Abs}_n) \ \& \ t_1, \dots, t_n \in \mathbf{ITer}_{S1} \Rightarrow P t_1 \dots t_n \in \mathbf{X}$
2. $F \in \mathbf{X} \Rightarrow \ulcorner \neg F \urcorner \in \mathbf{X}$. $: n \in \mathbb{N}$.
3. $F, G \in \mathbf{X} \Rightarrow \ulcorner (F \vee G) \urcorner, \ulcorner (F \wedge G) \urcorner, \ulcorner (F \rightarrow G) \urcorner, \ulcorner (F \leftrightarrow G) \urcorner \in \mathbf{X}$.
4. $F \in \mathbf{X} \ \& \ x_1, \dots, x_n \in \mathbf{IVar}_{S1} \Rightarrow \ulcorner \lambda x_1 \dots x_n F \urcorner \in \mathbf{Abs}_n$.
5. $F \in \mathbf{X} \ \& \ x \in \mathbf{IVar}_{S1} \Rightarrow \ulcorner \exists x F \urcorner, \ulcorner \forall x F \urcorner \in \mathbf{X}$.
6. $\mathbf{Fo}_S$ is the intersection of all sets $\mathbf{X}$ satisfying 1–5.

As with ι - and ε -expressions, we may give a more compact definition.

Definition 10.16 (Formulae with the lambda abstractor).

$$\begin{aligned} \mathbf{Abs}_n &\stackrel{\text{DF}}{=} \{ \ulcorner \lambda x_1 \dots x_n F \urcorner : F \in \mathbf{X} \ \& \ x_1, \dots, x_n \in \mathbf{IVar}_{S1} \} \\ \mathbf{Fo}_S &\stackrel{\text{DF}}{=} \{ P t_1 \dots t_n : P \in (\mathbf{Pre}_n \cup \mathbf{Abs}_n) \ \& \ t_1 \dots t_n \in \mathbf{ITer}_{S1} \} \cup \\ &\quad \{ \ulcorner \neg F \urcorner : F \in \mathbf{Fo}_S \} \cup \\ &\quad \{ \ulcorner (F \heartsuit G) \urcorner : F, G \in \mathbf{Fo}_S \ \& \ \heartsuit \in \mathbf{Con}_{S1}[2] \} \cup \\ &\quad \{ \ulcorner \bullet x F \urcorner : \bullet \in \mathbf{Qua}_{S1} \ \& \ x \in \mathbf{IVar}_{S1} \ \& \ F \in \mathbf{Fo}_S \} : n \in \mathbb{N} \end{aligned}$$

A *lambda abstract* is an expression of the form ' $\ulcorner \lambda x_1 \dots x_n F \urcorner$ ', which denotes the property of being an x such that F holds.

Task 10.7. Provide natural language expressions having the form of the following lambda abstracts:

- | | |
|---------------------------------------|---|
| 1. $\lambda x P^1 x$. | Regimented: $\lambda x P^1(x)$. |
| 2. $\lambda x \neg P^1 x$. | Regimented: $\lambda x \neg P^1(x)$. |
| 3. $\lambda x Q^2 x a$. | Regimented: $\lambda x Q^2(x, a)$. |
| 4. $\lambda x Q^2 a x$. | Regimented: $\lambda x Q^2(a, x)$. |
| 5. $\lambda x (P^1 x \wedge Q^1 x)$. | Regimented: $\lambda x (P^1(x) \wedge Q^1(x))$. |

Solution to task 10.7.

- | | |
|---|---|
| 1. $\lambda x P^1 x$. | Regimented: $\lambda x P^1(x)$. |
| Natural language expression: | |
| The property of being a president. | |
| 2. $\lambda x \neg P^1 x$. | Regimented: $\lambda x \neg P^1(x)$. |
| Natural language expression: | |
| The property of not being a president. | |
| 3. $\lambda x Q^2 x a$. | Regimented: $\lambda x Q^2(x, a)$. |
| Natural language expression: | |
| The property of loving Andrew. | |
| 4. $\lambda x Q^2 a x$. | Regimented: $\lambda x Q^2(a, x)$. |
| Natural language expression: | |
| The property of being loved by Andrew. | |
| 5. $\lambda x (P^1 x \wedge Q^1 x)$. | Regimented: $\lambda x (P^1(x) \wedge Q^1(x))$. |
| Natural language expression: | |
| The property of being both a philosopher and a mathematician. | |

Lambda abstraction allows us to apply complex predicates to a singular term as if it were a single predicate.

Task 10.8. Formalise the following natural language expressions using the corresponding lambda abstracts used in **task 10.7**:

1. Andrew is president.
2. Andrew is not a president.
3. Bertha loves Andrew.
4. Andrew loves Bertha.
5. Bertha is a philosopher and a mathematician.

Solution to task 10.8.

1. Andrew is president.

Representation: $\lambda x P^1 x a$.

Regimented: $(\lambda x P^1(x))(a)$.

2. Andrew is not a president.

Representation: $\lambda x \neg P^1 x a$.

Regimented: $(\lambda x \neg P^1(x))(a)$.

3. Bertha loves Andrew.

Representation: $\lambda x Q^2 x a b$.

Regimented: $(\lambda x Q^2(x, a))(b)$.

4. Andrew loves Bertha.

Representation: $\lambda x Q^2 a x b$.

Regimented: $(\lambda x Q^2(a, x))(b)$.

5. Bertha is a philosopher and a mathematician.

Representation: $\lambda x (P^1 x \wedge Q^1 x) b$.

Regimented: $\lambda x (P^1(x) \wedge Q^1(x))(b)$.

The length of the expressions of a formal language may also be a criterion for classifying formal language. The concept of the length of an expression e , or $|e|$, may be defined as follows.

Definition 11.1 (Length of an expression).

$$\begin{aligned} |c| &\stackrel{\text{DF}}{=} 1 \\ |ef| &\stackrel{\text{DF}}{=} |e| + |f| \\ &\quad : c \text{ is a character.} \end{aligned}$$

Note the following corollary:

Corollary 11.2. $|\epsilon| \stackrel{\text{DF}}{=} 0$.

Although expressions in principle have a length equal to some natural language, it is possible to think about expressions consisting of an infinitely long chain of characters. Think, e.g., of the number π expressed in decimal notation.

Based on this, we may distinguish between languages which only admit finitely long formulae (*finitary languages*) and languages which also admit infinitely long formulae (i.e., *infinitary languages*).

Definition 11.3 (Finitary language). A formal language $\mathbf{Fo}_S$ is **finitary** $\stackrel{\text{DF}}{=} \forall e \in \mathbf{Fo}_S \exists n \in \mathbb{N} : |e| \leq n$.

Definition 11.4 (Infinitary language). A formal language $\mathbf{Fo}_S$ is **infinitary** $\stackrel{\text{DF}}{=} \exists e \in \mathbf{Fo}_S \forall n \in \mathbb{N} : |e| > n$.

Most languages studied in this textbook, in particular $\mathbf{Fo}_{S0}$, $\mathbf{Fo}_{S1}$, $\mathbf{Fo}_A$, are finitary. Among finitary languages, we might distinguish between languages with formulae of limited length (which is the case of $\mathbf{Fo}_A$, all of whose formulae have length 3), and of unlimited or arbitrary length (which is the case of all modern languages which are finitary).

On the other hand, infinitary logics admit infinitely long formulae. Among such logics we find, for example, logics with infinitary relations, i.e., relations of infinite arity. In the case of infinitary languages, it is not possible to write down their formulae of infinite length, but it is possible to formulate certain rules of inference that apply to them.

Some may think that standard S1 is an infinitary logic in disguise. Consider the formula ' $\forall x Px$ ' as an abbreviation of the following

infinitely long formula:

$$Pa_1 \wedge Pa_2 \wedge \dots,$$

where each a_i is an individual constant of our language, and all individual constants of our language correspond to some a_i .

Despite the possible equivalence between those formulae in an infinitary logic, ' $\forall x Px$ ' itself is a formula of finite length.

Infinitary logics break the condition that a string should be a finite sequence of characters. However, these infinitely long formulae are usually handled via a finite shorthand as follows.

Let $\mathbf{A}$ be an infinite set of formulae. Their infinitary conjunction and disjunction may be abbreviated as follows:

$$\bigwedge_{A_i \in \mathbf{A}} A_i \stackrel{\text{DF}}{=} A_1 \wedge A_2 \wedge \dots \quad (\text{Infinitary conjunction})$$

$$\bigvee_{A_i \in \mathbf{A}} A_i \stackrel{\text{DF}}{=} A_1 \vee A_2 \vee \dots \quad (\text{Infinitary disjunction})$$

Some infinitary formulae are nevertheless very difficult to handle in this way.

In this chapter, we consider formal languages with various kinds of quantification. Thus, these are formal languages which must be of first-order or higher order.

12.1 Sort

A sort is a set of objects or entities that share the same type within a logical system. It may be understood as a category that groups together elements with similar properties, allowing us to distinguish between different kinds of objects when reasoning or quantifying in a logical language.

This notion organises the logical domain by ensuring that variables and predicates apply only to certain types of objects, according to their sort.

In S_1 there is only one sort, which encompasses all objects in the domain of reference. This means that no distinction is made between different kinds of entities: variables and predicates apply uniformly to all objects in the domain, regardless of their particular characteristics. For this reason, S_1 is said to be one-sorted (monosorted).

By contrast, many-sorted logics allow for multiple sorts, that is, different domains of objects. Thus, we may have one set of variables quantifying over numbers, another over persons, and so on, thereby permitting more specific relations between different kinds of objects within a single logical system.

In what follows, we will introduce a many sorted language MS , which is an approximation to how these languages may be defined (see SEP).

In order to construct MS , we need to assign each of our singular terms a sort. To do so, we need sort terms.

Let us use the set:

$$\mathbf{Stu} \stackrel{\text{DF}}{=} \{ 's', 't', 'u' \}$$

as our set of sort letters. Thus, we might define our set $\mathbf{STer}_{MS}$ of sort terms as follows.

Definition 12.1 (Sort terms of MS).

$$\mathbf{STer}_{MS} \stackrel{\text{DF}}{=} \{ s_i : s \in \mathbf{Stu} \wedge i \in \mathbf{Num}_{\mathbb{N}} \}$$

Hence, the expressions ' s ', ' s_1 ', ' $\dots$ ', ' t ', ' t_1 ', ' $\dots$ ', ' u ', ' u_1 ', ' $\dots$ ' will be sort terms. If we want just a two sorted language, we just might define: $\mathbf{STer}_{\mathbf{MS}} \stackrel{\text{DF}}{=} \{ 's', 't' \}$.

Now, in order to mark that a singular term is of a sort called ' s ', we will write ' s ' on top of a variable or constant which belongs to that sort. Thus, our individual terms will be defined as follows.

Definition 12.2 (Individual terms of MS).

$$\mathbf{ICon}_{\mathbf{MS}} \stackrel{\text{DF}}{=} \{ \overset{s}{a} : a \in \mathbf{ICon}_{S_1} \wedge s \in \mathbf{STer}_{\mathbf{MS}} \}$$

$$\mathbf{IVar}_{\mathbf{MS}} \stackrel{\text{DF}}{=} \{ \overset{s}{x} : x \in \mathbf{IVar}_{S_1} \wedge s \in \mathbf{STer}_{\mathbf{MS}} \}$$

$$\mathbf{ITer}_{\mathbf{MS}} \stackrel{\text{DF}}{=} \mathbf{ICon}_{\mathbf{MS}} \cup \mathbf{IVar}_{\mathbf{MS}}$$

Hence, the expressions ' $\overset{s}{x}$ ', ' $\overset{s}{x_1}$ ', ' $\dots$ ', ' $\overset{s}{y}$ ', ' $\overset{s}{y_1}$ ', ' $\overset{s}{z}$ ', ' $\overset{s}{z_1}$ ' are variables of the sort s ; the expressions ' $\overset{s_1}{x}$ ', ' $\overset{s_1}{x_1}$ ', ' $\dots$ ', ' $\overset{s_1}{y}$ ', ' $\overset{s_1}{y_1}$ ', ' $\overset{s_1}{z}$ ', ' $\overset{s_1}{z_1}$ ' are variables of the sort s_1 ; etc. Similarly, the expressions ' $\overset{s}{a}$ ', ' $\overset{s}{a_1}$ ', ' $\dots$ ', ' $\overset{s}{b}$ ', ' $\overset{s}{b_1}$ ', ' $\overset{s}{c}$ ', ' $\overset{s}{c_1}$ ' are constants of the sort s ; the expressions ' $\overset{s_1}{a}$ ', ' $\overset{s_1}{a_1}$ ', ' $\dots$ ', ' $\overset{s_1}{b}$ ', ' $\overset{s_1}{b_1}$ ', ' $\overset{s_1}{c}$ ', ' $\overset{s_1}{c_1}$ ' are constants of the sort s_1 ; etc.

Note here that the expressions ' $\overset{s}{a}$ ' and ' $\overset{t}{a}$ ' are two distinct singular terms, to which different denotations may be assigned – especially if ' s ' and ' t ' stand for distinct sorts.

Now we need to define our predicates. Note that some predicates are only applicable to some sorts. For instance, 'is green' cannot be said of a number, like 2.

Moreover, some relators (predicates of two or more places) require arguments of various sorts. For instance, 'reads' relates a person (in the left) to a text (in the left). In order to formalise this idea, we introduce the notion of a predicate's rank, represented by ' $\lceil \text{RankP} \rceil$ ', where P is a predicate.

Definition 12.3 (Rank of a predicate).

$$\begin{aligned} \text{Rank}(P) &\stackrel{\text{DF}}{=} \langle S_1, \dots, S_n \rangle \\ &: S_1, \dots, S_n \subseteq \mathbf{STer}_{\mathbf{MS}} \ \& \ P \in \mathbf{Pre}_n. \end{aligned}$$

Intuitively, $\text{Rank}(P)[i]$ is the set of sort terms which can occur in the i -th argument of the n -place predicate P – where $i \leq n$. In other words, $\text{Rank}(P)$ contains the ways in which an atomic formula with P can be constructed.

With tall these notions defined, the formal language $\mathbf{Fo}_{\mathbf{MS}}$ may be recursively defined as follows:

Preliminary 12.4 (Formulae of $\mathbf{Fo}_{\mathbf{MS}}$). Where $n \in \mathbf{Num}_{\mathbb{N}}$, the set $\mathbf{Fo}_{\mathbf{MS}}$ is recursively defined as follows:

1. $P \in \mathbf{Pre}_n \mathrel{\mathbb{M}} \mathbf{t}_1^{s_1}, \dots, \mathbf{t}_n^{s_n} \in \mathbf{ITer}_{\mathbf{MS}} \mathrel{\mathbb{M}} \text{Rank}(P) = \langle \mathbf{S}_1, \dots, \mathbf{S}_n \rangle \mathrel{\mathbb{M}} s_1 \in \mathbf{S}_1, \dots, s_n \in \mathbf{S}_n \Rightarrow \mathbf{Pt}_1^{s_1} \dots \mathbf{t}_n^{s_n} \in \mathbf{X} \text{ :.}$
2. $F \in \mathbf{X} \Rightarrow \lceil \neg F \rceil \in \mathbf{X}.$
3. $F, G \in \mathbf{X} \Rightarrow \lceil (F \vee G) \rceil, \lceil (F \wedge G) \rceil, \lceil (F \rightarrow G) \rceil, \lceil (F \leftrightarrow G) \rceil \in \mathbf{X}.$
4. $F \in \mathbf{X} \mathrel{\mathbb{M}} x \in \mathbf{IVar}_{\mathbf{MS}} \Rightarrow \lceil \exists x F \rceil, \lceil \forall x F \rceil \in \mathbf{X}.$
5. $\mathbf{Fo}_{\mathbf{MS}}$ is the intersection of all sets $\mathbf{X}$ satisfying 1–4.

As with ' $\mathbf{Fo}_{\mathbf{S0}}$ ', we might define ' $\mathbf{Fo}_{\mathbf{MS}}$ ' using ' $\stackrel{\text{DF}}{=}$ '.

Task 12.1. Define ' $\mathbf{Fo}_{\mathbf{MS}}$ ' using ' $\stackrel{\text{DF}}{=}$ '.

Definition 12.5 (Formulae of MS (answer to task 12.1)).

$$\begin{aligned} \mathbf{Fo}_{\mathbf{MS}} \stackrel{\text{DF}}{=} & \{ \mathbf{Pt}_1^{s_1} \dots \mathbf{t}_n^{s_n} : P \in \mathbf{Pre}_n \mathrel{\mathbb{M}} \mathbf{t}_1^{s_1}, \dots, \mathbf{t}_n^{s_n} \in \mathbf{ITer}_{\mathbf{MS}} \mathrel{\mathbb{M}} \\ & \text{Rank}(P) = \langle \mathbf{S}_1, \dots, \mathbf{S}_n \rangle \mathrel{\mathbb{M}} s_1 \in \mathbf{S}_1, \dots, s_n \in \mathbf{S}_n \} \cup \\ & \{ \lceil \neg F \rceil : F \in \mathbf{Fo}_{\mathbf{MS}} \} \cup \\ & \{ \lceil (F \heartsuit G) \rceil : F, G \in \mathbf{Fo}_{\mathbf{MS}} \mathrel{\mathbb{M}} \heartsuit \in \mathbf{Con}_{\mathbf{MS}}[2] \} \cup \\ & \{ \mathbf{Q} \vee F : \mathbf{Q} \in \mathbf{Qua}_{\mathbf{MS}} \mathrel{\mathbb{M}} x \in \mathbf{IVar}_{\mathbf{MS}} \mathrel{\mathbb{M}} F \in \mathbf{Fo}_{\mathbf{MS}} \} \\ & : n \in \mathbf{Num}_{\mathbb{N}} \end{aligned}$$

This defines a multi-sorted language.

Task 12.2. Formalise the following expressions using the sort terms ' s ' for books, ' t ' for courses, and ' u ' for persons. (You may assume that your predicate is of the appropriate rank.)

1. Every person has read some book.
2. Some student is enrolled in all courses.
3. Every book has at least one author.
4. Some course requires no book.

Solution to task 12.2.

1. Every person has read some book.

Representation: $\forall x^u \exists y^s R x^u y^s$.

2. Some student is enrolled in all courses.

Representation: $\exists x^u \forall y^t E x^u y^t$.

3. Every book has at least one author.

Representation: $\forall x^s \exists y^u A y^u x^s$.

4. Some course requires no book.

Representation: $\exists x^t \neg \exists y^s R x^t y^s$.

12.2 Quantificational singularity and plurality

We may also distinguish between quantificationally singular languages and quantificationally plural languages. This classification reflects how each language handles reference to, and reasoning about, individual objects as opposed to groups of objects.

Quantificationally singular languages, such as classical logic, focus primarily on individual entities. In these languages, quantification is carried out using singular terms and predicates, which allows us to make statements only about individual objects. For example, we may say:

Sentence 12.6. For every student, there is a book.

Representation: $\forall x (E^1x \rightarrow \exists y B^2yx)$.

Here, the quantifiers bind individual variables (' x ' and ' y '), indicating that each instance is treated separately. The logic is structured around predicates applied to singular terms, and reasoning typically revolves around individual objects rather than collections.

By contrast, quantificationally plural languages, such as plural logics, are designed to handle multiple objects simultaneously. These languages employ plural terms and plural quantifiers, which make it possible to formulate statements about groups or collections without referring to their individual members. For example, a plural logic might express:

Sentence 12.7. Some students are celebrating.

Representation: $\exists \underline{xx} (E^1\underline{xx} \wedge C^1\underline{xx})$.

In this example, the plural quantifier ' $\exists \underline{xx}$ ' allows us to refer to a group of students collectively. Plural logics facilitate reasoning about properties that apply to a group as a whole, without reducing such reasoning to claims about each individual.

12.3 Order

We can distinguish between zero-order, first-order, second-order systems, and so on. This is related to the scope of the quantifier.

For example, in propositional logics we do not have quantifiers, so they are of zero order.

Predicate logics, by contrast, have quantifiers whose scope ranges over individual variables; therefore, they are first-order logics.

Metalogical systems normally have quantifiers whose scope ranges over predicates typical of first-order logic, and thus they are second-order logics.

Aristotelian syllogistic is a difficult case to classify. One could argue that it is of second order, since its quantifiers bind terms which refer to categories, that is, they bind predicates. However, because it has no individual variables, such predicates do not apply to variables and do not satisfy the conditions of a second-order logic.

Since Aristotle's syllogistic can be reconstructed as a fragment of first-order logic, it may be more appropriate to say that it is a first-order logic.

Task 12.3. Formalise the following expressions using second order quantifiers.

1. Everything that is true is the case. (You may use '1' here.)
2. Everything that is necessary is possible.
(You may alethic modal operators here.)
3. Socrates has at least one property.
4. Someone has at least one property.

Solution to task 12.3.

1. Everything that is true is the case.
Representation: $\forall^2 p (p = 1 \rightarrow p)$.
2. Everything that is necessary is possible.
Representation: $\forall^2 p (\Box p \rightarrow \Diamond p)$.
3. Socrates has at least one property.
Representation: $\exists^2 P P s$.
4. Someone has at least one property.
Representation: $\exists^1 x \exists^2 P P x$.

Part IV

NON-STANDARD CONSEQUENCE RELATIONS

We shall now classify relations of logical consequence. Each principle or rule of logic enables us to distinguish between logical systems that satisfy it and those that do not.

We will examine some of these rules below, along with the systems that satisfy them and those that do not. The proposed subdivision is somewhat arbitrary, but it helps to organise the presentation of the rules.

Logical principles may be divided into two basic kinds: operational and structural principles.

Operational principles are the most well known as they involve logical operators, that is, expressions such as ‘ $\neg$ ’, ‘ $\rightarrow$ ’, ‘ $\forall$ ’, etc.

Structural principles, instead, specify which transformations of an inference preserve validity while abstracting entirely from the internal logical form of its formulae. They concern only the structural configuration of premises and conclusions.

In order to describe these principles, it is useful to introduce the following notation in which ‘ $\mathbf{Cn}_S$ ’ will denote a function rather than a relation.

Definition 14.1 (Consequence set).

$$\mathbf{Cn}_S(\mathbf{X}) \stackrel{\text{DF}}{=} \{F \in \mathbf{FoS} : \langle \mathbf{X}, \{F\} \rangle \in \mathbf{Cn}_S\}.$$

That is, $\mathbf{Cn}_S(\mathbf{X})$ is the set of all S-logical consequences of $\mathbf{X}$.

Let us consider the most well-known structural principles.

14.1 Extension aka Reflexivity

Say we have a set of formulae $\mathbf{X}$. In our notation, $\mathbf{Cn}_S(\mathbf{X})$ would be the set of S-consequences of $\mathbf{X}$.

Intuitively, all formulae in $\mathbf{X}$ should also follow from $\mathbf{X}$. This property is called *extension* or *reflexivity*.

Definition 14.2 (Extension aka Reflexivity).

$$\mathbf{Cn}_S \text{ is extensive}^* \stackrel{\text{DF}}{=} \mathbf{X} \subset \mathbf{Cn}_S(\mathbf{X}).$$

Extension, or reflexivity, states that every set of formulae is included in its own consequence set. In other words, any set of premises entails at least itself.

There seems to be no motivation for extension to fail. However, consider a logical system in which we want the premises of a valid inference to provide reasons for the conclusion which are independent of the conclusion itself. Thus, no inference of the form F / F should be valid under such system.

It is also noteworthy that Aristotle’s syllogistic – as well as any adequate extension of it – is not extensive.

*: The term *reflexive logic* is also appropriate. However, there are some logical systems called ‘non-reflexive’ which are related to properties of the identity symbol.

14.2 Idempotence

Consider a set of formulae X and its consequence set $\mathbf{Cn}_S(X)$. Since $\mathbf{Cn}_S(X)$ is also a set of formulae, we can also talk about $\mathbf{Cn}_S(\mathbf{Cn}_S(X))$, that is, of the set of S -consequences of X of the set of S -consequences of X .

Intuitively, if something follows from X , it should also follow from $\mathbf{Cn}_S(X)$. This property is called *idempotence*.

Definition 14.3 (Idempotence).

$$\mathbf{Cn}_S \text{ is idempotent} \stackrel{\text{DF}}{=} \mathbf{Cn}_S(X) = \mathbf{Cn}_S(\mathbf{Cn}_S(X)).$$

Idempotence guarantees that applying the consequence operation twice does not change the result. That is, once a set of consequences has been obtained, applying the consequence operation again will not produce any new conclusions, nor will it remove previous ones.

14.3 Increasingness and Monotonicity

Say we have two sets of formulae X and Y such that $X \subseteq Y$. It would be reasonable to think that, if something follows from X , it should also follow from Y , since every formula in X is also in Y .

This property is called *increasingness*.

Definition 14.4 (Increasing).

$$\mathbf{Cn}_S \text{ is increasing} \stackrel{\text{DF}}{=} X \subseteq Y \Rightarrow \mathbf{Cn}_S(X) \subseteq \mathbf{Cn}_S(Y).$$

This reflects the idea that adding more premises may expand the set of deducible conclusions.

This property is often presented as a property of derivation systems, which is commonly known as *monotony* or *monotonicity*.

Definition 14.5 (Monotonicity). $\mathbf{Cal}_S$ is monotonous or monotonic $\stackrel{\text{DF}}{=} A_1, \dots, A_n / F \in \mathbf{Der}_S \Rightarrow A_1, \dots, A_n, B / F \in \mathbf{Der}_S$.

Non-monotonic logics have been around for some decades now. Thus, it would make sense to do away with monotonicity when trying to capture defeasible reasoning; that is, cases in which, given a set of premises, we have a valid conclusion, but if we add more premises, the conclusion would no longer follow.

Consider, e.g., the following argument:

- All observed swans as of yesterday are white. $(\top_1)$
 $\therefore$ It is very plausible that all swans are white. $(\top_2)$

This seems to be valid enough. However, let us augment a premise to get the following argument:

- All observed swans as of yesterday are white. $(\top_1)$
 A black swan was observed today. $(\top_2)$
 $\therefore$ It is very plausible that all swans are white. $(\top_3)$

The sole addition of $(\top_2)$ has turned the valid argument $(\top)$ into the non-valid argument $(\top)$, which suggests a motivation to have non-monotonic or non-increasing logical systems.

14.4 Transitivity

Say we have three sets of formulae X, Y, Z such that $X \subseteq \mathbf{Cn}_S(Y)$ and $Y \subseteq \mathbf{Cn}_S(Z)$. It would be reasonable to conclude that $X \subseteq \mathbf{Cn}_S(Z)$.

This property is called *transitivity*.

Definition 14.6 (Transitivity).

$\mathbf{Cal}_S$ is transitive $\stackrel{\text{df}}{=} X \subseteq \mathbf{Cn}_S(Y) \ \& \ Y \subseteq \mathbf{Cn}_S(Z) \Rightarrow X \subseteq \mathbf{Cn}_S(Z)$.

Transitivity states that if a set of formulae X can be deduced from Y , and Y can be deduced from Z , then X can be deduced directly from Z .

This property is often presented as a property of derivation systems, which is commonly known as *cut*.

Definition 14.7 (Cut). $\mathbf{Cal}_S$ features cut $\stackrel{\text{df}}{=} X_1, \dots, X_m / A \in \mathbf{Der}_S \ \& \ A, Y_1, \dots, Y_n / B \in \mathbf{Der}_S \Rightarrow X_1, \dots, X_m, Y_1, \dots, Y_n / B \in \mathbf{Der}_S$.

Logics that do not feature the cut rule are usually known as *cut-free logics*, of which there are a number.

One reason to do away with transitivity is to investigate *default consequence* and not necessary consequence. This is the kind of consequence which is, so to speak, once inferential step away from its premises.

For instance, ‘Peter works at LMU’ is a default consequence of ‘Peter is a professor at LMU’; similarly, ‘Peter receives a salary from LMU’ is a default consequence of ‘Peter works at LMU’. However, ‘Peter receives a salary from LMU’ may not be regarded as a default consequence of ‘Peter is a professor at LMU’.

14.5 Further Properties

A consequence relation may be characterised in several ways as result of a combination of the properties defined above.

Definition 14.8 (Closure operation). $\mathbf{Cn}_S$ is a closure operation $\stackrel{\text{DF}}{=} \mathbf{Cn}_S$ is extensive, increasing, and idempotent.

Definition 14.9 (Tarskian consequence). $\mathbf{Cn}_S$ is Tarskian $\stackrel{\text{DF}}{=} \mathbf{Cn}_S$ is extensive, transitive, and monotonic.

Definition 14.10 (Substructural consequence). $\mathbf{Cn}_S$ is substructural $\stackrel{\text{DF}}{=} \mathbf{Cn}_S$ it does not feature at least one of the structural rules featured assumed by classical logic.

Operational principles, unlike structural principles, are principles concerning operators.

It is relatively easy to write very large lists of interesting operational principles. Some of them correspond to basic inference rules of diverse logical systems, and others to important derived rules or tautologies in those systems.

We will classify those principles into ...

15.1 Principles about the Conditional

The principles of the first group concern the conditional. However, as some logical systems have more than one conditional, we will use the placeholder ' $\multimap$ ' for greater abstraction.

The first such principle is known as *residuation*.

Definition 15.1 (Residuation). S is $\multimap$ -residual $\stackrel{\text{DF}}{=}$

$$A_1, \dots, A_n, B \ / \ C \in \mathbf{Cn}_S \Leftrightarrow A_1, \dots, A_n \ / \ B \multimap C \in \mathbf{Cn}_S.$$

Residuation is the property that allows us to convert the deduction of a formula C from a collection of premises $A_1, \dots, A_n$ together with a formula B into the deduction of a compound formula $B \multimap C$ from $A_1, \dots, A_n$.

S_0 and S_1 are ' $\rightarrow$ '-residual, since:

$$A_1, \dots, A_n, B \ / \ C \in \mathbf{Cn}_{S_0} \Leftrightarrow A_1, \dots, A_n \ / \ B \rightarrow C \in \mathbf{Cn}_{S_0}.$$

A better-known name for this principle is *the deduction theorem*.

Task 15.1 (Residuation).

- Determine whether the non-standard systems studied in this course are ' $\rightarrow$ '-residual.
- Out of curiosity, determine whether S_0 is ' $\leftrightarrow$ '-residual or ' $\vee$ '-residual or ' $\wedge$ '-residual (or none).

Some principles strongly related to residuation principles may be called *detachment principles*.

Definition 15.2 (Detachment). S is $\multimap$ -detaching $\stackrel{\text{DF}}{=}$ $A, A \multimap B \ / \ B \in \mathbf{Cn}_S$.

S0 and S1 are clearly ' $\rightarrow$ '-detaching, since:

$$A, A \rightarrow B / B \in \mathbf{Cn}_S.$$

This specific detachment principle is known as or *modus ponendo ponens* (MPP).

Task 15.2 (Detachment).

- Determine whether the non-standard systems studied in this course are ' $\rightarrow$ '-detaching.
- Out of curiosity, determine whether S0 is ' $\leftrightarrow$ '-detaching or ' $\vee$ '-detaching or ' $\wedge$ '-detaching (or none).

15.2 Detonation Principles

The principles of this group concern a phenomenon of most logical system which may be called *embracingness* or *detonation*. These principles usually concern the negation, for which we will use the placeholder ' $\bullet$ '.

The following are some denotation principles.

Definition 15.3 (Explosion). S is $\bullet$ -explosive $\stackrel{\text{DF}}{=} A_1, \dots, A_n / B \in \mathbf{Cn}_S \wedge A_1, \dots, A_n / \neg B \in \mathbf{Cn}_S \Rightarrow A_1, \dots, A_n / C \in \mathbf{Cn}_S$.

S0 and S1 are clearly ' $\neg$ '-explosive. This specific explosion principle is known as *ex contradictione sequitur quodlibet* (ECQ).

Task 15.3 (Explosion).

- Determine whether the non-standard systems studied in this course are ' $\neg$ '-explosive.
- Given a system $\mathbf{C0}_n$ (with $1 < n < \omega$), determine whether it is: ' $\neg^{(n-1)}$ '-explosive, or ' $\neg^{(n)}$ '-explosive, or ' $\neg^{(n+1)}$ '-explosive.
- Out of curiosity, determine whether S0 is ' $\neg\neg$ '-explosive.

A system which fails to be ' $\neg$ '-explosive is ' $\neg$ '-paraconsistent.

Definition 15.4 (Paraconsistency). S is $\bullet$ -paraconsistent $\stackrel{\text{DF}}{=} S$ is not $\bullet$ -explosive.

In paraconsistent systems it is not possible to derive anything whatsoever from two opposing premises A and $\neg A$.

Task 15.4 (Paraconsistency).

- Determine whether the non-standard systems studied in this course are ‘ $\neg$ ’-paraconsistent.
- Given a system $C0_n$ (for $n > 1$), determine whether it is: ‘ $\neg^{(n-1)}$ ’-paraconsistent, or ‘ $\neg^{(n)}$ ’-paraconsistent, or ‘ $\neg^{(n+1)}$ ’-paraconsistent.
- Out of curiosity, determine whether $S0$ is ‘ $\neg\neg$ ’-paraconsistent.

We can define a strong notion of paraconsistency:

Definition 15.5 (Strong paraconsistency). S is strongly $\bullet$ -paraconsistent^{DF}
 $\equiv \forall A_1 \dots \forall A_n \forall B \forall C (A_1, \dots, A_n / B \in \mathbf{Cn}_S \wedge A_1, \dots, A_n / \bullet B \in \mathbf{Cn}_S \wedge A_1, \dots, A_n / C \notin \mathbf{Cn}_S)$.

Moreover, we can define a more abstract version of the idea of explosion, which could well be considered a structural-type rule.

Definition 15.6 (Detonation). S is detonating or embracing*^{DF}
 $\equiv \exists M (M \subset \mathbf{Fo}_S \wedge \mathbf{Cn}_S(M) = \mathbf{Fo}_S)$.

Note that even a strongly paraconsistent logic can be detonating, as there can be several sets of formulae able to generate arbitrary conclusions.

In this regard, it is worth mentioning a type of system that, although useless, is often mentioned as a limiting case of a logical system.

Definition 15.7 (Trivial logic). S is **trivial**^{DF} $\equiv \forall M \mathbf{Cn}_S(M) = \mathbf{Fo}_S$.

It is worth mentioning here the other systems have attempted to avoid explosion by introducing a notion of relevance. In this case, it would not be sufficient for the premises to entail the conclusion according to the standard rules, but the former should also be relevant to the latter.

The very notion of relevance is nevertheless difficult to explicate. A suggested principle has been that of *variable sharing* between premises and conclusion. Logical systems in which this relevance from premises to conclusion is pursued are known as *relevance logics* or *relevant logics*.

*: The term ‘detonation’ comes from Scotch and Jennings (1989), who argued that a logic without this property would imply ‘a completely new conception of inference, which violates the classical theory of structural rules’. The term ‘embracing’ comes from Karl Popper (1943), who argued that, even if we rejected $\neg$ -explosion, logical systems still need to be embracing in the sense mentioned above.

15.3 Classical or Aristotelian principles

There are some principles typically associated to Aristotle. However, it is debatable to what extent he defended these principles, or whether he regarded them as logical ones.

Definition 15.8 (Non-contradiction). S is non-contradictory $\stackrel{\text{DF}}{=} \forall \mathbf{M} \vdash \neg(A \wedge \neg A) \in \mathbf{Cn}_S(\mathbf{M})$.

The denomination 'paraconsistent logic' was at first used to denominate systems which aren't non-contradictory. However, it was later used to denominate systems which are non-explosive.

Definition 15.9 (Excluded middle). S satisfies the *principle of excluded middle* or *tertium non datur* $\stackrel{\text{DF}}{=} \forall \mathbf{M} \vdash (A \vee \neg A) \in \mathbf{Cn}_S(\mathbf{M})$.

A system which does not satisfy this principle is called *paracomplete*. Among them are many-valued logics (with more than two truth values), fuzzy logics, intuitionistic logics, among many others.

Further Aristotelian principles are those concerning connexivity and identity, i.e., for all t , $t = t$.